2011 ROOM REVERB

The 2011 Room Reverb

How it works, why it sounds the way it does, and where it is going

A living technical book
Written for Idle Hanz, in plain language
and prepared for handover to Ben Supper

Draft of 5 October 2026 · describes build 51
Chapters 1–8 written · Chapters 9–12 vision stubs
Labels: VERIFIED · ASSUMED · LOCKED · PLANNED · DESIGNED · OPEN

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Glossary

How to read this book — start here if you are not an engineer

For Idle, first. Draft 2026-10-05.

This book explains how the 2011 Room Reverb works: what you built in 2011, what has been added since, and what is coming. It is written so you never have to take anything on trust, and so that one day someone else (Ben) can pick it up without the chat history.

The short route (about an hour)

1. **Chapter 2, §2.0** "The whole picture in plain English": one page on how a clap in a room becomes direct sound, early echoes and a tail, and how the program builds each.
2. **Chapter 2, §2.7**: a one-page preview of how Surfaces will shape the sound.
3. **Chapter 5, §5.0 and §5.6a**: the engine as 32 corridors that feed each other, and *why* a filter has to be inside the loop to change how long the tail lasts.
4. **Chapter 6, §6.0**: the bouncing ball and the tuned sponges, then the summary table in §6.5a and the auto-treat explanation in §6.6.

That is enough to understand how Surfaces will shape the sound.

The full route

Read Chapters 1 to 8 in order. In each chapter, after every dense section, there is a quoted box called **"In plain English"**. If a section loses you, skip to its box, then go back if you want the detail. In Chapter 6 each of the eight steps starts with a **"Plain first"** note before any formula.

When you meet a word you don't know

Look it up in `99_glossary.md`. Every term used in Chapters 1–8 has a one-line everyday meaning, followed by the precise version.

The four labels and what they promise

Label	What it means for you
VERIFIED	Someone measured it or read it in a named file. You can check it.
ASSUMED	A judgement or something not re-checked. Treat it as a good guess.
LOCKED	You approved it, but it is not in the program yet.
PLANNED	An idea or a later design. Nothing is decided in detail.

You will also see DESIGNED (worked out and prototyped on the research machine, not in the program) and OPEN (not decided or not known). If a sentence has none of these labels and describes something in the future, that is a mistake in the book: please flag it.

Five ideas that unlock most of the book

1. **Mirrors for walls.** Each echo is a mirror copy of the instrument behind a wall. Its distance gives its delay and its level.
2. **Corridors that feed each other.** The tail comes from 32 delay lines whose outputs are mixed and sent back round, a little quieter each trip.
3. **Once vs every trip.** A filter before the loop changes tone; a filter inside the loop changes how long each pitch rings.
4. **Bites per bounce.** Each surface swallows a share of the sound at each bounce, differently at each pitch. Tuned panels swallow one bass band.
5. **Nothing changes by accident.** Reference recordings ("goldens") and automatic tests catch any change to the sound that you didn't approve.

What this book is not

It is not the work plan. The order of the Build sessions is in `../NEXT_SESSION_PLAN.md`, and the book follows it. The book is also never ahead of the code: a chapter is updated only after a session has landed and been checked.

The 2011 Room Reverb rebuild — the living technical book

Status: draft, 2026-10-05 (plain-language pass added the same day). Nothing in this book is finished. It describes build 51 (WORKLET_VER 51, after S-MIX, D-MIX-J and S-AUDIT) and marks everything after that as `LOCKED`-not-merged or `PLANNED`.

Who it is for

Idle Hanz first: so the technology never runs away from the person who designed it. Ben Supper later: so someone who already thinks in auralisation can pick it up without the chat history.

How to read it

- **Not an engineer? Start with `HOW_TO_READ.md`.** It gives a one-hour route (Ch. 2 §2.0 and §2.7, Ch. 5 §5.0 and §5.6a, Ch. 6 §6.0, §6.5a, §6.6) that is enough to understand how Surfaces will shape the sound.
- After every dense section of Chapters 1–8 there is an **"In plain English"** box; in Chapter 6 each design step opens with a **"Plain first"** note before any formula. Worked illustrations are labelled as such and computed in the book, never presented as project settings.
- Start with **Chapter 1** (why this exists) and **Chapter 2** (the signal path). Every later chapter hangs off Figure 2.2.
- **Chapter 6 (Surfaces, panels and auto-treat)** is the key chapter for the new technology. Read Chapter 5 first.
- **Chapter 8** explains why a "bit-exact" claim can be trusted.
- Every chapter opens with a one-paragraph purpose and its source paths, and every claim is tagged:
- `VERIFIED`: read in code, a cited source, or measured, as cited.
- `ASSUMED`: judgement, a source seen second-hand, or not re-checked.
- `LOCKED`, **not merged**: Idle decided it; Build has not shipped it yet.
- `PLANNED`: an idea or design, not in the app.
- `DESIGNED` (Ch. 6): prototyped and pre-built on the box, not merged. `OPEN`: not decided or not known.
- Diagrams are mermaid (render in any mermaid viewer, VS Code, or GitHub). Where frequency matters, colours run red (low) to violet (high). Existing PNGs are copied into `figures/`; no plots were invented for this book.

Chapters

#	File	Status
1	`01_purpose_and_history.md`	`DONE` (draft)
2	`02_signal_flow.md`	`DONE` (draft)
3	`03_play_vs_render.md`	`DONE` (draft)

#	File	Status
4	`04_decisions_and_gates.md`	DONE (draft)
5	`05_fdn_and_early.md`	DONE (draft); Smooth and Movement sections are plans
6	`06_surfaces_and_panels.md`	DONE as far as the prototypes go (with a plain-English walk-through, §6.0); the shipped behaviour arrives with SL1–SL7b
7	`07_mics_and_mixer.md`	DONE (draft)
8	`08_testing_and_goldens.md`	DONE (draft)
9	`09_nonrectangular_rooms.md`	STUB, expanded (PLANNED)
10	`10_studio_one_shared_room.md`	STUB, expanded (PLANNED)
11	`11_variants_and_export.md`	STUB, expanded (PLANNED)
12	`12_handover_to_ben.md`	STUB, expanded (PLANNED)
—	`99_glossary.md`	DONE (draft): every term in Ch. 1–8
—	`HOW_TO_READ.md`	Front piece for non-engineers

The full plan, including chapters not yet started, is in `00_outline.md`. How much is written and when it is updated is in `STATUS.md`.

Relation to the roadmap

The book does not replace the roadmap. The run order is `../NEXT_SESSION_PLAN.md` (T1 → SL7b); the decisions are `../roadmap/DECISIONS_LOCKED_2026-10-04.md` and `../DECISIONS_LOCKED_2026-09-28.md`. Paths in this book are relative to the handoff folder (`build_support_2`) unless they start with `/workspace/` (box-only research).

How it grows

A chapter is updated **after** a locked session ships and its HANDOVER is mirrored to the box, never ahead of the code. Each update moves claims from " LOCKED , not merged" to VERIFIED with the new source path, and records the date in `STATUS.md`.

Book status

Date: 2026-10-05 (Mon). First draft in the morning; plain-language enrichment pass the same day.

How much is written

- **Chapters 1–8** are drafted from build 51 and the box research, now about 25,000 words including the plain-English material (about 15,000 words of the original technical text plus about 10,000 of plain-language explanation, worked illustrations and analogies).
- **Front piece** `HOW_TO_READ.md` and a `DONE` **glossary** (`99_glossary.md`, about 3,700 words, every term in Ch. 1–8).
- **Chapters 9–12** are expanded stubs (about 500–800 words each): still `PLANNED`, with a plain-English picture of the end-state vision and no invented algorithms.
- **Honest estimate:** about 40–45 % of the eventual book, and close to all of what can be written truthfully today. Most of the rest depends on code that does not exist yet: Smooth (T4), Movement (T11), Surfaces in the loop (SL2/SL3), directional tails (SL4), treatment rows (SL5), auto-treat (SL6) and the default switch (SL7a/b).
- **Chapter 6**, the key chapter, is about half done: the design, the prototype numbers, the auto-treat logic and a full plain-English walk-through are written. How the shipped version behaves, sounds and tests cannot be written until SL2–SL6.
- **Readability target:** Idle should be able to read Ch. 2 (§2.0, §2.7) and Ch. 6 (§6.0, §6.5a, §6.6) alone and understand how Surfaces will shape the sound. This has not yet been checked with Idle.

Update triggers (after the session ships and its HANDOVER is on the box)

After	Update
T1, T2, T3, T3b, T4b	Ch. 7 (desk, MS, ORTF C off); Ch. 8 (golden change at T4b)
T4 (Smooth)	Ch. 5 §5.4 and its plain box, Ch. 2 figures, Ch. 8 golden chain
P-D7	Ch. 5 small-room decay figures, Ch. 3 §3.3
T7–T10	Ch. 7 (Split, room spots, drag)
T11 (Movement)	Ch. 5 §5.5, Ch. 3 §3.5, Ch. 8 statistical gates
SL1	Ch. 6 Room result readout
SL2 / SL3	Ch. 6 core, Ch. 2 §2.5/§2.7, Ch. 3 (Surfaces in Render / Play)
SL4	Ch. 6 directional tails and calibration
SL5 / SL6	Ch. 6 treatment rows and auto-treat (the chapter Idle most needs)
SL7a / SL7b	Ch. 2, 6, 7, 8: Natural mode, Flat parity suite, new defaults, U87
Each locked session	Ch. 4 decision table; glossary; plain-English boxes of any changed section

Known source gaps (to close before the next pass)

- The S-AUDIT HANDOVER section and current GOLDEN_META are on the PC only; the newest mirrored HANDOVER is `roadmap/eq_plan/src/HANDOVER.md` (through D-MIX-J).

- No written history of 2011 → Sep 2026 on the box (Ch. 1 is thin there).
- Surfaces fitter convergence is not specified in any source.
- Speed of sound: Surfaces prototype 343 m/s vs engine 340 m/s.
- The Smooth decay-correction cause and formula are not yet known (T4 computes them).

Backup

The pre-enrichment versions of every chapter are at `/workspace/book_backup_pre_enrich_20261005/` on the box (outside the handoff folder).

- 2026-10-07: Ch. 11 gained a researched section 'Virtual speakers as virtual microphones' (PLANNED , ~3,000 words, 24 references); notes in research/virtual_speaker_mics_RESEARCH.md.

Outline — full plan of the book

> Draft 2026-10-05, updated after the plain-language pass the same day. Front piece: `HOW_TO_READ.md` (for non-engineers). Status key: DONE = written from shipped work and sources (still a draft); **PARTIAL** = written, but a large part waits on locked sessions; **STUB** = a page under 400 words that holds the idea; PLANNED = no file yet.

Front matter

File	Status
`HOW_TO_READ.md` — start here if you are not an engineer	DONE (draft)
`README.md` — map, tags, how the book grows	DONE (draft)

Every DONE chapter (1–8) now carries "In plain English" boxes after its dense sections; Ch. 2 has §2.0 (whole picture) and §2.7 (Surfaces preview); Ch. 5 has §5.0 (corridors) and §5.6a (why filters must be inside the loop); Ch. 6 has §6.0 (bouncing ball, tuned sponges), "Plain first" notes on all eight steps and §6.5a (summary table).

Part I — What it is

#	Chapter	Status	Grows after
1	Purpose and history (2011 Reaktor → web/wasm)	DONE	Idle's own history notes (none on the box yet)
2	Signal flow (source → early → FDN → mics → out; Play/Render; Flat vs Surfaces/Natural)	DONE	T4, SL2/SL3, SL7a/b
3	Play versus Render	DONE	SL3, J2 C migration
4	Decisions and gates	DONE	every locked decision

Part II — How it works

#	Chapter	Status	Grows after
5	Early field and FDN tail (Smooth, Movement, EQ split)	DONE for build 51; Smooth/Movement PARTIAL	T4 (Smooth), T11 (Movement), P-D7
6	**Surfaces, panels and auto-treat** (key chapter)	PARTIAL: prototype facts DONE, shipped behaviour waiting	SL1, SL2, SL3, SL4, SL5, SL6, SL7a/b
7	Microphones and the mix desk	DONE for build 51	T1, T2, T3, T3b, T4b, T5, T7–T9, SL7b
8	Testing, goldens and parity	DONE	each golden change; SL7a parity suite
—	Air and humidity (ISO 9613-1, the air shelf, no air on the FDN return)	PLANNED (facts scattered in Ch. 2, 5, 8)	SL7a Natural mode
—	Source radiation (J1: talking head, trumpet, cabinets; SRC_RAD)	PLANNED	none scheduled; facts in BIG_JOBS

#	Chapter	Status	Grows after
—	Binaural in depth (SADIE II, IC-matched tail, mirror table)	PLANNED (summary in Ch. 7)	S6 / J12 binaural gaps
—	Two sources and the shared tail (mirror symmetry, D-MIX-J)	PLANNED	—
—	Structure mining and 2011 parity (how the Reaktor patch was reverse-read)	PLANNED	sources to be gathered
—	CPU budget (2.667 ms block, ABBA method, what each feature costs)	PLANNED (method in Ch. 8)	SL3, T11
—	Output strip and limiter (D-AUD-3, 24-bit Render, Render length)	PLANNED	S-AUDIT HANDOVER mirrored

Part III — Where it could go

#	Chapter	Status
9	Non-rectangular rooms	STUB, expanded with plain-English vision (PLANNED)
10	One shared room in Studio One (incl. partitioned convolution)	STUB, expanded with plain-English vision (PLANNED)
11	Variants and a reverb designer (code export, 3-D delay, cut-down room, game/film binaural)	STUB, expanded with plain-English vision (PLANNED)
12	Handing the technology to Ben Supper	STUB, expanded with plain-English vision (PLANNED)

Appendices

#	Appendix	Status
99	Glossary (every term in Ch. 1–8, everyday + precise)	DONE (draft)
—	Decision index (every D-, K-, H- id with chapter links)	PLANNED
—	Source index (which file proves which claim)	PLANNED
—	Figure list	PLANNED

Chapter 1 — Purpose and history

> **Status:** DONE **(draft 2026-10-05).** Covers what exists up to build 51 (S-AUDIT, 3–4 Oct 2026) plus the decisions locked on 4 Oct 2026. Nothing here describes unmerged work as finished. **Tags:** VERIFIED = read in a source file named next to the claim, or measured on the box/PC and recorded there. ASSUMED = judgement, a design choice, or something only reported second-hand. **Paths** are relative to the handoff folder (`build_support_2/` on the box, `handoff_build_support_2\` on the PC) unless they start with `/workspace/`.

Chapter purpose. This chapter says why the program exists, where it came from, and how the work has been organised so that the person who owns it — Idle Hanz — never loses sight of how it works. It is the chapter to read first and the one to hand to anyone new (for example Ben Supper, see Chapter 12). It sets out the one idea that runs through the whole book: there is a **2011 instrument** that must stay reproducible, and there are **modern upgrades** that are allowed to change the sound only when Idle has listened and said yes.

1.1 What the program is

The *2011 Room Reverb* is a rebuild of a room reverb that Idle built in Native Instruments Reaktor in 2011. The original ensemble files are `ensemble\2011-room-reverb.ens` and `ensemble\20 October 2011 room reverb.ens`, described as "the same 887 KB family", and they are never edited (VERIFIED, `roadmap/eq_plan/src/HANDOVER.md`, "What this project is").

The 2011 ensemble models a rectangular room in two parts that work together (VERIFIED, HANDOVER "2011 wet path (locked)"):

1. **Image sources.** The source is mirrored in the walls. Each mirror image is a reflection; its distance to the microphones gives a delay (`delay_ms = d / 340 × 1000`) and a level (`1/r²`).
2. **A feedback delay network (FDN).** 32 delay lines are mixed through an unscaled ±1 Hadamard matrix and fed back. In the 2011 design the first read of each line *is* one image: "First read at each image time **is** that image. Hadamard is mix/invert only." (VERIFIED, HANDOVER.)

The output is picked up by three microphones (left, centre, right), folded to stereo with the Mixer rule `L + 0.5·C`, `R + 0.5·C` — the "477 fold" (the rule is VERIFIED, HANDOVER hard-locks table, row "Mixer fold … VERIFIED (477)"; that 477 is the number of the Structure screenshot that proves it is ASSUMED from how the HANDOVER cites screenshots).

Today the program is a desktop application on Idle's PC (VERIFIED, HANDOVER "What this project is"):

- a **FastAPI** server (`software\app_server.py`, launched by `software\Launch Tweak App.bat`) serving a browser UI at `http://127.0.0.1:7860`, built with Canvas and Three.js;
- **Play live**: real-time audio in the browser. The page posts parameters to `/api/live_params`; an AudioWorklet runs `live_stereo`, a C engine compiled to WebAssembly;
- **Render**: an offline Python path (`render_pipeline.py`) that writes a WAV file (24-bit since S-AUDIT, VERIFIED `roadmap/THURSDAY_LIST.md`, "Done since 3 Oct").

The legacy Gradio app `tweak_app.py` is reference only (VERIFIED, HANDOVER).

The UI has seven tabs: Room, Source, Mic, Mix, Space, Surfaces, Polar (VERIFIED, `ui_audit_1004/UI_AUDIT.md` §D).

> **IN PLAIN ENGLISH.**
> In 2011 Idle built a reverb in Reaktor (Native Instruments' modular sound-design software) that imagines a real rectangular room. It works out where the echoes off each wall would come from, by treating the walls as mirrors, and how late and how quiet each echo arrives (further away = later and quieter: twice the distance, a quarter of the level). It then feeds those echoes into a network of 32 recirculating delay lines that keeps them going and blends them into a smooth fading tail. Three virtual microphones listen and are mixed to stereo. Today the same idea runs as a program on Idle's PC with a browser screen. **Play** lets you hear every change live; **Render** writes a finished audio file. The original 2011 files are kept untouched as the reference.

1.2 Two lists kept apart: fundamentals and upgrades

The HANDOVER keeps two lists apart, and this book does the same (VERIFIED, HANDOVER "What this project is"):

1. **Fundamentals** — the audio core that should match the Reaktor Structure: Prepare geometry, image macros, distances, Air Abs `1/r²`, Diffuser Delay plus Hadamard feedback, 32 voices, `Out = In × (1/32) × Gain` on the 2011 path, no `1/√N` on the matrix, Mixer L/C/R fold plus RMS/AGC makeup.
2. **Upgrades** (tagged ASSUMED in the HANDOVER): free mic placement, modern toe-in and ORTF, Surfaces cards, measured mic colour, 3D views, seeded jitter, the Mix desk, Decay lock, the Play-live worklet, a stereo file as two sources, and so on.

The fundamentals were recovered by reading the Reaktor Structure screen by screen ("Structure mining"). The hard locks that came out of it are listed in the HANDOVER, each with a reference number in brackets (ASSUMED: the Structure screenshot or note that proves it), for example speed of sound `c = 340` (VERIFIED Const), `MCZ = L − MZ`, Diffuser Delay max 1000 ms (472, 476), Humidity Wet 100–700 default 400 (486), Diffusion Master 0…−0.5 default −0.25 (485) (VERIFIED, HANDOVER "Hard locks"). Some Structure questions remain open, for example "Dffs room-scale knob IC Default unread" and the To V 17–52 voice map (VERIFIED, HANDOVER "Still open (Structure)").

The engine keeps the 2011 behaviour reachable through a `reaktor_parity` path (the "Hall" button) and a Decay-lock-off path, where the loop runs `Out = In × (1/32) × Gain` and the matrix stays plain H (VERIFIED, HANDOVER "S4b", "Engine notes"). The desktop default, **Decay lock on**, is an upgrade: the loop runs at `g = 1/√N` (about 0.177) with a per-line loss `line_t60` (VERIFIED, HANDOVER).

> **Why this matters for the future.** The decision locked on 4 Oct 2026 (D-SL10) keeps this split permanent: when the defaults change in session SL7b, "Flat (2011)" stays a selectable mode and its references become a **frozen parity test** (`tests/refs_flat2011/`) (VERIFIED, `roadmap/DECISIONS_LOCKED_2026-10-04.md`). That test is not built yet (LOCKED, session SL7a).
>
> **In plain English.** There are two kinds of things in the program. **Fundamentals** are the parts that must behave exactly as in Idle's 2011 Reaktor patch; they were worked out by reading the patch screen by screen, and each one is "locked" (for example the speed of sound is 340 m/s, because that is what the 2011 patch uses). **Upgrades** are everything added since: free mic placement, wall materials, measured microphone sound, 3-D views, the mixing desk and so on. Upgrades may change the sound only when Idle decides. The "Hall" button (`reaktor_parity`) gets you back to the 2011 behaviour, and when the defaults change in the future the 2011 sound will be kept as a permanently selectable, test-protected mode.

1.3 How the work is organised

Three roles do the work (VERIFIED as a working pattern from `START_HERE.md`, `NEXT_SESSION_PLAN.md`; the description of roles is ASSUMED wording):

- **Idle** decides. Every change to the sound is Idle's decision, usually after an A/B listen. Decisions are written down with a time stamp and a number (Chapter 4).
- **Build sessions** change `software\` on the PC, "one prompt at a time" (VERIFIED, `START_HERE.md`: "Never touch `software\` from these docs: Build sessions change it, one prompt at a time.").
- **The box** (a Linux research machine) does research, prototypes, measurements, A/B renders and pre-built tested code, and writes the prompts. It works on copies; the PC `software\` is not touched from the box (VERIFIED, repeated in every box report, e.g. `roadmap/DECISIONS_LOCKED_2026-10-04.md` line 6).

Every session ends with the tests green and the Render golden hash recorded (VERIFIED, `roadmap/THURSDAY_LIST.md` §5 "Rules for every session"). A session that changes goldens changes nothing else (VERIFIED, `NEXT_SESSION_PLAN.md` rules).

> **IN PLAIN ENGLISH.**
> Idle decides; nothing that changes the sound goes in without his yes, usually after listening to an A/B comparison. **Build sessions** are the only things that change the actual program on Idle's PC, one carefully scoped task at a time. **The box** is a separate research computer that experiments on copies, measures, prepares A/B audio and writes the instructions for Build. Every session must end with all automatic tests passing and with a fingerprint of the reference sound recorded (the "golden hash", Chapter 8), so any accidental change would be caught.

1.4 Timeline (what is VERIFIED from dated notes)

The dates below come from dated sections in the HANDOVER copies and roadmap files. Earlier history (2011 to September 2026) is not documented in the sources read for this draft; see "Open" at the end.

Date (2026, BST)	What happened	Source
22 Sep	HANDOVER started as the single read-first file	HANDOVER header

Date (2026, BST)	What happened	Source
23–24 Sep	Mic brief v2.6 locked; measured mic colour Phases 1–3 (Render, Mic tab, Play)	HANDOVER "Mic, air, and wall cards"
25 Sep	Stereo file as two sources: each source has its own direct pattern and its own FDN	HANDOVER "Two-source image"
26 Sep	Play becomes one C block (`live_stereo`) instead of ~50 ConvolverNodes plus a JS loop	HANDOVER "NOW — Play is one C block"
27 Sep	"Refresh day" list: sample rate follows the stream; denormal guard	HANDOVER "NEXT — refresh day"
28 Sep	K1–K4 and items 5–7 locked (yaw convention, air default, source default, FDN-first faders, diffusion −0.25, headroom trim, colour per rate)	`DECISIONS_LOCKED_2026-09-28.md`
1 Oct	Big-jobs roadmap; S1 Foundations (blob v2, ISO 9613-1 air, mirror fix B, binaural vendoring, goldens)	`BIG_JOBS_ROADMAP_2026-10-01.md` §5
2 Oct	S2/S3 source radiation; S4/S4b Surfaces materials, Solo/Mute, H·D column signs, DC restore; Shared tail; S7 "z^{-1} in both" (Play = Render loop)	BIG_JOBS §0, §5; HANDOVER S4b/S7
3 Oct	S-UI2; S5 (D-UI26 = A, face colour on the tail inject); S5 fix-up; S-MIX desk matrix; D-MIX-J; S-AUDIT (build 51, `WORKLET_VER` 51, wasm `6ebecc333b50`)	HANDOVER copy 4 Oct 06:16; `THURSDAY_LIST.md`
4 Oct	EQ plan locked (Smooth + Movement; FWHT rotation rejected); UI audit; D18–D24, D-UI29–38; Surfaces in the loop D-SL1–9 after Idle's listen ("sounds good, certainly a lot better than flat"); ST-1, ST-2, N11, D-SL10: one default switch	`roadmap/eq_plan/EQ_PLAN.md` §4f; `DECISIONS_LOCKED_2026-10-04.md`
8 Oct (Thu)	Build usage resets 10:28 BST; Thursday sessions T1 → SL7a begin	`NEXT_SESSION_PLAN.md`

IN PLAIN ENGLISH.

In two weeks, from 22 September to 4 October 2026, the program went from a first handover note to build 51: measured microphones, live playback in one fast engine, two-source stereo, accurate air absorption, wall materials, a proper mixing desk, and a long list of fixes, each one decided and tested. On 4 October the next big steps were approved: Smooth and Movement for the tail, and Surfaces-in-the-loop for real wall-dependent decay. Build work resumes on Thursday 8 October.

1.5 Where it is going (pointer, not a plan)

The run order T1 → SL7a, and the gated SL7b, live in `NEXT_SESSION_PLAN.md`. This book **references** that plan and never replaces it. In short (VERIFIED from that table):

- T1–T3b, T7–T10: desk, mics and UI, with no change to the main golden (T3b changes MS references only).
- T4 (Smooth) and T4b (ORTF C off): the two sound changes on Thursday that regenerate the main golden.
- T11: Movement (delay modulation), default 0.
- SL1–SL6: Surfaces in the loop, as an opt-in mode with Flat (2011) bit-exact.
- SL7a: preparation and a final four-way A/B. SL7b: the single default switch, only after Idle's written OK.

Beyond that roadmap are ideas Idle has described: non-rectangular rooms, a Studio One plug-in where every track is a source in one shared room, a reverb designer that exports code or variants, and a handover to Ben Supper. These are PLANNED and appear in this book only as stubs (Chapters 9–12).

IN PLAIN ENGLISH.

The work plan from Thursday has five phases (as summarised for Idle on 5 Oct from `NEXT_SESSION_PLAN.md`): make the desk and mics easy to use; two first sound upgrades (Smooth, ORTF without the centre mic); freer mic placement and tidier screens, ending with Movement; Surfaces as an optional mode; and finally, after Idle has listened and agreed, a single switch that makes the new sound the default while keeping the 2011 sound available. Further ideas (unusual room shapes, a Studio One plug-in, a reverb designer, handing the technology to Ben Supper) are kept in this book as clearly labelled future chapters.

1.6 How to keep this book honest

- A chapter is updated **after** the session that changes its subject has landed and been verified, not before (see `README.md` and `STATUS.md`).
- Every number is copied from a source file and the file is named. If a number is not in a source, the book says "not specified" or "open".
- Unbuilt features are labelled `LOCKED` (decided, not merged), `PLANNED` (idea or later session) or `OPEN` (needs a decision).

IN PLAIN ENGLISH.

The book only describes things after they have been built and checked. Every number comes from a named file. Anything not yet built is labelled as decided-but-not-built (`LOCKED`), an idea (`PLANNED`) or undecided (`OPEN`), so you can always tell what is real today.

Open / not yet documented

- The story from 2011 to September 2026 (when the Python and web rebuild started, which tools were tried) is not in the sources read for this draft. Idle may want to add it in his own words.
- An older public document (`/workspace/docs-publish/docs/HOW_IT_WORKS.md`) gives the geometric level as `1/r`; the HANDOVER locks `1/r²` (`VERIFIED`, 256). The HANDOVER is canonical; the older document is out of date on this point.

Chapter 2 — Signal flow

> **Status:** DONE **(draft 2026-10-05).** Describes build 51 (S-AUDIT tree, `WORKLET_VER` 51) and marks every later stage as LOCKED or PLANNED. **Tags:** VERIFIED / ASSUMED as in Chapter 1. **Diagram key:** solid boxes = in the build 51 tree; dashed boxes = LOCKED or PLANNED, not merged.

Chapter purpose. This chapter follows one sound from the source to the speakers, stage by stage, so that every later chapter has a place on the map. It shows the 2011 wet path that the rebuild must keep reproducible, the modern chain as it runs today, how Play and Render split, and where the planned Natural and Surfaces decay modes plug in. If you only remember one picture from the book, make it Figure 2.2.

2.0 The whole picture in plain English

Clap your hands in a real room and listen closely. First you hear the clap itself, straight from your hands to your ears: the **direct sound**. A few thousandths of a second later come a handful of distinct echoes off the nearest walls, floor and ceiling: the **early reflections**. After that the echoes pile up so fast that you can no longer tell them apart; they melt into a smooth, fading wash: the **tail** (the reverb proper). How long that wash lasts, and whether it is boomy or bright as it fades, is what makes a bathroom sound different from a carpeted studio.

The program builds those three parts on purpose, in that order:

1. **Direct sound** — the source, delayed and made quieter by its distance to each microphone.
2. **Early reflections** — 32 echoes worked out from the room's shape, as if each wall were a mirror (Chapter 5 explains the "mirror" trick).
3. **The tail** — the same 32 echoes are fed into a recirculating network of 32 delay lines, the **FDN** (feedback delay network). Each line is like a corridor the sound runs down; at the end, the sound from all corridors is shuffled together and sent round again, a little quieter each time. Round after round, the echoes multiply into the smooth wash.

Three virtual **microphones** (left, centre, right) listen to all of it. A small **mixing desk** balances them, and the result goes to the speakers (**Play**, live) or into a file (**Render**, offline).

One more idea is needed to understand the future of the program. *Where* a filter sits decides what it can do. A filter applied **once**, before the sound enters the recirculating network, can only change the tail's tone, like tinted glass in front of a lamp. A filter **inside** the network acts again on every round, so its effect piles up over time and changes how fast each pitch range dies away. Today the walls are mostly "tinted glass". The planned **Surfaces** mode moves the walls inside the network, so that a carpeted room really does stop ringing in the treble sooner than a tiled one (§2.5, §2.7 and Chapter 6).

2.1 The 2011 wet path (locked)

The HANDOVER keeps the 2011 wet path as a locked diagram (VERIFIED, `roadmap/eq_plan/src/HANDOVER.md`, "2011 wet path (locked)"):

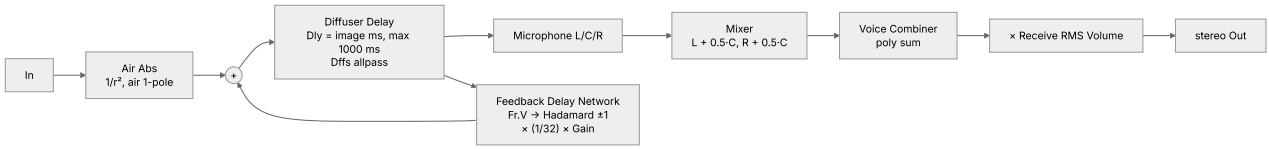


Figure 2.1. The 2011 wet path, redrawn from the HANDOVER's ASCII diagram. "× (1/32) × Gain" is the lock-off / `reaktor_parity` rule; with Decay lock on the desktop uses `g = 1/√N` and a per-line `line_t60` loss.

Three rules from the same section matter everywhere (VERIFIED , HANDOVER):

- "First read at each image time **is** that image." Each of the 32 voices is one image source, and its delay line length is that image's delay. The FDN's first pass therefore *is* the early field; there is no gap or lump between early and late "by construction" (VERIFIED , `roadmap/sound_tricks/SOUND_TRICKS.md` , "Already done or locked", citing `fdn_delays_from_taps` , `fdn.py` 785–814).
- "Hadamard is mix/invert only."
- "Mixer has **no** Delay."

IN PLAIN ENGLISH.

Figure 2.1 is Idle's 2011 design in one line. Sound goes **in**, is made quieter with distance ($1/r^2$: twice as far, a quarter of the level) and slightly duller by the air. It then enters the **Diffuser Delay**: 32 delay lines, each exactly as long as the travel time of one echo, each with a small "smearing" filter (the **allpass** or **Dffs**, which spreads a click out in time without changing its tone). The microphones listen to those lines. The lines also feed the **FDN**, which shuffles all 32 together with a fixed pattern of plus and minus signs (the **Hadamard** matrix), turns them down, and sends them back to the start. The three mics are folded to stereo as left + half the centre, right + half the centre. The three rules say: the first time you hear each line, *that is* the echo (so early echoes and tail join seamlessly); the shuffle only mixes and flips signs, it adds no delay or filtering; and the mixer adds no delay either.

2.2 The modern chain today (build 51, Flat (2011) decay)

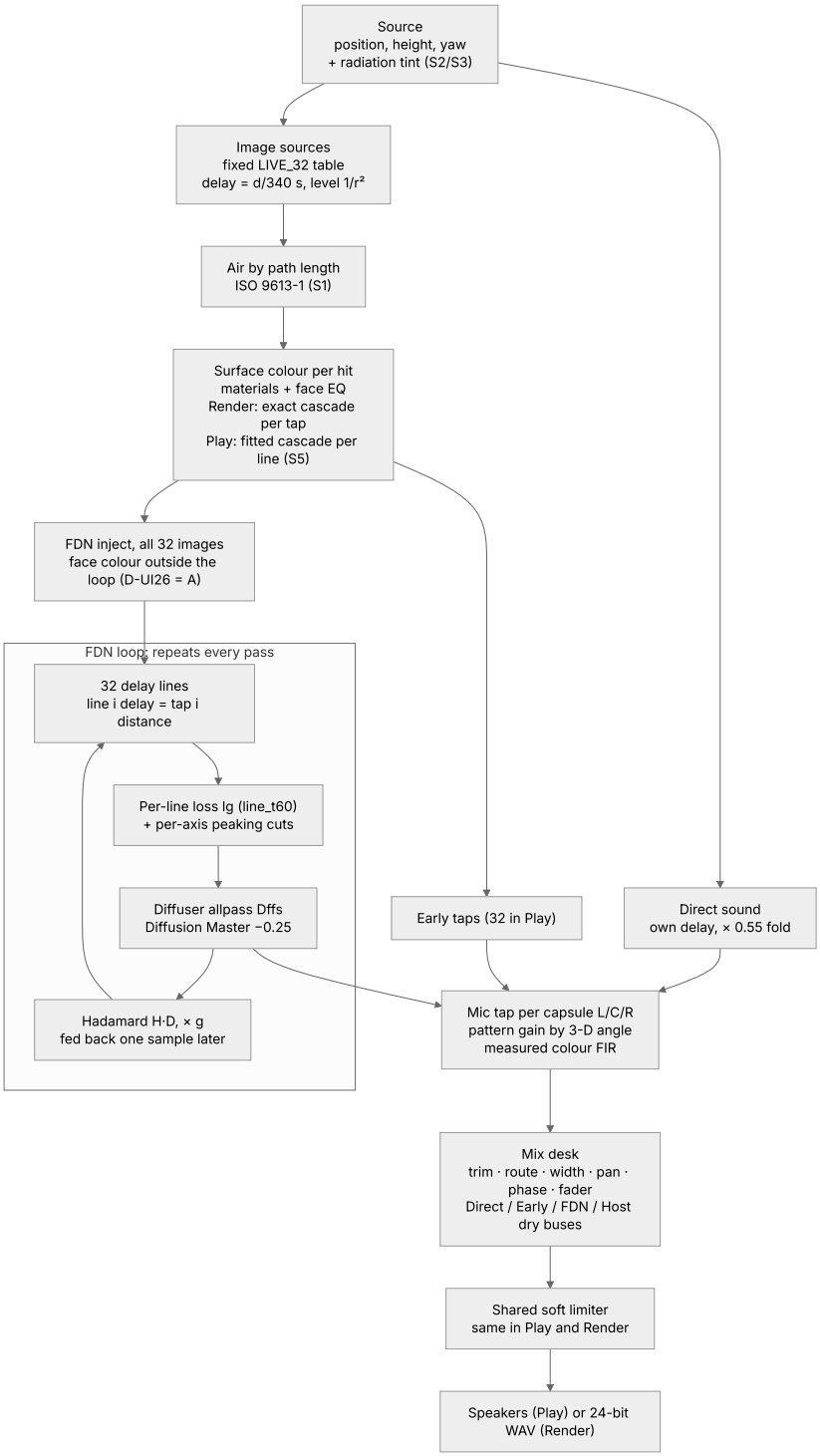


Figure 2.2. Signal chain in build 51. The order of the boxes inside the loop is schematic: see Chapter 5 for exactly where `lg`, the peaking cuts and the allpass sit in each engine.

Stage by stage:

1. **Source.** A position in metres and a height; since S2/S3 also a facing direction and a radiation pattern (instruments beam), with a tail that stays yaw-invariant (VERIFIED , `BIG_JOBS_ROADMAP_2026-10-01.md` J1 "Done in S2/S3"). The K3 default source is x 2.6 m, height 1.2 m (VERIFIED , `DECISIONS_LOCKED_2026-09-28.md` K3).

2. **Image sources.** A fixed index table, not "the loudest 32 at the mic". `LIVE_32` uses reflection orders 0–4 as 1 / 6 / 11 / 10 / 4 images; the direct image is voice 3, line index 2 (VERIFIED , `roadmap/nonrect/NONRECT_RESEARCH.md` §1 citing `early_field.py:257` ; HANDOVER hard locks). Mic patterns change tap *gains* only; they never change which images are chosen (VERIFIED , `roadmap/THURSDAY_LIST.md` U9).
3. **Air.** ISO 9613-1 air by path length on the early images and the FDN inject (S1, J6). The direct sound stays air-free (VERIFIED , BIG_JOBS J6; `3_phase_acceptance_tests/README.md` , `test_p2_air.py` : "Direct is air-free in Play and Render").
4. **Surfaces.** Each face A–F has a material card (absorption per octave) and a face EQ. Each reflection is coloured by the faces it hits (a serial per-hit cascade). Since S5 (D-UI26 = A) the face colour also reaches the tail, through a per-line feed-forward filter on the FDN inject, *outside* the loop (VERIFIED , HANDOVER "S5 — D-UI26 = A"). Being outside the loop, it changes tail **tone and level, not decay time** (VERIFIED reasoning, `roadmap/eq_plan/EQ_PLAN.md` "plan in five lines" item 3).
5. **FDN loop.** 32 lines, Hadamard mixing with fixed column signs (H·D), one diffuser allpass per line, a per-line loss set by the Decay dial, and three per-axis peaking cuts inside the loop. Details in Chapter 5.
6. **Mic tap.** Each capsule listens to the direct sound, the early taps and the FDN output. "Mic is the listen tap off the diffuser output, in parallel with feedback. It is not on the inject, and the coloured signal is not written back." (VERIFIED , HANDOVER "Mic, air, and wall cards".) The pattern gain uses the 3-D angle between capsule aim and the arriving ray. Details in Chapter 7.
7. **Desk.** Three mic strips (L, C, R) and four buses (Direct, Early, FDN, Host dry), with trim, route, width, pan, phase and fader (VERIFIED , HANDOVER "S-MIX"; `roadmap/THURSDAY_LIST.md` U7). There is no EQ on the desk (VERIFIED , EQ_PLAN §5).
8. **Limiter and output.** A shared soft limiter, the same in Play and Render, landed in S-AUDIT (VERIFIED , `roadmap/THURSDAY_LIST.md` "Done since 3 Oct"). Render writes 24-bit WAV.

The 28 Sep K4 decision made the default mix **FDN-first**: the FDN fader up, Direct and Early faders down, with the early bus ready to be raised at once without a stale replay (VERIFIED , `DECISIONS_LOCKED_2026-09-28.md` K4). In build 51 the client defaults are still `fader_direct: 0, fader_fdn: 0.5, fader_early: 0, fader_dry: 0` (VERIFIED , `ui_audit_1004/app/software/static/js/app.js` line 112). How 0.5 maps to dB on the S-MIX desk is not stated in the sources read; the HANDOVER describes the faders as "dB, 0 dB at mid-throw" (VERIFIED , older one-source live-mix note), so 0.5 = 0 dB is ASSUMED .

> **IN PLAIN ENGLISH.**
>
> Figure 2.2 is the same journey with today's extras. A **source** (an instrument at a spot in the room, facing a direction) produces a **direct sound** and **32 mirror-image echoes**. Each echo is dulled by the **air** it travels through (a longer path loses more treble) and coloured by the **walls** it bounces off (each wall has a material and an EQ). The echoes are heard as **early reflections** and also sent into the **FDN loop**, where they circulate: each round they lose a little level (set by the **Decay** dial), get smeared by the diffusers, and are shuffled together. The **microphones** listen to the direct sound, the early reflections and the circulating tail, each according to where it points and what kind of mic it is. The **desk** balances everything, a **limiter** stops it clipping, and you hear it or get a file. Important for later: today the wall colour reaches the tail only on the way *in* (outside the loop), so the walls change the tail's tone but not how long it lasts. The Decay dial alone sets the length, the same for bass and treble.

2.3 Two sources

A stereo file can be split into two sources: file left → source A, file right → source B. Each source has its own image set, its own direct pattern and its own 32-line tank; "Do not fold B's tail back into A's tank" (VERIFIED, HANDOVER "Two-source image"). Mirror places B at `(W − sx, sy, sz)` (VERIFIED, same section).

The **Shared tail** option (Play only, default off, stereo setups only) runs one network for all mics to save CPU: 2.001 ms → 1.507 ms per block, a 24.7 % saving (VERIFIED, HANDOVER "Shared tail").

> **IN PLAIN ENGLISH.**
> A stereo recording can be treated as two instruments in the room: its left channel at one spot and its right channel at another. Each gets its own echoes and its own tail, so the two sides stay distinct. "Mirror" places the second one at the mirror-image spot across the room. The optional "Shared tail" runs one tail for both in Play to save about a quarter of the processing time.

2.4 Play and Render: one design, two engines

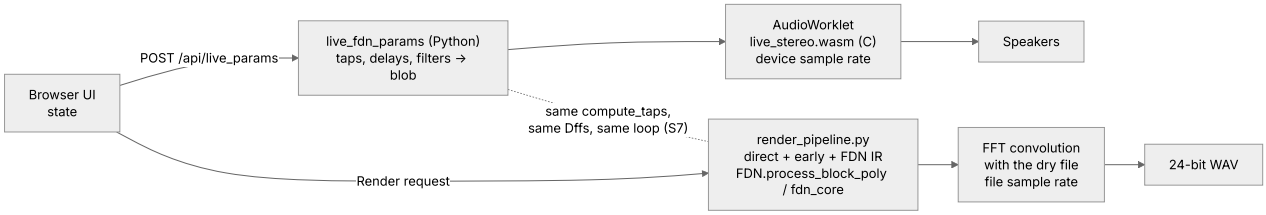


Figure 2.3. Play and Render share the geometry and coefficient code in Python; the sample loop runs in C/wasm for Play and in Python (or `fdn_core.so`) for Render. Chapter 3 explains what is identical and what still differs.

Key facts (VERIFIED unless marked):

- Play uses the same `compute_taps` as Render (THURSDAY_LIST U9, citing `render_pipeline.py` 2029; Render 1433/1462).
- Since S7 both loops store `g·H·y` and apply it on the next sample ("z^{-1} in both"); the FDN-only Play-vs-Render null is −71.8 dB (HANDOVER S7 and S5 fix-up).
- Render builds an impulse response and convolves the dry file with it (`fft_convolve`), so it can only carry time-invariant processing (EQ_PLAN §4e).
- Play runs at the sound card's rate; Render at the file's rate. "They null sample-for-sample only when the file rate and the device rate are the same number." (HANDOVER "Fix 1".)

> **IN PLAIN ENGLISH.**
> There are two engines. **Play** runs live in the browser and must finish each tiny slice of audio (2.667 thousandths of a second) on time, so it is written in C and compiled for the browser (**wasm**). **Render** has no deadline: it builds the room's complete "echo fingerprint" (the **impulse response**, what you would record if you fired a starting pistol) and then applies it to your file in one go (**convolution**). Both are fed by the same Python code that works out the geometry and the filter settings, and since session S7 their recirculating loops are built the same way, so their tails match to within −71.8 dB (far below hearing). Chapter 3 explains what still differs.

2.5 Where the decay modes plug in

Today every line loses the same amount per pass at every frequency: **Flat (2011)**. Two in-loop modes are designed and locked but **not merged** (VERIFIED status, `roadmap/DECISIONS_LOCKED_2026-10-04.md`):

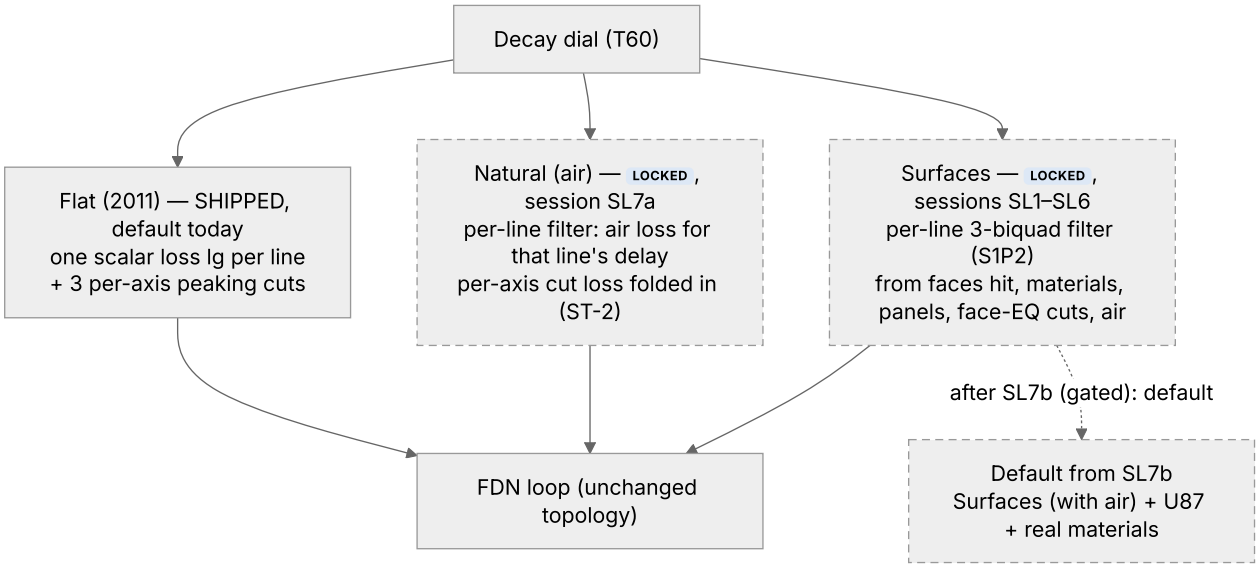


Figure 2.4. Decay modes. The FDN topology (Hadamard, 3 × 32 lines, mic and source directivity) does not change in any mode; only the per-line loss changes (VERIFIED , `roadmap/sound_tricks/SURFACES_IN_LOOP.md` §2 step 7).

- **Flat (2011)** stays selectable for ever, and its references are planned as a frozen bit-exact parity suite (LOCKED , D-SL10).
- **Natural (air)**: "Each pass round the loop now loses a little more treble than bass, by exactly the amount a sound wave loses in air while travelling that line's delay length." (VERIFIED quote, SOUND_TRICKS §1; status LOCKED as ST-1/ST-2, built in SL7a.)
- **Surfaces**: the decay per band "is therefore *not drawn*. It **emerges** from the treatment." (VERIFIED quote, SURFACES_IN_LOOP §1; LOCKED as D-SL1–9.) Chapter 6 explains it.
- **One default switch** (SL7b) makes Surfaces (with air), the U87 Ai Cardioid mics and the real default materials the default in one re-recording, only after Idle's written OK on the SL7a A/B (LOCKED , D-SL10, D-UI38 amended).

IN PLAIN ENGLISH.

A **decay mode** decides how much each of the 32 corridors loses on every round, at each pitch. - **Flat (2011)**, today: every corridor loses the same amount at every pitch, so bass and treble fade at exactly the same speed. The Decay dial sets that speed. Real rooms don't behave like this, but this is the 2011 sound and it stays available forever. - **Natural (air)**, planned for SL7a: each round, each corridor also loses the treble that air would absorb over that corridor's length. Treble now fades a bit faster than bass, as in any real space. - **Surfaces**, planned for SL1–SL6: each corridor gets a small tone control worked out from the walls its echo actually hits (materials, absorber panels, cuts in the face EQ) plus the air. The fade per pitch range is no longer set by hand; it *follows from* what the room is made of. A bare brick room rings long and bright; a treated studio dies quickly and evenly. Nothing else in the engine changes: same 32 corridors, same shuffle, same mics. Only the "loss per round" changes. After Idle has listened and agreed in writing, one session (SL7b) makes Surfaces the default.

2.6 Where EQ lives (and where it cannot)

The EQ plan's rule of thumb (VERIFIED , `roadmap/eq_plan/EQ_PLAN.md` §1 and "plan in five lines"):

- **Colour EQ** (air, materials, face EQ) sits on Surfaces, *before* the FDN.
- A cut on the FDN inject "cannot build up over passes. The FDN is LTI, so it acts exactly like a post EQ on the tail. It is a level control, not a decay control."
- Only filters **inside the loop** change decay time per band: the existing per-axis cuts, the planned resonance cuts (dB ∝ delay), and the planned Natural/Surfaces filters.
- One **Output EQ** is planned on the Mix tab after the desk (D-UI24 = A, session S-OUT, not scheduled for Thursday). There is no strip EQ.

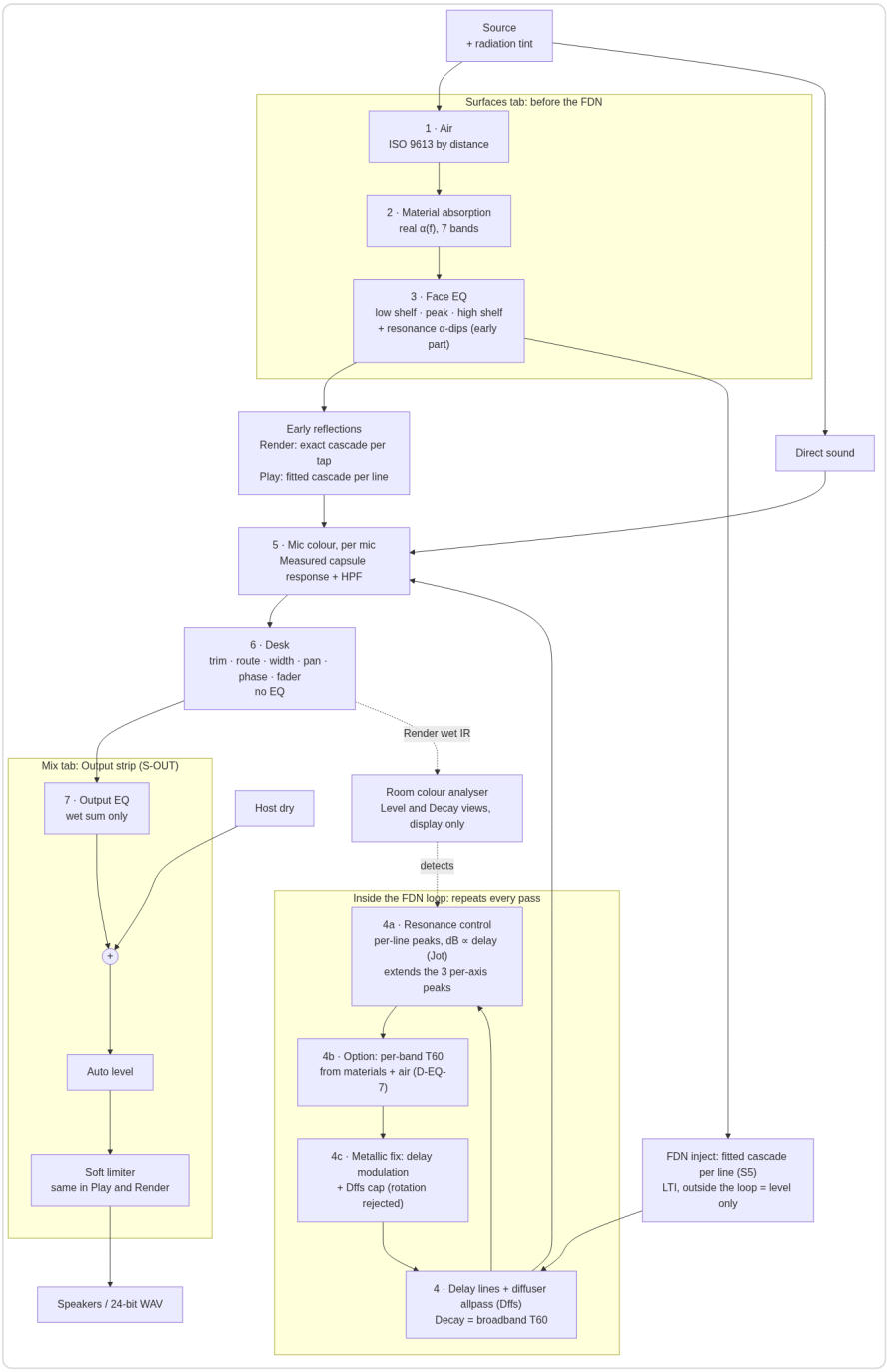


Figure 2.5. The EQ plan's signal-order diagram (copied from `roadmap/eq_plan/signal_order.png`). Note: it includes stages that are not built — 4b per-band T60, 4c delay modulation, 4a resonance control and 7 Output EQ. FWHT rotation, shown there as rejected, stays rejected.

> **IN PLAIN ENGLISH.**
> Think of the recirculating loop as a hall of mirrors with a tinted pane. - A tint **in front of** the hall (before the FDN) colours the light once: the tail comes out with a different tone, but it fades just as fast. That is why an EQ cut on the inject is "a level control, not a decay control". - A tint **on one of the mirrors** (inside the loop) acts on every bounce, so its effect multiplies: a 0.2 dB cut per round becomes several dB after half a second. That is the only way to make, say, the treble die sooner than the bass. - An EQ **after** the desk shapes the finished mix, like the tone control on a hi-fi. **LTI** ("linear, time-invariant") means a process that treats every moment the same way and simply adds signals up; a whole LTI chain can be squashed into one fixed filter, which is why order doesn't matter outside the loop but does matter inside it.

2.7 Preview: how Surfaces will shape the sound (plain English)

Status: LOCKED, DESIGNED and pre-built on the box, not merged (sessions SL1–SL6; default only after SL7b). Full detail in Chapter 6.

Picture the sound of one echo as a ball that bounces round the room. At every bounce the surface it hits takes a bite out of it. A hard plaster wall takes a tiny bite; a thick absorber panel takes a big one. The bite also depends on pitch: a foam or mineral-wool panel bites hard at mid and high pitches but hardly touches the bass; a **tuned panel** (a membrane or a perforated "Helmholtz" board) is like a sponge that only soaks up one narrow band of bass, around the frequency it is built for. Air takes its own small bite of the treble all the time.

In Surfaces mode the program does that bookkeeping for each of the 32 corridors:

1. For each wall it works out how much it absorbs at each pitch, from the base material plus any panels you add.
2. From the whole room it predicts a decay time per pitch band (a classical formula; shown in a "Room result" panel).
3. For each corridor it looks at the walls that corridor's echo hits and turns the bites into "lose this many dB per round at this pitch".
4. It sets the corridor's small tone control (three filter stages) to match, checks that nothing could ring forever, and after one test render nudges any octave that came out too long or short.

The result is that the decay per pitch band is not a curve someone drew: it **comes out of** the room's materials and treatment, the way it does in a real room.

Auto-treat is a helper button. You say how long you want the room to ring (the Decay dial); it suggests a list of treatment — first broadband panels spread over four walls, then up to four tuned bass panels aimed at the worst boomy bands — and shows before/after. Nothing changes until you accept or edit the list. With Decay lock on, the dial still sets the mid-band length and the materials set the shape around it.

Open / gaps

- Fader value → dB mapping on the S-MIX desk (0.5 = 0 dB is ASSUMED).
- **Resolved while writing:** the sources disagree on whether Play follows Render's Hadamard size chips (the HANDOVER's older "Engine notes" say "Play live N locked to 32"; BIG_JOBS lists "Play `hadamard_n`

(t17)" as done in S3). The build 51 code settles it: `live_fdn_params` "honours hadamard_n 8, 16, and 32. 64 and 128 stay on the 32-line network" (VERIFIED , `ui_audit_1004/app/software/render_pipeline.py` 1940–1962). The HANDOVER engine note is out of date.

Chapter 3 — Play versus Render

> **Status:** DONE **(draft 2026-10-05).** Build 51 facts plus the locked plans that touch parity (Movement, Surfaces, Natural). Numbers are copied from the named files.

Chapter purpose. The program has two engines that must sound like one room: **Play**, which runs live in the browser, and **Render**, which writes a file. This chapter explains why there are two, what is guaranteed identical between them today, what still differs and why, and which planned features put that parity at risk. It is the chapter to read before any change that touches the FDN loop.

3.1 Why two engines

- **Play** has to finish every audio block in time. The block budget is 128 samples at 48 kHz, i.e. 2.667 ms (VERIFIED, HANDOVER S5 "Gate 0.8 × 2.667 = 2.133 ms"; `BUDGET_MS` stays `128/48`, HANDOVER S5 fix-up). It runs as one C function, `live_stereo` in `software/engine/live_stereo.c`, compiled to WebAssembly and called from an AudioWorklet (VERIFIED, HANDOVER "NOW — Play is one C block").
- **Render** has no deadline. It is Python (`render_pipeline.py`, with `engine/fdn.py` and an optional C helper `fdn_core.so`), builds an impulse response and convolves the dry file with it (VERIFIED, EQ_PLAN §4e: "Today's Render builds one IR and convolves the dry file with it (`render_pipeline.py` `fft_convolve`)").

Both are driven from the same Python geometry and coefficient code: Play's parameters come from `live_fdn_params`, which uses "Same geometry / inject / Dffs / cut-only peaking as the offline IR" (VERIFIED, docstring, `ui_audit_1004/app/software/render_pipeline.py` 1940–1950).

> **IN PLAIN ENGLISH.**
> Live sound has a deadline. The sound card asks for the next 128 samples (2.667 thousandths of a second at 48 kHz) and if the program is late you hear a click. So **Play** is one compact piece of C code running inside the browser. **Render** has all the time it wants: it calculates the room's complete echo fingerprint (the **impulse response**, what a microphone would record after a single click in the room) and then applies it to the whole file (**convolution**: every sample of the file triggers a copy of the fingerprint, and all copies are added up). Both engines get their settings from the same Python code, so they start from the same room.

3.2 What is identical today (VERIFIED)

Part	Status in build 51	Evidence
Image selection	Both use `compute_taps` and the same image tables	THURSDAY_LIST U9
FDN loop structure	"z^{-1} in both": Play and Render both save `g·H·y` and add it on the next sample when any Dffs is set; the no-Dffs path is same-sample in both	HANDOVER S7 (D3)
Loop trims	Equal −1.70 dB F3 trim on both paths (D3b)	HANDOVER S7
FDN-only null (t18)	−71.8 / −71.8 dB after the S5 fix-up (gate −30 dB; target −60 dB)	HANDOVER S5 fix-up D-S5c-3
Tail T30, std room 1.8 s	Play and Render 1.70, 1.72, 1.80, 1.80, 1.80 s (500 Hz–8 kHz), ratio 1.000	HANDOVER S7

Part	Status in build 51	Evidence
Face colour on the tail	Render inject uses Play's fitted cascade (`fit_line_cascade` , same K and f32 rounding); worst tail band difference 0.019 dB on the default scene	HANDOVER S5 fix-up D-S5c-3
Mirror	`test_mirror_exact` : Render −232.3 dB, Play max-abs 0 (gate ≤ −120 dB)	HANDOVER D-MIX-J
Limiter	One shared soft limiter, same in Play and Render	THURSDAY_LIST "Done since 3 Oct"

IN PLAIN ENGLISH.

Today the two engines agree on everything that matters for the tail: which echoes are used, how the recirculating loop is built, the level trims, the decay times (identical to three decimal places in the standard room), the wall colour on the tail, mirror symmetry and the limiter. "Null" is the test used: play both versions, flip one upside down and add them; whatever is left is the difference. −71.8 dB means the difference is about 4,000 times quieter than the signal (computed here: 10^(71.8/20) ≈ 3,900).

How "z^{-1} in both" was found

Before S7, Render solved the diffuser and Hadamard on the same sample, which needs a matrix solve, while Play fed back one sample later. The F5 research (2 Oct) showed that Play's diffuser "is already a true allpass", that Idle's proposed restructure "is what Play already does", and that the fix was to give **Render** Play's loop (VERIFIED , `research_f5_allpass/REPORT.md` §0). The alternative — porting the same-sample solve into Play — gave the same parity at **+18.5 % Play CPU**, estimated to fail the CPU gate, whereas "z^{-1} in both" cost 0 ms on Play and made Render's loop 14 % faster (VERIFIED box, PC estimate ASSUMED , same §0). Idle accepted D3/D3b; S7 landed it.

The lesson for the book: when Play and Render disagree, prefer changing Render to match Play, because Play carries the CPU constraint (ASSUMED as a general rule; it is what happened in S7 and in D-S5c-3).

IN PLAIN ENGLISH.

"z^{-1}" is engineering shorthand for "one sample later". Before session S7 the two engines built the loop slightly differently: Render worked out the feedback on the same sample, which needs heavy maths, while Play fed it back one sample later. Research showed Play's way was already correct and cheaper, so Render was changed to match Play. The general lesson: when they disagree, change Render, because Play is the one with the deadline.

3.3 What still differs (and why)

1. **The early field.** Render's early field is "the full geometric image set with the absolute cascade"; Play's is "32 FDN-timed taps with the relative fit" (VERIFIED , HANDOVER S5 "Misses"). Harness test t13 still XFAILs on the early field: L energy +3.25 dB, R +3.37 dB (VERIFIED , HANDOVER S7). The 0–80 ms full-mix gap for all-faces HS+6 was 0.964 dB, reported but not gated (VERIFIED , HANDOVER S5 fix-up D-S5c-4). Listed as known open gate L11 (VERIFIED , THURSDAY_LIST §3).
2. **Sample rate.** Play runs at the sound card's rate (Chrome picks the device rate; a 44.1 kHz file is resampled to 48 kHz on a 48 kHz device); Render runs at the file's rate. "They null sample-for-sample only when the file rate and the device rate are the same number." (VERIFIED , HANDOVER "Fix 1".) In build 51, `live_fdn_params` designs delays, the decay-lock mean, air, the shelf and peaking at the posted rate when one is sent (VERIFIED , docstring lines 1946–1949).
3. **44.1 kHz interpolation (D-INT, open as N6).** On a 44.1 kHz device all Play delays become fractional and `read_lane` interpolates linearly, giving "−0.05 to −2.4 dB per pass at 10 kHz, uneven per line". Computed, not rendered: the 10 kHz T60 would fall from 1.8 s to about 0.63 s in the default room

(VERIFIED as computed, EQ_PLAN §2). "Parity tests run at 48 kHz, so they cannot see it" (ASSUMED, same row). The recommendation is to round delays to whole samples at the device rate (EQ_PLAN §6, D-INT); not scheduled on Thursday.

4. **Small-room decay in Play (D7 / D21).** The long-standing `--slow` fail: the small room's 1 kHz band decays in 0.732 s against a 0.8 s dial (−8.5 %) (VERIFIED, THURSDAY_LIST N10). D21 (locked) says: re-measure Play on the PC after T4; if Play ≈ Render (≈ −4 % at 1 kHz) accept and widen the tolerance; if Play is still ≈ −8 %, treat it as a live-tank bug, and report before any change (VERIFIED, DECISIONS_LOCKED_2026-10-04 D21). Session P-D7, a PC step with no code.
5. **Colour at other rates.** Measured mic colour has SOS tables for 44.1 and 48 kHz only; "other rates turn colour **off with a visible notice**" (VERIFIED, DECISIONS_LOCKED_2026-09-28 item 7).

> **IN PLAIN ENGLISH.**
> Five known differences remain. (1) The early echoes: Render uses the complete set, Play only the 32 that also feed the tail, so Play's early part is about 3 dB louder; a known open item. (2) Play runs at the sound card's sample rate, Render at the file's, so they match sample-for-sample only when those are the same. (3) On a 44.1 kHz sound card Play's delays fall between samples and the in-between values are estimated in a way that dulls the treble on each trip; a fix (rounding to whole samples) is recommended but not scheduled. (4) The small room's mid band decays a little fast in Play; it will be re-measured on the PC before anything is changed. (5) Measured mic colour only exists for 44.1 and 48 kHz; at other rates it switches off and says so.

3.4 The CPU budget

Play's limits are measured on the PC with harness test t19 (two directional sources, colour on). The gate is the **median**, not the mean, since S7 (VERIFIED, HANDOVER S7 "CPU gate"). The ceilings and recent numbers:

Measure	Ceiling	Recent value	Source
t19 shared tail off	2.40 ms (0.90 × 2.667)	median 2.22415 ms (after D-MIX-J)	HANDOVER S5 fix-up, D-MIX-J
t19 shared tail on	2.133 ms (0.80 × 2.667)	median 1.74235 ms	HANDOVER D-MIX-J
FDN fader at 0	—	2.17 ms (tank keeps running for unmute)	THURSDAY_LIST N17

Acceptance uses a 12-round paired ABBA comparison against the previous wasm (VERIFIED, HANDOVER S5 fix-up D-S5c-2). Headroom is therefore small — the EQ plan estimates Output EQ 0.003 + delay modulation 0.015 + up to three extra in-loop slots 0.141 ≈ 2.38 ms, "It fits, only just" (ASSUMED estimates, EQ_PLAN §5).

> **IN PLAIN ENGLISH.**
> Play has a fixed time budget per slice of audio, 2.667 thousandths of a second, and the hardest test case (two directional sources with mic colour on) already uses about 2.2 of them on Idle's PC. The safety ceiling is 2.40 (90 %). Because timings on a computer jitter, new and old versions are timed alternately (old, new, new, old, twelve times) and the middle value (**median**) is compared. The margin is small, which is why every new feature has to say what it costs.

3.5 Planned features that touch parity

Feature	What it does to parity	Status
Smooth (Dffs cap 0.7)	Applied where Dffs is made (`tap_dff`), "used by both the offline FDN and `live_fdn_params`. Play and Render then get identical values from one source."	LOCKED, session T4 (`roadmap/eq_plan/EQ1a_PROMPT.md`)
Movement (delay modulation)	Time-varying, so it cannot live in a single IR. "Render runs the dry signal through the loop when Movement > 0; at Movement 0 it keeps today's IR + convolution path." LFO phase indexed by absolute sample from a seed, one shared C function, so sample-exact is possible if both start at t = 0 (ASSUMED design)	LOCKED, session T11 (EQ_PLAN §4c, §4f)
Surfaces in the loop	One Render calibration pass; "Play uses the Render-designed coefficients" (D-SL4). Box SL3 cost: +0.031 ms per block (wasm). Flat Play bit-exact	LOCKED, SL2–SL4 (`roadmap/prebuild/sl/README.md`)
Natural (air)	Same per-line hook as Surfaces, Render + Play; stop if it needs > +0.05 ms of Play CPU over Flat on top of Surfaces' budget	LOCKED, SL7a
C migration (J2)	"one C core for Play, Render and later the plug-in, so Play and Render can no longer drift"	PLANNED, not started (BIG_JOBS J2)
Lean convolution mode (Idle's idea)	A partitioned convolution of a rendered IR for many stacked instances. "A convolution engine is LTI, so it cannot carry modulation exactly"; options: no modulation, or a few cross-faded IR variants	PLANNED idea, L6 (EQ_PLAN §4e; THURSDAY_LIST L6)

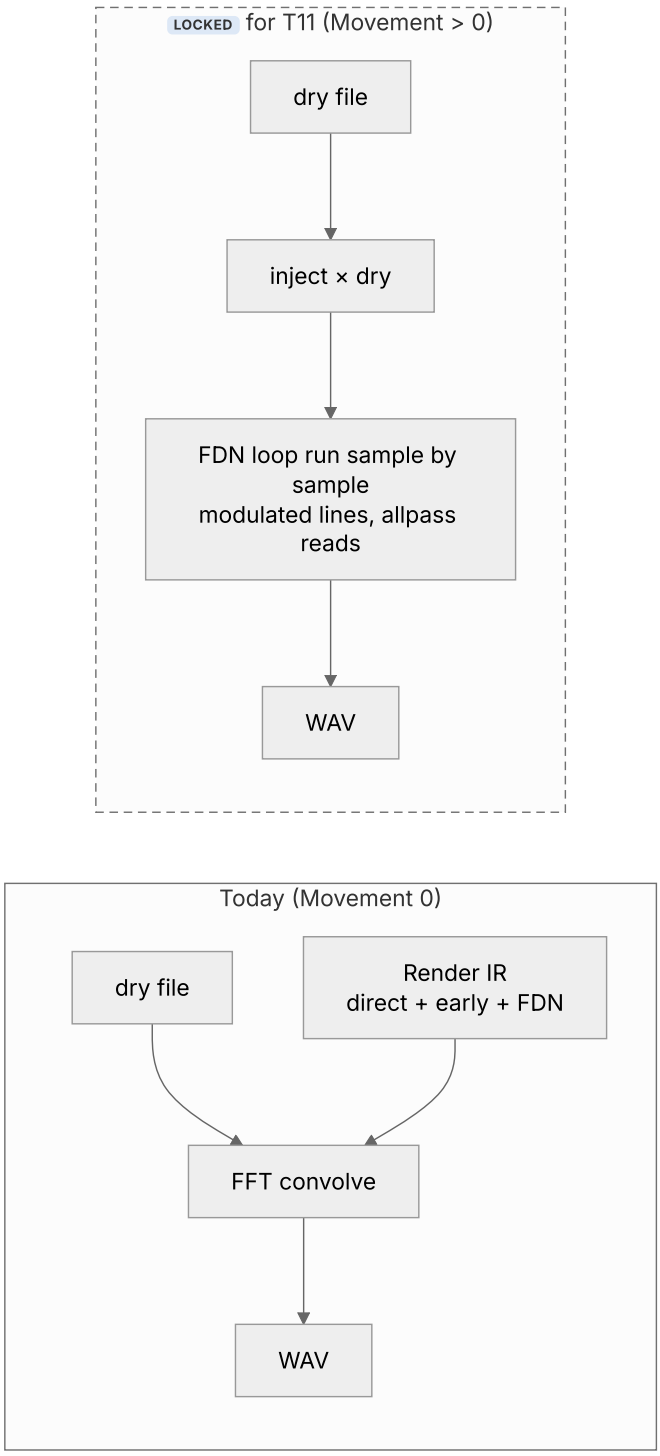


Figure 3.1. Render today versus Render with Movement. Only time-varying features force the loop path; LTI features (cuts, Dffs cap, EQ) keep the IR path (VERIFIED, EQ_PLAN §4e).

IN PLAIN ENGLISH.

Each planned feature has a plan for keeping the two engines in step. Smooth is calculated in one place for both. Surfaces is designed once in Render and then copied into Play. Movement is the hard one: a tail that changes over time cannot be captured in a single fingerprint, so Render will run the loop sample by sample whenever Movement is turned up. Further ahead, moving everything into one shared C engine would make drift impossible. A convolution-only plug-in (Idle's idea) would be very efficient but could not carry Movement exactly.

3.6 Rules of thumb (for future sessions)

- Any change to the loop must keep t18 (FDN-only null) at or near −71.8 dB and `test_mirror_exact` at ≤ −120 dB (VERIFIED as gates, `roadmap/eq_plan/EQ1a_PROMPT.md` G6).
- Default-off features must be **bit-exact** when off, in both engines (VERIFIED as the pattern of every locked prompt, e.g. EQ1a G1, SL2 "Flat default bit-exact null test").
- Never put a 32×32 matrix multiply back in the audio thread; never drop Play below 32 lines (VERIFIED , HANDOVER "NOW — Play is one C block").
- Do not raise `--perf-scale` or loosen `BUDGET_MS` to pass t19 (VERIFIED , HANDOVER S5 / S7).

> **IN PLAIN ENGLISH.**
> Four rules protect the match: the tails must keep nulling deeply and mirror tests must stay perfect; any new feature, when switched off, must give exactly the old sound in both engines; never put back the heavy per-sample maths or reduce Play's 32 lines; and never pass the CPU test by loosening the test.

Open / gaps

- Render length (D-AUD-2) was locked as "Render length follows Decay + short end fade (default 1.8 s unaffected: old cut at −93 dB; Decay 4 s T30 3.40 → 3.98 s, VERIFIED)" (`BIG_JOBS_ROADMAP_2026-10-01.md` , D-AUD list) and is listed as landed in S-AUDIT; the PC measurements from the S-AUDIT session itself are not on the box.
- No source states how Play and Render will be checked against each other at 44.1 kHz once D-INT is fixed.

Chapter 4 — Decisions and gates

> **Status:** DONE **(draft 2026-10-05).** Register of how decisions are made and checked, with the decisions locked up to 4 Oct 2026 20:41 BST. Tags copied from `roadmap/DECISIONS_LOCKED_2026-10-04.md` and `DECISIONS_LOCKED_2026-09-28.md` where present.

Chapter purpose. This project only works because nothing changes the sound without Idle knowing. This chapter explains the machinery that guarantees that: how a decision is proposed, tagged, locked and numbered; what a "gate" is and which gates every session must pass; which ideas were tried and rejected (so nobody rediscovers them); and which decisions are still open. The full current run order stays in `NEXT_SESSION_PLAN.md`; this chapter only explains the rules behind it.

4.1 How a decision is made

1. **Research on the box.** A question is researched, prototyped and measured on a copy of the software. The report gives options with a recommendation in bold, and says for each whether it changes the sound and the goldens (VERIFIED pattern, e.g. `roadmap/sound_tricks/SURFACES_IN_LOOP.md` §9: "Decisions for Idle (every sound change; recommendation in bold)").
2. **Listening.** Sound changes come with A/B audio, loudness-matched, with a listening checklist (VERIFIED, e.g. `roadmap/sound_tricks/surfaces_ab/LISTENING_CHECKLIST.md`, `roadmap/sound_tricks/ab/`).
3. **Idle locks it.** A lock is written with the time and Idle's words where available, e.g. "Idle accepted all 14 recommendations from the 16:32 research report at **16:43 BST, 4 Oct 2026**", and the Surfaces lock at 20:37 after "sounds good, certainly a lot better than flat" (VERIFIED, `roadmap/DECISIONS_LOCKED_2026-10-04.md`).
4. **A session lands it.** Each locked decision names the session that will merge it and its golden impact. "Nothing below is merged yet." (VERIFIED, same file, line 5.)
5. **HANDOVER block.** The first Thursday session pastes the HANDOVER block from the lock file into `software\docs\HANDOVER.md`, so the decisions travel with the code (VERIFIED, `START_HERE.md` step 1).

The two tags

- VERIFIED = "measured or tested on the box (build 51 copy)" or "measured by a script/test in this folder".
- ASSUMED = "judgement" (VERIFIED definitions, DECISIONS_LOCKED 2026-10-04 line 6 and 2026-09-28 line 7).

Some reports add finer labels (for example "Build-reported", "Computed, not rendered" in EQ_PLAN §8, or "FORUM" in the Studio One design). This book keeps them when quoting.

> **IN PLAIN ENGLISH.**
> Every sound change follows the same path: the research machine tries it on a copy and measures it; Idle gets level-matched A/B audio and a checklist; Idle says yes or no, and the yes is written down with the time and his words; a named Build session later puts it into the program. Every claim is marked VERIFIED (measured or read in a file) or ASSUMED (a judgement), so you always know how solid it is.

4.2 Decision families (numbering)

Family	Meaning	Example	Source
K1–K4, 5–7	28 Sep engine decisions, implemented and tested on the box tree	K1 one yaw convention flip `YAW_X_SIGN = +1`	`DECISIONS_LOCKED_2026-09-28.md`
D1–D17	Earlier engine decisions (2 Oct, BIG_JOBS §6), e.g. D3/D3b z^{-1} in both, D11–D17 Solo/Mute rules	D16 Solo keeps every image touching the face	BIG_JOBS §6; HANDOVER S4b
D18–D24	4 Oct sound/engine	D20 ORTF centre mic off by default	DECISIONS_LOCKED 2026-10-04
D-UI1–D-UI38	UI decisions (some change sound, e.g. D-UI26, D-UI30)	D-UI37 Mirror / Linked / Split control	same; BIG_JOBS §0.0
D-SL1–D-SL10	Surfaces in the loop and the single default switch	D-SL5 auto-treat suggests rows	same
ST-1–ST-10	Sound tricks (ST-1, ST-2 locked; ST-3–ST-10 open)	ST-1 natural decay on by default	same; SOUND_TRICKS
D-EQ, D-RES, D-DFF, D-INT	EQ plan decisions	D-DFF locked as the Smooth switch	EQ_PLAN §6
D-NR1–D-NR8	Non-rectangular rooms (all open)	D-NR3 FDN delays frozen per preset	NONRECT_RESEARCH §6
N, U, R, L, B, P, Q	Thursday list item ids: needs-decision, Idle's UI notes, ready, later, audit bugs, polish, questions	N18 → D20	THURSDAY_LIST; UI_AUDIT

Numbers are never reused: "D-UI27–31 and D1–D17 were already used, so these take **D18–D24** ... and **D-UI32–D-UI38**" (VERIFIED, DECISIONS_LOCKED 2026-10-04 line 4).

> **IN PLAIN ENGLISH.**
> Decisions get short labels so they can be referred to precisely: K for the 28 September engine decisions, D for engine and sound decisions, D-UI for screen decisions, D-SL for Surfaces, ST for sound tricks, D-NR for non-rectangular rooms, and so on. A number is never reused, so "D20" always means the same thing.

4.3 Locked but not merged (as of 5 Oct 2026)

All of these are LOCKED and land on or after Thursday 8 Oct (VERIFIED, DECISIONS_LOCKED 2026-10-04; NEXT_SESSION_PLAN).

Decision	In one line	Session	Main golden
D18	MS presets decode at unity, −3 dB trim on M and S (gain 0.708)	T3b	MS refs only
D19	MS direct sound uses the pattern spikes, Render + Play	T3b	MS refs only
D20	ORTF = centre mic off by default; "ORTF + C" preset keeps today's sound	T4b	YES
D21	Re-measure Play's small room on the PC, then accept/widen or fix (report first)	P-D7	none
D22	"Physical" realism options opt-in; 2011 stays default	later	NO
D23	Gate the tank after the tail dies (−96 dB); low priority	later	NO
D24	Shared velvet-noise smear, default off, measure CPU first	later	NO
D-UI29, D-UI31	MS presets a/b/c (default b); L/R desk buttons with phase on every strip	T3, T2	NO (T3 adds new MS tests)

Decision	In one line	Session	Main golden
D-UI30	U87 Ai Cardioid on L, C and R instead of SM57	SL7b (gated; T6 merged)	YES
D-UI32–34	Surfaces / Polar / Space redesigns	later	NO
D-UI35–37	Room size on Room only; "On stand"/"Room mic" + solo-in-place; Mirror/Linked/Split + wall drag "stop"	T10, T2, T7/T8	NO
D-UI38 (amended)	U87 default only after Idle's written OK on the SL7a final A/B	SL7b	YES
Smooth (D-DFF)	Diffuser gain cap 0.7, default ON, decay-corrected	T4	YES
Movement	Delay modulation knob, default 0	T11	NO (new Movement refs)
D-SL1–9	Surfaces in the loop (Chapter 6)	SL1–SL6	NO until SL7b
ST-1, ST-2, N11/D4	Natural decay in the loop; cut loss folded; frequency-dependent decay becomes default	SL7a, SL7b	YES in SL7b
D-SL10	One re-recording session switches Surfaces + natural decay + U87; Flat (2011) frozen as a parity test	SL7a, SL7b	YES in SL7b only

IN PLAIN ENGLISH.

This table is the list of things Idle has already approved that are not in the program yet. The right-hand column says whether each one will change the reference recordings ("goldens"). Most do not. The ones that do (ORTF without the centre mic, Smooth, and the final default switch to Surfaces with U87 mics) are deliberate sound changes and each happens in a session of its own.

4.4 Gates: what every session must prove

A **gate** is a measurable condition a session must meet before its work counts. Gates are written into each prompt with a **stop rule**: if a gate cannot be met, the session ships the safe variant or reverts, and reports (VERIFIED pattern, e.g. `roadmap/eq_plan/EQ1a_PROMPT.md` "Stop rule").

Rules for every session (VERIFIED, `NEXT_SESSION_PLAN.md` "Rules"; THURSDAY_LIST §5):

- One prompt = one Build session in a new window.
- Work above the **cut line** first and commit; go below only if the suite is green with time left.
- "A stop condition ends the session cleanly with a report, never a half-merge."
- "A session that changes goldens changes nothing else."
- End with the full suite green, the Render golden hash recorded in HANDOVER, and a synced snapshot.

Standing numeric gates (VERIFIED, quoted from the prompts and HANDOVER):

Gate	Threshold	Where it comes from
Main Render golden	bit-exact unless the session is a declared golden session	THURSDAY_LIST §5 "Goldens column"
Feature off	max-abs 0 against the pre-session tree, Render and Play	EQ1a G1; SL2/SL3 "Flat bit-exact"
t18 FDN-only Play–Render null	≤ −60 dB, target −71.8 dB	EQ1a G6
`test_mirror_exact`	≤ −120 dB	EQ1a G6; HANDOVER D-MIX-J

Gate	Threshold	Where it comes from
t19 CPU (PC, paired)	shared-off ≤ 2.40 ms, shared-on ≤ 2.133 ms	EQ1a G3; HANDOVER S5 fix-up
Decay	Smooth: octave T30 within ±5 % of Smooth off; mid T30 within ±5 % of the dial	EQ1a G2
Play T30 vs dial	±8 % (phase test P0.2)	`3_phase_acceptance_tests/README.md`
Surfaces	default room ≥ 16/23 ⅓-octaves within ±10 % after auto-treat	SURFACES_IN_LOOP §10 S-SL6
Listening	Idle's written OK on the A/B for any default switch	D-UI38, D-SL10

Chapter 8 explains the test suites behind these numbers.

IN PLAIN ENGLISH.

A **gate** is a pass/fail test a session must meet before its work counts, like an MOT for the software. If a session cannot pass, it must stop cleanly, ship the safe version or undo the change, and report; it may never leave half a change in place. The standing gates check that the reference sound is unchanged unless the session was allowed to change it, that new features switched off change nothing, that Play and Render still agree, that the processing still fits the time budget, that decay times are accurate, and that any change to the default sound has Idle's written OK.

4.5 Tried and rejected (do not repeat)

Idea	Why rejected	Source
FWHT stage rotation against metallic ringing	Named peaks moved only 0–4 dB, peakiness unchanged (7.21 → 7.27 / 7.12 dB); "Idle listened: no audible difference"	EQ_PLAN §4e
F5 same-sample loop in Play	Same parity as z^{-1} in both, but +18.5 % Play CPU, PC estimate 2.43 ms > 2.133 ms gate	`research_f5_allpass/REPORT.md` §0
Six ways of building the Direct sound from the FDN	Each listed as REJECTED (hash, clamps, silent buses)	HANDOVER "Failed attempts (do not repeat)"
A 32×32 matrix multiply per sample in the audio thread	It was blowing the audio thread; replaced by one-sample feedback	HANDOVER "Same-sample mix"
The `0.68/√32` loop cap	"Do not put that cap back"	HANDOVER "Loop"
Uniquifying delays	"Do not uniquify delays" (it is a lock; co-prime nudges conflict with it)	HANDOVER; EQ_PLAN §4b
Choosing the "loudest 32" reflections at the mic	Not rejected outright, but "No change recommended without Idle's call": delays would jump with pattern/yaw and mirror invariance and parity goldens would break	THURSDAY_LIST U9
A stricter "2× everywhere" guard on the Surfaces line filters	Over-triggered: lines fell back and HF ran +100–250 % long	SURFACES_IN_LOOP §7
Raw directional decay with s′ = 0.15	Tails +10–27 % long and a double slope	SURFACES_IN_LOOP §6

IN PLAIN ENGLISH.

These ideas were tried, measured and turned down, with the reason written down so nobody spends time on them again. Examples: rotating the shuffle pattern made no audible difference; a more exact loop in Play cost too much processing; making every delay length unique conflicts with a 2011 lock; and too strict a safety check on the Surfaces filters made the treble ring far too long.

4.6 Still open (for Idle)

Before SL4–SL6 (VERIFIED , `NEXT_SESSION_PLAN.md` ; details in `roadmap/prebuild/sl/README.md` "Findings"):

1. Helmholtz Q: the physical model gives about 1.5–2, against the ASSUMED 3.
2. D-SL3: keep the per-axis cuts below 30 Hz (treated plan 16/23) or drop them all (15/23).
3. The SL4 mild double slope in the untreated plaster room: directional or room average.
4. Treatment also colours the early reflections in Surfaces mode.

Other open items (VERIFIED as open in THURSDAY_LIST §2 and DECISIONS_LOCKED_2026-09-28 "Still open"): S-OUT Auto level (N2) and master headroom (N4); resonance control (N5); D-INT 44.1 kHz interpolation (N6); keeping face EQ when a material is picked (N8); C migration order (N12); licence of the TU-Berlin source presets before any public release (N13); confirm the 1.8 s decay by ear; whether to trim direct/early (whole mix +1.61 dB); the K2 distance unit after the 2011 captures; ST-3–ST-10; D-NR1–8; Q2–Q4, Q6 from the UI audit.

> **IN PLAIN ENGLISH.**
> These are the questions waiting for Idle. The four Surfaces questions come up before sessions SL4–SL6; the rest are smaller items or later ideas. None of them blocks Thursday's first sessions.

Open / gaps

- The full text of D1–D17 and D-UI1–D-UI28 is spread over BIG_JOBS §6 and the UI wish-list report; a single decision index (appendix) is PLANNED for a later update of this book.

Chapter 5 — The early field and the FDN tail (Smooth, Movement, the EQ split)

> **Status:** DONE **(draft 2026-10-05)** for what is in build 51. Smooth and Movement are LOCKED, **not merged** (sessions T4 and T11); they are described from their locked prompts and box prototypes, not as shipped behaviour.

Chapter purpose. This chapter opens the engine: how the 32 reflections are chosen and timed, how they become the 32 lines of the feedback delay network, what happens to the sound on every pass round the loop, and why the tail sounds the way it does today (including its metallic ringing). It then explains the two locked fixes — **Smooth** and **Movement** — and the "EQ split": which filters can change decay time and which can only change level. Chapter 6 builds on this to explain Surfaces. §5.0 explains the engine without formulas, and §5.6a walks slowly through why decay-shaping filters must sit inside the loop.

5.0 The engine in plain English (read this first)

No formulas here; §5.1–§5.7 give the precise version with sources.

Mirrors for walls. To find where the echoes in a rectangular room come from, pretend each wall is a mirror. Stand at the microphone and look at a wall: you "see" a copy of the instrument behind it. That copy is an **image source**. The sound from it travels in a straight line to the mic, exactly as long as the real bouncing path. Mirrors of mirrors give echoes that bounce twice, three times and so on. For each image the program knows the distance, so it knows how late the echo arrives (distance ÷ the speed of sound, 340 m/s) and how quiet it is (`1/r²`: twice as far, a quarter of the level). The program uses a fixed list of 32 images: the direct sound plus 31 echoes of one to four bounces.

From 32 echoes to a tail: corridors that feed each other. Real reverb has thousands of echoes per second. Calculating them all would be far too slow, so the 2011 design uses a trick, the **feedback delay network (FDN)**. Imagine 32 corridors of different lengths. Each corridor is exactly as long as the travel time of one of the 32 echoes, so the first time a sound comes out of corridor 7, that *is* echo 7. At the far end of the corridors, all 32 outputs are poured into a mixer that blends them with a fixed pattern of pluses and minuses (the **Hadamard matrix**: every output gets a bit of every input, some flipped upside down) and pours the blend back into the starts of all 32 corridors. So echo 7 comes back as part of every corridor, comes out again at 32 different later times, gets mixed again, and so on. After a few rounds there are hundreds, then thousands of overlapping echoes: the smooth tail.

Why it fades. On every trip each corridor turns the sound down a little (the **loss**). The Decay dial sets how much. A long corridor is visited less often, so it must lose more per trip; the program sets each corridor's loss so that all of them fade at the same speed in dB per second. If the losses were zero the tail would ring for ever; if anything gained level, it would grow out of control. That is why "stability" comes up so often.

The smearing filter. Each corridor also has a **diffuser** (an **allpass** filter, controlled by **Dffs**). It does not change the tone; it smears each click slightly in time, so the echoes blur into a smoother wash sooner. Turned up too far, it can ring at its own pitches.

The ringing problem. Today's tail has some metallic, pitched ringing in the mids, measurably more "peaky" than real noise. **Smooth** (session T4) and **Movement** (session T11) are the two approved fixes (§5.4, §5.5).

The one rule to remember. Anything done to the sound *before* it enters the corridors happens once: it changes the tail's tone or level, never its length. Anything done *inside* the corridors happens on every trip, so it piles up and changes how fast the tail dies. That is why per-pitch decay (Natural and Surfaces modes) needs filters *inside* the loop (§5.6).

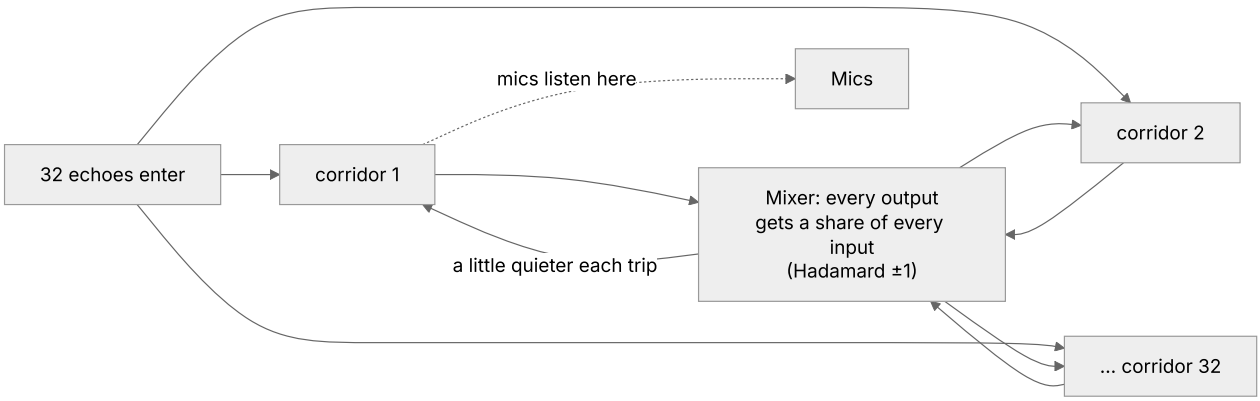


Figure 5.0. The corridors picture of the FDN (explanatory drawing; the exact order of loss, diffuser and mixer is in §5.2).

5.1 The early field: image sources

Geometry. The room is a rectangle W × H × L metres. Images are mirror copies of the source, built with the 2011 mirror macros (`2W-X`, `-X`, `4L-Z` and so on; "full-R; never half-W") (VERIFIED, HANDOVER hard locks). For each image and each capsule:

- distance `d` is the 3-D Euclidean distance;
- delay is `(d / 340) × 1000` ms with `c = 340` m/s (VERIFIED, HANDOVER hard locks, "VERIFIED Const", "(420–433)");
- geometric level is `1/r²` — "Air Abs geometry is `1/r²` VERIFIED — do not change it" (VERIFIED, HANDOVER "Cause").

Which 32. Images come from a fixed table, `LIVE_32` in `engine/early_field.py` (line 257 in the build 51 copy), using reflection orders 0–4 as 1 / 6 / 11 / 10 / 4 (VERIFIED, `roadmap/nonrect/NONRECT_RESEARCH.md` §1). Variants exist for `reaktor_parity` (`LIVE_32_REAKTOR`) and binaural (`BINAURAL_REAR32`, with 5 rear images); other line counts come from `enumerate_scaled_grid` (VERIFIED, THURSDAY_LIST U9). The direct image `(0,0,0)` is voice 3, line index 2 (VERIFIED, HANDOVER "Live voices"). `compute_taps` picks the table and then only computes gains per capsule from the pattern: "Patterns change gains only. A reflection in a fig-8 null keeps its line" (VERIFIED, U9).

What colours each reflection (in order; VERIFIED, `3_phase_acceptance_tests/README.md` `test_p2_air.py`: "Render stage order image → air → surfaces → mic"):

1. **Air** by path length (ISO 9613-1, session S1).
2. **Surfaces**: the material and face EQ of every face the path hits, as a serial cascade (`stages([A,B]) == stages(A)+stages(B)`, VERIFIED same README). Render applies the exact cascade per tap; Play uses a fitted cascade per line (VERIFIED, EQ_PLAN §1 diagram; HANDOVER S5).

3. **Jitter**: a seeded path-length jitter `Δd` per reflection (the Surfaces "diffusion" layer, kept separate from the Structure Dffs, VERIFIED, `3_phase_acceptance_tests` P0.3). Since D-MIX-J it is mirror-invariant: mirror pairs share one random draw (VERIFIED, HANDOVER D-MIX-J).
4. **Mic**: pattern gain and measured colour (Chapter 7).

Solo and Mute on a face act on the early field only (D11, D12, D15, D16): Solo keeps every image that touches the face, Mute drops every image that touches it, and the FDN inject is unaffected (VERIFIED, HANDOVER S4b).

> **IN PLAIN ENGLISH.**
> The program finds echoes by mirroring the source in the walls, and uses a fixed list of 32 of them (always the same images, wherever the mic points). Each echo is then made duller by the air it travels through, coloured by every wall it bounces off, given a tiny random change in path length so the pattern sounds less rigid, and finally heard by each mic according to where that mic points. Soloing a wall lets you hear only the echoes that touch it; muting a wall removes them (early echoes only; the tail is unaffected). The direct sound and the reflections use "voices": 32 parallel copies of the same processing, one per image. **Reflection order** means the number of bounces: order 0 is the direct sound, order 1 has bounced once, and so on.

5.2 From reflections to tail: the FDN

One line per reflection. "FDN line i delay = tap i distance" (`fdn_delays_from_taps`, `engine/fdn.py` 785–814) (VERIFIED, THURSDAY_LIST U9). All 32 early images inject into the FDN (VERIFIED, `test_locks.py` in package 3). The default source position was chosen so that the default scene has 29 distinct delays per bus (was 20 / 12 / 20) (VERIFIED, DECISIONS_LOCKED_2026-09-28 K3). Delays are deliberately **not** made unique: "Do not uniquify delays" (VERIFIED, HANDOVER).

> *Plain first:* on each sample, a corridor takes what is arriving (new input plus the mixed feedback from the previous sample), reads its own delayed sound with the loss and its fixed dip applied, smears the two together in the allpass way, and writes the result back into the corridor. The formulas below are the exact recipe.

One pass round the loop (Play, build 51). The F5 research read the C code (VERIFIED, `research_f5_allpass/REPORT.md` §1):

- `s = xin + fb`, where `fb = g·(H·D·y)[n−1]` is stored for the next sample;
- `y = −d·s + r`, where `r` is the delay read with the loss `lg` and the peaking cuts applied inside `read_peaked`;
- the line writes `s + d·y`.

So each line is a delay wrapped in a Schroeder allpass with coefficient `d` (the Dffs): per line `Y/S = (lg·z⁻ᴰ − d) / (1 − d·lg·z⁻ᴰ)`, with $|T| = 1$ when lossless (VERIFIED, same report §2). Since S7, Render runs the same structure (Chapter 3).

The matrix. An unscaled ±1 Hadamard (Sylvester construction; "construction ASSUMED", no OCR sign table) applied by a fast Walsh–Hadamard transform `fwht32` (VERIFIED, HANDOVER hard locks; package 3 `test_locks.py`). Since S4b the 32-line loop is **H·D**, with a fixed mirror-safe diagonal of column signs, which fixed a comb in the default room; `reaktor_parity` keeps plain H (VERIFIED, HANDOVER S4b). Shared-tail mode flips entries 3 and 24 of that diagonal (locked) (VERIFIED, HANDOVER "Shared-loop signs").

> *Plain first:* two ways of setting how fast the tail fades. In the 2011 way, the whole mixer is turned down by one gain knob. In today's default ("Decay lock on"), the mixer is kept at a neutral level and each corridor gets its own loss per trip, sized so that every corridor drops by 60 dB in the time on the Decay dial.
>
> *Worked illustration (computed here):* with the dial at 1.8 s, a 20 ms corridor loses 0.67 dB per trip and a 40 ms corridor 1.33 dB per trip. The first makes 50 trips a second, the second 25, so both lose 33.3 dB per second and reach −60 dB at 1.8 s together.

The loop gain. Two laws (VERIFIED, HANDOVER "Engine notes"; `engine/fdn.py` 862–870 in build 51):

- **2011 / `reaktor_parity` / Decay lock off:** `g = Gain / N`, i.e. `Out = In × (1/32) × Gain`, clamp `0.99/√N`.
- **Decay lock on (desktop default):** `g = 1/√N` (about 0.177) and a per-line loss `lg = 10^(−3·dᵢ/(sr·T))` where `dᵢ` is the line delay in samples and `T` the Decay dial — that is $-60 \cdot \tau_i / T$ dB per pass, so every line decays at the same rate in dB per second (VERIFIED, `fdn.py` line 870).

The diffuser. Each line's Dffs comes from the surface "diffusion" sum times **Diffusion Master**, clipped to ±1 (`dff_after_master`, VERIFIED `fdn.py` 211–218). Diffusion Master defaults to the Structure default −0.25 and is clamped to [−0.5, 0] (VERIFIED, DECISIONS_LOCKED_2026-09-28 item 5; `fdn.py` 200–208).

Per-axis peaking cuts (inside the loop). Every line carries one cut-only RBJ peak at the first axial mode of X, Y or Z (line i % 3), −1.5 dB per pass, Q 1.5, whatever its length (VERIFIED, EQ_PLAN §2; `live_peaking_banks` in `fdn.py` 581–609). Play restores each peak's `b2` in double precision so DC gain is exactly 1; that stops long decays from growing (VERIFIED, HANDOVER S4b "DC restore"). The EQ plan measured that these cuts over-cut the bass: default room octave T20 with / without them 31.5 Hz 0.58 / 1.53 s, 63 Hz 0.97 / 1.65 s at a 1.8 s dial (VERIFIED, EQ_PLAN §2).

Level trims. Each loop change that altered loudness got a fixed makeup row in `LOOP_LEVEL_TRIM_DB`: F2 line absorption −2.81 dB (both paths, only while Decay lock carries the decay), F3 play-rate delays −1.70 dB (both paths, always), F5 0 (VERIFIED, `fdn.py` 244–248). Trims only ever reduce level.

Denormals. After the input stops, filter states could sink into subnormal floats and multiply CPU cost; the guard flushes air and shelf states under about 1e-20 and adds a tiny offset to the loop (VERIFIED, HANDOVER "Fix 2"; SOUND_TRICKS "Already done": `ls_flush` 1e-20 at `live_stereo.c` 501, plus an `LS_DC` offset).

IN PLAIN ENGLISH.

Each of the 32 echoes gets its own corridor, exactly as long as that echo's delay. On every trip: the corridor's output is turned down (loss), passed through one small fixed dip (each corridor in turn gets the dip for the room's lowest length, width or height resonance), smeared by the allpass diffuser, mixed with all other corridors by the ±1 Hadamard pattern (with a fixed set of sign flips that removed a comb-like colouring), and fed back one sample later. The **Diffusion Master** knob (default −0.25) scales all the smearing filters together. Two housekeeping details: whenever a loop change made the tail louder, a fixed level trim was added so loudness did not jump; and a tiny guard stops the processor slowing down when the sound has died to almost nothing (**denormals** are extremely small numbers that some processors handle very slowly). **Sample**: one measurement of the waveform; at 48 kHz there are 48,000 a second. **RBJ peak**: a standard EQ bell filter. **Q**: how narrow a filter is (higher Q = narrower).

5.3 How today's tail sounds (measured)

Decay against the dial. At diffusion −0.25 the Play T30 sits within ±8 % of the dial in 14 of 15 band × scene values; the exception was the small room at 500 Hz (+10.5 %) (VERIFIED, DECISIONS_LOCKED_2026-09-28 tail table). After the S4b column signs the small room's worst band moved to 1 kHz (−8.5 % → −3.5 % in one measurement; still listed as −8.5 % in THURSDAY_LIST N10), which is why D21 asks for a PC re-measure (VERIFIED, BIG_JOBS §0.G; THURSDAY_LIST N10).

Flat decay across bands. "The tail decays at the same rate in every band" — `lg` is one scalar per line (VERIFIED, SOUND_TRICKS "Short answer", citing `fdn.py` 866–872 and `live_stereo.c` 1893–1907). Real rooms lose treble faster. This is what Natural and Surfaces modes address (Chapter 6).

Metallic ringing. The EQ plan's measurements (VERIFIED, EQ_PLAN §2, §4a):

- Late-tail peakiness (std of fine-bin dB, 0.3–1.3 s): **engine 7.20 dB** against **noise 5.53 dB** (textbook Rayleigh 5.57).
- Strongest lines at 724, 1285, 781 and 2008 Hz, +18.5 to +21 dB over the local median — "in the mid band, **not at room modes, and not from the banks**."
- Box room (3 × 2.4 × 4 m) is worse: line peaks 19.6 dB against noise 8.6–8.8, and 15 of 32 diffuser gains at the −0.98 clamp.
- Likely causes: uneven modal excitation by a plain Hadamard, diffuser gains at the clamp, and duplicate delays (ASSUMED, EQ_PLAN §4a).
- Room modes do **not** ring in the tail beyond chance: "FDN modes are not room modes" (VERIFIED, EQ_PLAN §2).

> **IN PLAIN ENGLISH.**
> Measured today: the tail's length matches the Decay dial to within 8 % almost everywhere; the small room's mid band is the exception still being checked. Bass and treble fade at exactly the same speed (unlike a real room). And the tail has a few pitched "rings" in the mids, around 700 Hz to 2 kHz, standing about 20 dB above their neighbours. They are not the room's own resonances and not caused by the fixed dips; the likely causes are the shuffling pattern, diffusers turned up to their limit and corridors with identical lengths (an educated guess, marked ASSUMED). **Peakiness** is the measure used: how much the fine detail of the tail's spectrum jumps up and down. Real random noise scores about 5.5 dB; the engine scores 7.2 dB.

5.4 Smooth (EQ-1a) — LOCKED, session T4

Status: LOCKED by Idle 4 Oct 10:13; not merged. Source: `roadmap/eq_plan/EQ1a_PROMPT.md`, EQ_PLAN §4f.

- **What:** cap the diffuser gain |Dffs| at **0.7**, **default ON**. Off = today's engine, bit-exact.
- **Where:** where Dffs is made (`tap_dff` in `render_pipeline.py`), so Play and Render get identical values from one source.
- **Decay correction:** the uncapped-to-capped change lengthened octave T30 by +4–8 % at 63–500 Hz with HF unchanged. The correction must be computed, not hand-tuned, and folded into an existing per-line coefficient (`lg`), "likely" because the allpass group delay changes the effective loop time (ASSUMED cause, stated as such in the prompt).
- **Forced off** in `reaktor_parity`, flat cards and the mirror/null test modes.
- **Gates:** G1 bit-exact off; G2 T30 within ±5 % (default, box, carpet rooms); G3 t19 CPU; G4 peakiness at least 0.4 dB lower (prototype −0.64 dB); G5 regenerate only the goldens the Smooth default changes; G6 t18 ≤ −60 dB, mirror ≤ −120 dB.

- **Stop rule:** if G2 and G3 cannot both pass, ship Smooth default **OFF** (no golden change) or revert.

Prototype numbers (VERIFIED, EQ_PLAN §4e, `proto_mod/`): the Dffs cap ("dcap") alone took peakiness from 7.21 to 6.57 dB.

> **IN PLAIN ENGLISH.**
> The diffusers' smearing strength (Dffs) can range from 0 to nearly ±1. Near ±1 a diffuser starts to behave like a tuned pipe and rings. In one test room 15 of 32 diffusers sat at the −0.98 limit. **Smooth** simply never lets any diffuser go beyond 0.7. In the prototype, capping made the bass (63–500 Hz) ring 4–8 % longer, probably because the smearing filters change the effective timing of each corridor (the prompt marks this cause as ASSUMED). So Smooth also applies a calculated correction, folded into each corridor's loss, to keep the decay where the dial says. It is on by default and can be turned off to get today's sound exactly. In the prototype the cap alone cut peakiness from 7.21 to 6.57 dB. If the session cannot keep both the decay accuracy and the processing cost within limits, Smooth ships switched off.

5.5 Movement (EQ-1b) — LOCKED, session T11

Status: LOCKED; not merged; the biggest Thursday session. Source: EQ_PLAN §4b, §4e, §4f; `roadmap/eq_plan/EQ1b_PROMPT.md`.

- **What:** a Movement knob on the Room tab, **default 0**. 0 = off; 50 % = "mod10" (0.10 ms peak); 100 % = "mod15" (0.15 ms peak); linear in between. The 50 % midpoint is the plan's reading of Idle's request and is listed as needing a quick yes (ASSUMED, THURSDAY_LIST N14).
- **Frenette set:** sine LFOs at 0.7 Hz × 0.6–1.4 per line, phases k·112.5°, on 16 of 32 lines, with **first-order allpass reads** on the modulated lines only (linear interpolation showed a −24 dB HF tilt and T30 −14 % in the box prototype).
- **Render must run the loop** when Movement > 0 (Chapter 3).
- **Forced off** in parity, flat and mirror/null modes.
- Prototype (VERIFIED, `proto_mod/`): peakiness 7.21 → mod10 6.51, mod15 6.28, mod10 + dcap 6.07 dB (noise 5.53); named peaks down 0–9 dB; pitch wobble on a sustained 440 Hz tone 0.2–0.3 cents rms.

> **IN PLAIN ENGLISH.**
> A pitched ring happens when a corridor-and-mixer pattern repeats exactly, over and over. **Movement** keeps 16 of the 32 corridors very slightly changing length, each swinging slowly (about once every 1.4 seconds, varied per corridor) by at most 0.10–0.15 thousandths of a second. That is about 3.4–5 cm of path length (computed here at 340 m/s), like the walls breathing by a few centimetres. The rings can no longer lock in, so they smear out. The pitch wobble this causes on a held note was measured at 0.2–0.3 cents (a cent is a hundredth of a semitone), far too small to hear as vibrato (ASSUMED judgement). Because the tail now changes over time, Render can no longer use one fixed "echo fingerprint" and has to run the loop sample by sample when Movement is on. The knob defaults to 0, so nothing changes unless you turn it up. **LFO** = low-frequency oscillator, a slow wobble generator.

5.6 The EQ split: level versus decay

The single most useful idea in the EQ plan (VERIFIED, EQ_PLAN "plan in five lines" and §3c). In plain words first: where you put a filter decides whether it changes the tail's *tone* or its *length*. The table sorts every filter by place; §5.6a explains why, step by step.

Where a filter sits	What it can do	Examples
Before the FDN (on the inject)	Level and tone of the tail only. "LTI and outside the loop: the same dB at every moment"	Air, materials, face EQ, D-UI26 face colour
Inside the loop, same dB on every line	Changes decay, but unevenly: "neighbouring modes decay at different rates"	Today's per-axis cuts (kept for 2011 parity only)
Inside the loop, dB ∝ line delay (Jot)	Changes decay per band homogeneously: c_i = $60 \cdot d_i \cdot (1/T_{target} - 1/T_{now})$ dB per pass	Planned resonance control (N5); Natural; Surfaces
After the desk	Overall tone of the wet sum	Planned Output EQ (S-OUT, D-UI24 = A)

The formula in the third row reads: the cut per trip c_i equals 60 dB × the corridor's length in seconds × (1/the decay you want − 1/the decay you have now). Worked example from the plan: shortening a band from 2.4 s to 1.8 s needs 0.17 dB per pass on a 20 ms line and 0.33 dB on a 40 ms line, adding up to 0.8 dB at 0.1 s, 4.2 dB at 0.5 s and 8.3 dB at 1 s — "transients are barely touched and the long tail is re-timed" (VERIFIED, EQ_PLAN §3b, §2).

5.6a Why the filters must be inside the loop — step by step

This is the idea that the whole Surfaces chapter rests on, so here it is slowly.

1. **A filter before the loop acts once.** Suppose you cut 3 dB of treble from the echoes before they enter the corridors. Every echo that enters is 3 dB less bright. Everything the loop then makes from those echoes is also 3 dB less bright, at the start of the tail and at the end. The tail is duller, but it fades at exactly the same speed. In the language of the plan: "the same dB at every moment" — a level control, not a decay control.
2. **A filter inside the loop acts on every trip.** Now put a small treble cut on a corridor itself. The sound passes through it every time round. After one trip it has lost that cut once, after ten trips ten times. The longer the tail goes on, the more treble has been removed: the treble dies sooner while the start of the sound is barely touched.
3. **The plan's worked example** (VERIFIED, EQ_PLAN §3b): to shorten one band from 2.4 s to 1.8 s, a 20 ms corridor needs only 0.17 dB of cut per trip. That adds up to 0.8 dB after 0.1 s, 4.2 dB after 0.5 s and 8.3 dB after 1 s: "transients are barely touched and the long tail is re-timed".
4. **Why the cut must depend on corridor length.** A long corridor is visited less often than a short one. If every corridor had the same cut per trip, short corridors would remove more per second than long ones, and the tail would fade unevenly (the plan's words: "neighbouring modes decay at different rates"). So the right recipe, from Jot, gives each corridor a cut in proportion to its length: twice as long, twice the cut per trip (the 40 ms line needs 0.33 dB where the 20 ms line needs 0.17 dB).
5. **Never boost inside the loop.** A boost on every trip adds energy every time. Enough of it and the loss per trip at that pitch turns into a gain, and the tail grows instead of fading. That is why Surfaces only lets EQ **cuts** into the loop.

Natural and Surfaces modes are both just smarter versions of point 4: a per-corridor tone control whose cut per trip follows the corridor's length and, for Surfaces, the walls that corridor's echo hits.

5.7 Shipped versus Thursday

Item	Status
LIVE_32 image set, $1/r^2$, c = 340, air, per-hit surfaces, jitter	SHIPPED (build 51)

Item	Status
32-line H·D loop, one-sample feedback in both engines, Decay lock law, per-axis cuts, DC restore, trims	SHIPPED
D-UI26 face colour on the inject (tone, not decay)	SHIPPED (S5)
Smooth (Dffs cap 0.7, default ON)	LOCKED → T4 (golden YES)
Movement (default 0)	LOCKED → T11
D-INT 44.1 kHz rounding	OPEN (N6)
Resonance control (in-loop, dB ∝ delay, ≤ 4 rows)	OPEN (N5)
Natural (air) mode	LOCKED → SL7a; default in SL7b
Surfaces mode	LOCKED → SL1–SL6; default in SL7b
Tank gating after the tail dies (D23), velvet smear (D24)	LOCKED, later

IN PLAIN ENGLISH.

Already in the program: the mirror-image echoes, the 32-corridor tail with its shuffle, losses, diffusers and fixed dips, and wall colour on the way into the tail. Approved and coming: Smooth (T4) and Movement (T11). Approved for the Surfaces sessions: per-pitch decay from air (Natural) and from walls and panels (Surfaces). Still undecided: rounding delays at 44.1 kHz, and a "resonance control" for taming individual rings.

Open / gaps

- The exact Smooth decay-correction method is for Build to compute in T4; no formula exists yet in the sources.
- The SURFACES_IN_LOOP mode table gives the default-room axial modes as 28.6 / 57.2 / 21.4 Hz. Those match c = 343 m/s, and the Surfaces prebuild indeed sets `C_AIR = 343.0` (VERIFIED, `roadmap/prebuild/sl/engine/surf_target.py` line 20), whereas the engine's locked value is `c = 340` (about 28.3 / 56.7 / 21.3 Hz, computed here). This mismatch is OPEN for SL1/SL2 to settle (see Chapter 6 §6.11).

Chapter 6 — Surfaces, panels and auto-treat

> **Status: DONE as a description of the design (draft 2026-10-05). The feature itself is NOT merged.** Surfaces-in-the-loop is researched, prototyped, listened to and locked (D-SL1–D-SL10), and its code is pre-built and tested on the box; it lands in Build sessions SL1–SL6, with the default switch in SL7b only after Idle's written OK. Everything below that is not in build 51 is marked DESIGNED (box prototype / prebuild) rather than shipped. **Main sources:** `roadmap/sound_tricks/SURFACES_IN_LOOP.md` (research, 4 Oct 17:21–18:20 BST), `roadmap/prebuild/sl/README.md` and `roadmap/prebuild/sl/engine/*.py` (tested prebuild, 4 Oct 21:50 BST), `roadmap/DECISIONS_LOCKED_2026-10-04.md`.

Chapter purpose. This is the chapter Idle asked for most: how the panels, absorbers, materials and face EQ all work together to give the tail its frequency response, and exactly what the automation (auto-treat) does. It separates what the Surfaces tab does today from the in-loop design, walks through the design step by step with its formulas, shows how auto-treat chooses panels, gives the measured accuracy, and lists what is still unknown. Read Chapter 5 first: the key fact is that only filters *inside* the loop can change decay time. **If you are not an engineer, start with §6.0**, which tells the whole story without formulas; every design step in §6.5 then opens with a "Plain first" note, and §6.5a sums the steps up in one table.

6.0 The whole idea in plain English (read this first)

This section uses no formulas. Everything in it is restated with sources in §6.1–§6.11.

The bouncing ball. Follow one echo round a room as if it were a ball. Every time it hits a surface, the surface takes a bite of its energy. Between bounces, the air nibbles away a little of its treble. After enough bounces the ball has nothing left: the echo has died. How fast that happens is the room's **decay time**. Acousticians call it **T60**: the time it takes the sound to fall by 60 dB, roughly from "loud" to "gone".

Bites depend on pitch. Each material bites differently at different pitches. Its **absorption coefficient α** (alpha) says what fraction of the sound energy a surface swallows at each bounce, from 0 (a perfect mirror) to 1 (an open window). Plaster has a small α at every pitch. Carpet and mineral-wool panels have a large α in the mids and treble but a small one in the bass. So a room full of soft furnishings dies quickly in the treble but can still boom in the bass.

Tuned sponges. Bass is hard to absorb with soft panels, so studios use **tuned panels**. A **membrane panel** is a thin board over an air gap: it vibrates like a drum skin at one pitch and soaks up the bass there. A **Helmholtz panel** is a perforated board over a cavity: the air in the holes bounces on the air behind like a mass on a spring, again at one pitch (the same effect as blowing across a bottle). Both are like sponges that only soak up one narrow band and ignore the rest. Choosing their size and depth chooses the band.

What the program does today. In build 51 the walls already have materials and an EQ, and they colour the echoes and the tone of the tail. But the *length* of the tail is set by the Decay dial alone and is the same at every pitch. In the bouncing-ball picture: the walls tint the ball, but the ball always loses the same fixed share per second, whatever it hits.

What Surfaces mode will do (LOCKED, DESIGNED, not merged). It lets the walls take their real bites. The tail is made by 32 recirculating corridors (the FDN, Chapter 5). Each corridor stands for one echo path that hits particular walls. Surfaces mode gives each corridor its own small tone control, set so that on every trip round the loop the corridor loses, at each pitch, what its walls and the air would really take. Because that loss repeats on every trip, it shapes how fast each pitch range dies. Nobody draws the decay curve; it **comes out of** the materials and treatment, as in a real room.

The yardstick and the helper. Studio designers aim for a decay that is even across the mids, rising slightly in the deep bass. In this project that is the **Mäkivirta target**: flat at the mid-band time, rising by 0.2 s from 300 Hz down to 50 Hz, following air alone above 2 kHz (VERIFIED, SURFACES_IN_LOOP §3 table). **Auto-treat** compares the room's predicted decay with that target and suggests a shopping list of panels to bring it close: broadband panels spread over four walls first, then up to four tuned panels aimed at the boomiest bass bands. It only *suggests*. Nothing changes until you accept or edit the list.

Why bother? On a measured real room (BRAS CR2), Flat (2011) came out 20–105 % too long per octave and kept the treble far too long. Surfaces mode came within about −2 to +26 % and got the treble-to-mid balance right (VERIFIED, §6.8). And when Idle listened to the A/B files his verdict was "sounds good, certainly a lot better than flat".



Figure 6.0. The bouncing-ball picture behind Surfaces mode (an explanatory drawing, not a code diagram).

6.1 What Surfaces does today (build 51, SHIPPED)

- **Six faces, A–F.** The UI orientation is A Left (−X), B Front (+Z), C Right (+X), D Back (−Z), E Floor (−Y), F Ceiling (+Y) (ASSUMED orientation, `/workspace/docs-publish/docs/HOW_IT_WORKS.md` §4; D = back wall −Z is VERIFIED by the D17 hint text, HANDOVER S4b). The prebuild groups them in axis pairs x = A/C, y = E/F, z = B/D (VERIFIED, `surf_target.py` line 24).
- **Material cards** with absorption per octave: 12 real materials with role bans (no carpet on the ceiling), plus porous-panel-with-air-gap presets (VERIFIED, BIG_JOBS J9 "Done in S4"). The *default* card is not a real material ("α 0.05 → 0.22, ASSUMED plaster-ish") (VERIFIED, EQ_PLAN §5).
- **Face EQ** per face: low shelf, peak, high shelf (VERIFIED, EQ_PLAN §5).
- **Solo / Mute** per face, early field only (D11–D17, Chapter 5).
- **Air card** on Surfaces (VERIFIED, THURSDAY_LIST L15).
- **Face colour on the tail (D-UI26 = A).** The absorption shape relative to the card mean, plus face EQ, filters each line's FDN inject in both engines. It is outside the loop, so "T60 stays the Decay" (VERIFIED, BIG_JOBS §0.0 D-UI26; HANDOVER S5).

So today the walls colour the tail but **do not shorten it per band**. Every line loses the same dB per second at every frequency: "Flat (2011) gives the same T in every band, whatever the walls are made of." (VERIFIED, SURFACES_IN_LOOP Figure 2 caption.)

IN PLAIN ENGLISH.

Today each of the six surfaces (left, front, right, back, floor, ceiling = A–F) has a **material card** (how much it absorbs in each octave) and a **face EQ** (bass shelf, mid peak, treble shelf). You can solo or mute a surface to hear its early echoes. These settings colour the early reflections and, since session S5, the tone of the tail too. They do **not** change how long the tail lasts at each pitch: the Decay dial does that, equally for all pitches. "Octave" here means a band of pitches whose top is twice its bottom (for example 500–1000 Hz).

6.2 Where the in-loop design stands

Stage	Status	Evidence
Research and formulas	Done, 4 Oct	SURFACES_IN_LOOP §1–§8
Box prototype + 26 A/B WAVs	Done; Idle listened: "sounds good, certainly a lot better than flat"	`sound_tricks/proto2/` , `surfaces_ab/` ; lock 20:37 BST
Decisions	D-SL1–D-SL9 locked 20:37; D-SL10 20:41	DECISIONS_LOCKED_2026-10-04
Pre-built engine, Play C, UI, tests	Done on the box: fast 36/36; slow SL2 4/4, SL5/SL6 2/2, SL4 4 pass + 1 xfail; Flat bit-exact 9/9 scenes	`prebuild/sl/README.md` "Test results"
Merged into `software\`	**No**	sessions SL1–SL6
Default	**No** — Flat (2011) stays default until SL7b, gated on Idle's written OK	D-SL1, D-SL10

IN PLAIN ENGLISH.

The design is finished on paper, prototyped, listened to and approved by Idle, and its code is pre-written and tested on the research machine. It is **not** in the program on Idle's PC yet. Six Build sessions (SL1–SL6) will add it as an *option*, while today's sound stays the default and is protected by tests. Only after a final listening comparison and Idle's written OK does one session (SL7b) make it the default.

6.3 The idea in one paragraph

"The proposed **Surfaces** mode gives each line its own small loss filter (3 biquads), designed from the physics of its own image path: which faces it hits; what each face is made of; what treatment is on the face (porous panels, tuned membrane or Helmholtz panels); the face EQ (cuts only); air. The decay curve per band is therefore *not drawn*. It **emerges** from the treatment. The Mäkivirta target becomes the yardstick that **Auto-treat** uses to size panels, not a curve imposed on the tail." (VERIFIED quote, SURFACES_IN_LOOP §1.)

IN PLAIN ENGLISH.

Each of the 32 corridors in the tail gets its own small tone control, worked out from the physics of the walls its echo bounces off, the panels on those walls, any cuts in the face EQ, and the air. The decay per pitch band is not drawn by hand; it emerges from the room. The target curve is only used by Auto-treat, to decide what panels to suggest. "Biquad" means one standard second-order filter stage, the basic building block of most digital EQs; three biquads make one small, cheap tone control.

6.4 The pieces and how they connect

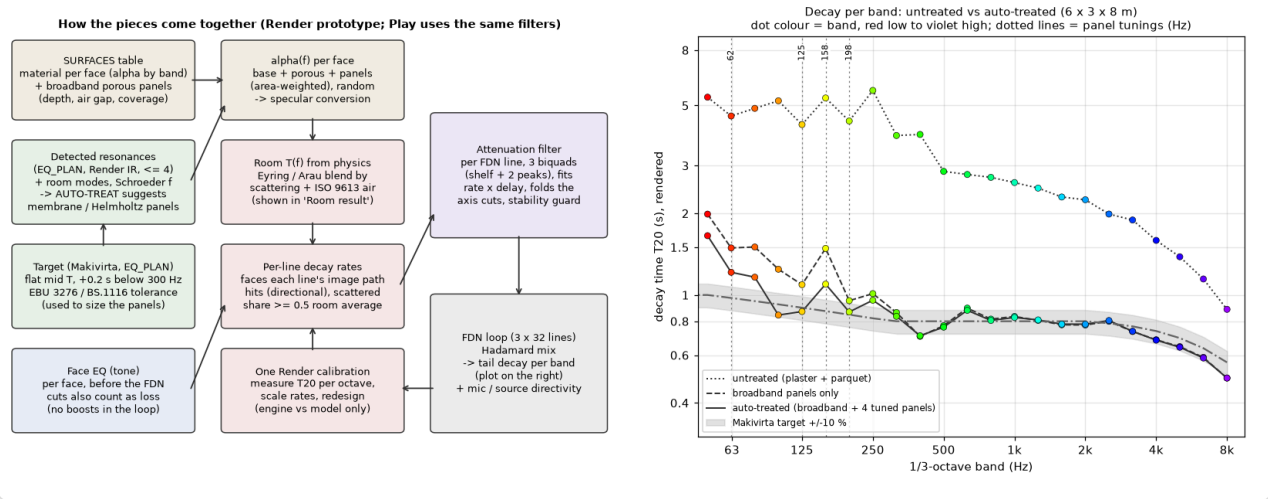


Figure 6.1 (copied from `roadmap/sound_tricks/surfaces_in_loop_overview.png`). Left: how the pieces come together. Right: T20 per ⅓-octave in the default 6 × 3 × 8 m room, rendered by the engine. Untreated (plaster + parquet) runs at about 2.6 s mid and about 5 s bass; broadband panels alone leave 50–160 Hz at +35–100 %; auto-treat puts 16 of 23 bands inside the Mäkivirta ±10 % band. Dot colour is the band, red (low) to violet (high).

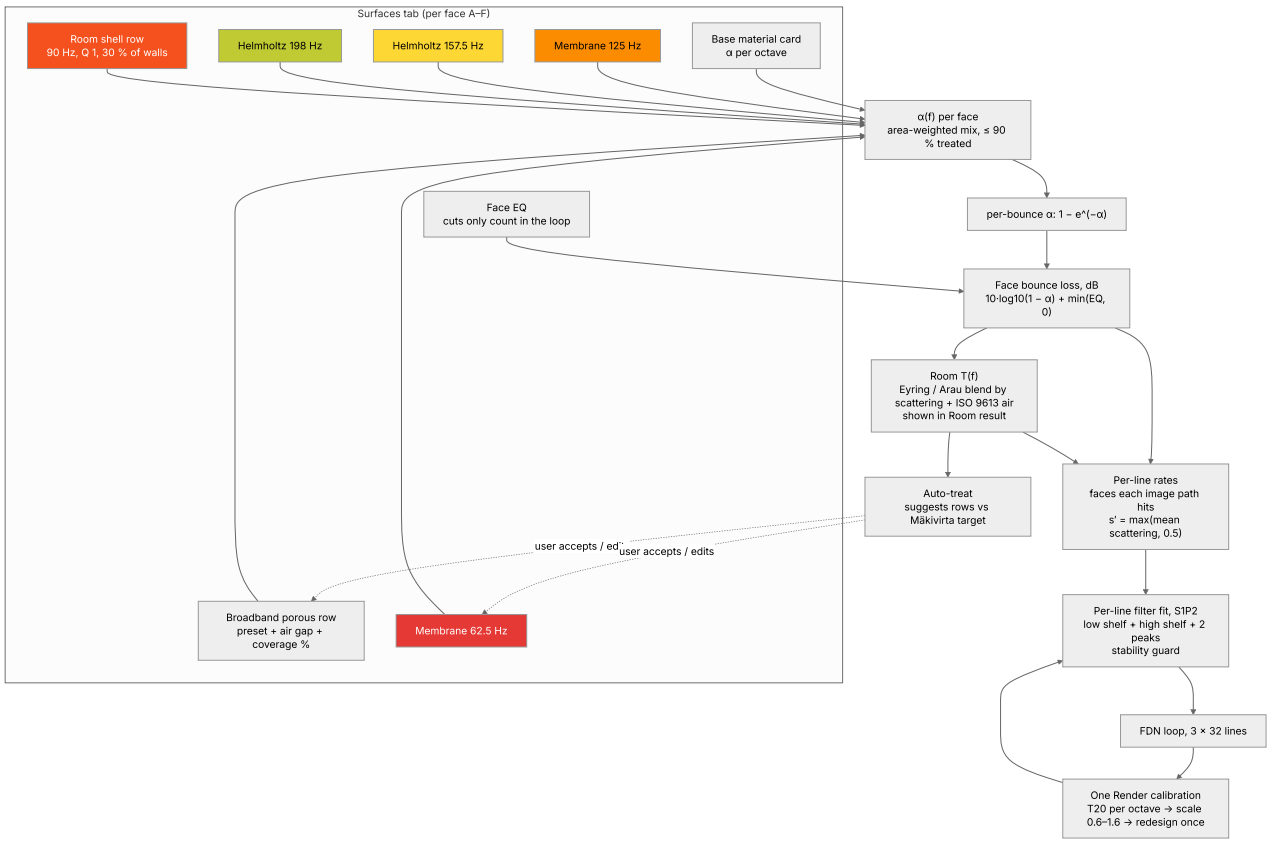


*Figure 6.2. Panels and materials feeding the loop (**DESIGNED**, not merged). Tuned-panel boxes are coloured by frequency, red = low → violet = high, using the four panels of the default-room plan (§6.7). The porous row works mostly from about 200 Hz up and is left uncoloured.*

IN PLAIN ENGLISH.

Read Figure 6.2 from top to bottom. Everything you put on a wall (base material, broadband panels, tuned panels, a "room shell" allowance for doors and windows, face-EQ cuts) is combined into one absorption-per-pitch curve for that wall. That curve is turned into "dB lost per bounce". From all the walls together the program predicts the room's decay per band (shown to you), and for each corridor it works out a loss rate from the walls that corridor's echo hits. A fitting step sets each corridor's tone control to match. One test render checks the result and nudges it once. Auto-treat sits on the side: it reads the predicted decay and suggests panels, which you can accept or edit.

6.5 Step by step

All steps below are DESIGNED (box prototype, then prebuild). Formulas are VERIFIED as written in the cited files; values marked ASSUMED are assumed in the source too.

Step 1 — α(f) per face.

> *Plain first:* for each wall, add up what is on it. If 20 % of a wall is covered by a panel and the rest is plaster, the wall's absorption at each pitch is 20 % "panel" plus 80 % "plaster". Panels may cover at most 90 % of a wall, because real walls have doors, sockets and edges. Each kind of treatment has its own pitch curve: broadband panels absorb from the mids up, tuned panels absorb one bass band. The formulas below predict a tuned panel's pitch (f0) from its weight and air gap: heavier boards or deeper gaps tune it lower.
>
> *Worked illustration (numbers chosen for the example and computed here, not a project setting):* a board of 5 kg per square metre over a 0.1 m gap gives f0 ≈ 60/√(5 × 0.1) ≈ 85 Hz, or about 71 Hz with mineral wool in the gap. With the ASSUMED membrane curve (Q 2, αmax 0.6) it absorbs 0.6 at 85 Hz, about 0.16 at 1.5 × that pitch and only 0.06 an octave above: a sponge for one band.

Each face's absorption is the area-weighted mix of its base material, its porous coverage and any panels, as a continuous α(f) (SURFACES_IN_LOOP §2). A face holds at most 90 % treatment ("doors, sockets, edges") (VERIFIED, `surf_panels.py` `MAX_FACE_COVER = 0.9`). Treatment models (SURFACES_IN_LOOP §3; `surf_panels.py`):

- **Membrane panel:** f0 ≈ 60/√(m·d), or ≈ 50/√(m·d) with mineral-wool fill (m in kg/m², d in m) — Cox & D'Antonio ch. 7 (formula VERIFIED).
- **Helmholtz (perforated) panel:** f0 = (c/2π)·√(ε/(t′·d)), t′ = t + 2·0.85·a (formula VERIFIED).
- **Resonant α curve:** α(f) = αmax / (1 + Q²(f/f0 − f0/f)²); membrane Q 2, αmax 0.6; Helmholtz Q 3, αmax 0.7; off resonance the panel face counts as α ≥ 0.05 (ASSUMED values, "needs datasheet or impedance-tube fit").
- **Porous plus air gap:** the app's library presets 1″+1″ gap, 2″, 2″+2″, 4″+4″ (library VERIFIED in the app); the prebuild adds a Miki model with the Paris random-incidence integral as a check (VERIFIED, prebuild README; porous presets within 0.012 α of the library).
- **Room shell (D-SL6):** doors, glazing and stud walls as a broad 90 Hz membrane, Q 1, on 30 % of each wall, αmax 0.35 in the prebuild (ASSUMED values, `surf_panels.room_shell_rows`). On by default in Surfaces mode, removable.

Step 2 — per-bounce loss.

> *Plain first:* absorption figures in catalogues are measured in a special test room and tend to overstate high absorption, so they are softened first with the conversion 1 − e^(−α). Then "fraction absorbed" becomes "dB lost per bounce". Face-EQ **cuts** add to the loss; face-EQ **boosts** are ignored inside the loop, because a boost on every trip would add energy each time and could make the tail grow instead of fade.
>
> *Worked illustration (computed here):* a library α of 0.10 becomes 0.095 per bounce, a loss of 0.43 dB; α 0.30 becomes 0.26, a loss of 1.30 dB; α 0.60 becomes 0.45, a loss of 2.61 dB. A surface with α 0.10 in the bass and 0.60 in the treble therefore takes about six times as many dB from the treble on every bounce, so the treble dies roughly six times faster (before air and the other walls are counted).

Library α is random-incidence (chamber) data; it is converted to a per-bounce α with 1 − e^(−α), which "tames the HF overstatement of high-α library data in small rooms" (ASSUMED , Kuttruff). Each face's loss per bounce is $10 \cdot \log_{10}(1 - \alpha)$ plus the face EQ's **cuts only**: "No boosts in the loop: they can only add energy and break stability." (VERIFIED , SURFACES_IN_LOOP §2; `surf_target.face_bounce_db` .)

Step 3 — the room's own T(f).

> *Plain first:* before touching the tail, the program predicts the decay time per band for the whole room with classical formulas. The idea behind all of them: a big room rings longer (sound travels further between bounces), and a room with more absorption rings shorter. **Eyring's** formula does this for an evenly absorbing room. **Arau-Puchades'** handles rooms where one direction (say floor-to-ceiling) is much deader than the others. Because real rooms scatter sound partly in all directions, the two answers are blended according to how much the surfaces scatter. Air absorption is added. The result is shown in the "Room result" panel, so you can see what your treatment should do before you render.

Eyring with area-weighted $\bar{\alpha}$ plus air, $T_E = 0.161 \cdot V / (-S \cdot \ln(1 - \bar{\alpha}) + 4 \cdot m \cdot V)$; Arau-Puchades = area-weighted geometric mean of the per-axis Eyring T; the two are blended by the mean scattering s: $T = \exp(s \cdot \ln T_E + (1 - s) \cdot \ln T_{Arau})$ (VERIFIED formulas; the blend is ASSUMED , SURFACES_IN_LOOP §3; code `surf_target.room_target`). With Decay lock on, the 500 Hz–1 kHz mean is anchored to the dial and the other bands keep the material ratios; with lock off, the materials set the absolute length and the dial *shows* it (DESIGNED , SURFACES_IN_LOOP §8).

Step 4 — per-line rates (directional, D-SL2).

> *Plain first:* the room has one predicted decay per band, but the tail has 32 corridors, each standing for an echo path that hits particular walls. A corridor whose echo bounces between the treated ceiling and floor should die faster than one bouncing between bare side walls. So each corridor's loss rate is a mix: part "room average" (because real sound scatters and spreads its energy over all walls) and part "my own walls". The design makes the room-average part at least half ($s' \geq 0.5$), because a stronger directional effect made the tail sound like two decays one after the other (a "double slope") in the prototype. κ (kappa) is just a scale factor that keeps the average of all corridors equal to the room's prediction.

Each line's loss rate in dB/s is $\text{rate}_i(f) = \text{air} + s' \cdot (\text{room surface rate}) + (1 - s') \cdot \kappa \cdot (\text{line i's own faces' loss per metre of its path})$, where κ is normalised so the delay-weighted mean equals the room target, and **s' = max(mean scattering, 0.5)** (VERIFIED , SURFACES_IN_LOOP §2 step 5; `surf_loop.line_rates`). The floor of 0.5 exists because the raw directional variant ($s' = 0.15$) ran +10–27 % long with a double slope (VERIFIED , §6). The fallback, "room average", gives every line the same rate 60/T(f).

Step 5 — the line filter (S1P2, D-SL7).

> *Plain first:* now each corridor needs a tone control that removes the right number of dB per trip at every pitch. A longer corridor is visited less often, so it must lose more per trip to fade at the same speed (loss per trip = time per trip × loss rate). The tone control has three stages: a bass shelf and a treble shelf (gentle tilts) and two adjustable bumps. A computer **fitting** routine tries settings, measures how far the result is from the wanted decay at 56 pitches, and adjusts, rather like tuning a guitar string by ear but with arithmetic ("least squares" = make the sum of squared errors as small as possible). It also adds a heavy penalty to any setting that would let the corridor lose too little somewhere, because that pitch would ring on far too long.

Three biquads per line: a first-order low shelf times a first-order high shelf packed into one biquad, plus two peaks. The wanted loss per pass is $-\tau_i \cdot \text{rate}_i(f)$ dB; the loss already in the loop (`lg` and any kept per-axis cut) is subtracted first (VERIFIED, `surf_loop.design_line`). How the fit works in the prebuild (VERIFIED from code):

- target points: 56 log-spaced frequencies from 20 Hz to 20 kHz;
- residual: the relative T error at each point, plus a stability penalty (weight 50) wherever the total loop gain on a dense grid from 0.01 Hz to Nyquist rises above a margin that keeps T ≤ 2× the longest target;
- solver: bounded non-linear least squares (`scipy.optimize.least_squares`), at most 600 function evaluations, starting from shelves at 200 Hz and 5 kHz and peaks at 700 and 2500 Hz.

Step 6 — the guard.

> *Plain first:* a safety inspection for every corridor. If its fitted tone control would let any pitch ring more than three times longer than the longest wanted decay, or if it misses its target badly, it is replaced by a simpler control (one stage), and if that also fails, by the plain Flat-style loss. In every prototype render the loop always lost energy at every pitch (it could never grow), with margin.

After the fit, a line is rejected if its loop gain anywhere implies T > 3× the longest target (or its octave error exceeds 50 %); it is refitted with one shelf biquad (S1), and if that also fails it falls back to the plain broadband loss (VERIFIED, SURFACES_IN_LOOP §7; `surf_loop.design_bus`). Stability on every prototype render: max loop gain −0.006 to −0.40 dB from DC to Nyquist, max pole ≤ 0.99902, every IR finite with peak ≤ 0.53 (VERIFIED, §7).

Step 7 — the 2011 per-axis cuts (D-SL3).

> *Plain first:* the 2011 engine has small fixed dips inside the loop at the lowest resonance of each room dimension; each corridor carries one of the three (Chapter 5). In Surfaces mode the walls and panels take over that job, so those dips are dropped, except for the very deepest resonances (21–29 Hz in the default room) where no practical panel exists. Doing it this way made the corridors match their targets better than keeping the dips and fitting on top of them.

In Surfaces mode the in-loop cuts are **replaced** by materials and panels; engine cuts stay only for the 21–29 Hz modes where no practical panel exists. Replacing gave 0 guard hits and median line error about 5 %, against 14–23 % when folded (VERIFIED, §5, §9). The prebuild reads this as "keep cuts below 30 Hz, fold them in; drop the rest" (`KEEP_CUT_BELOW_HZ = 30`) — an interpretation still listed for Idle (§6.10).

Step 8 — one Render calibration (D-SL4).

> *Plain first:* the formulas are good but not perfect, and the engine has its own small quirks. So the program renders the room once, measures how long each octave actually rang, and if an octave came out, say, 10 % too long it makes that octave's loss 10 % stronger (never more than a factor of 0.6–1.6), then redesigns the tone controls one time. This corrects the engine against the model; it does not "fix" the room to sound like something it isn't. Play then uses the same corrected settings as Render. **T20** is a way of measuring decay time from the first 20 dB of the fade (−5 to −25 dB), scaled up to 60 dB.

Render once, measure T20 per octave (Schroeder slope −5 to −25 dB), scale each band's loss by T20_measured / T_target clamped to 0.6–1.6, and redesign once. "This corrects engine-against-model error only; the room stays the room." Play uses the Render-designed coefficients. Over 12 renders × 7 octaves, bands within ±10 % of target went from 65/84 to 76/84 (VERIFIED, §2 step 8, §9; `surf_loop.calibration_factors`).

6.5a Summary of the eight steps in plain English

Step	What it does, in one sentence
1	Adds up everything on each wall into one "absorption per pitch" curve.
2	Turns absorption into "dB lost per bounce", counting EQ cuts but never boosts.
3	Predicts the whole room's decay per band with classical formulas plus air, and shows it to you.
4	Gives each of the 32 corridors a loss rate: at least half room average, the rest from its own walls.
5	Fits each corridor's three-stage tone control to lose the right dB per trip at every pitch.
6	Checks every corridor for safety; falls back to something simpler if needed.
7	Drops the old 2011 fixed dips except the deepest, because the walls now do that job.
8	Renders once, measures, nudges each octave once, and hands the result to Play.

6.6 How auto-treat works (the automation)

Auto-treat is a button on Surfaces: "Auto-treat to target T_mid = [dial]". It **suggests** Treatment rows; nothing counts until the user accepts, edits or discards them (D-SL5; VERIFIED, SURFACES_IN_LOOP §4; prebuild `surf_autotreat.auto_treat`: "The scene is never modified").

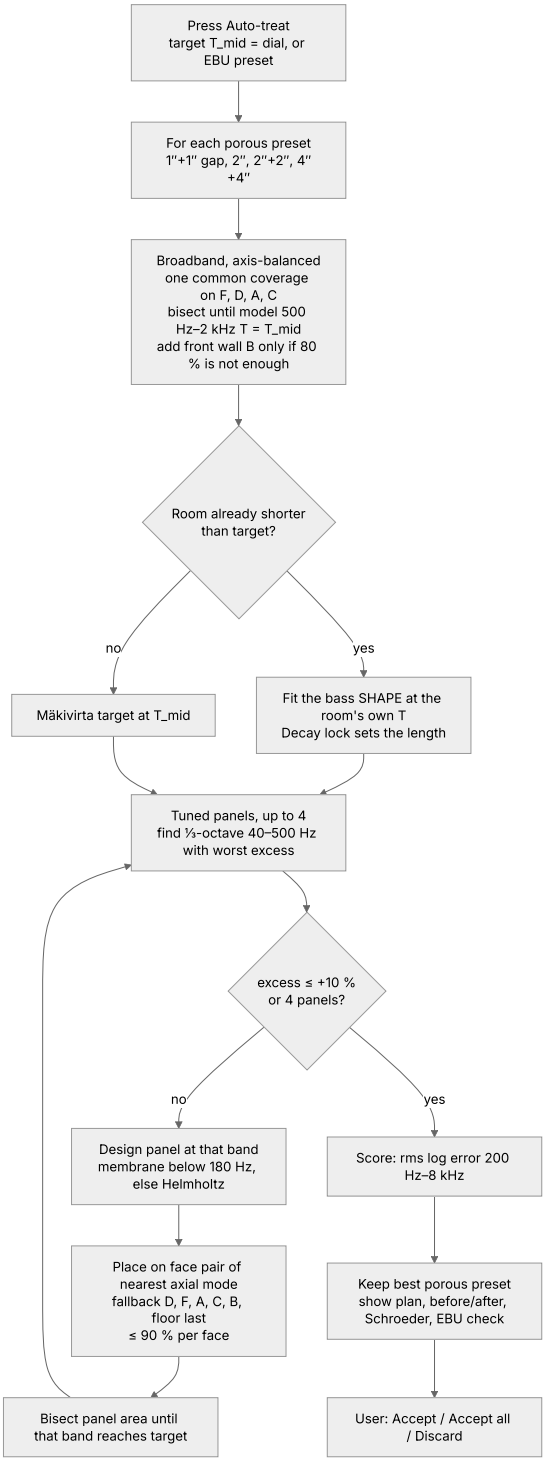


Figure 6.3. Auto-treat (DESIGNED; algorithm from SURFACES_IN_LOOP §4, details VERIFIED from `prebuild/sl/engine/surf_autotreat.py`).

Details from the prebuild code (VERIFIED):

- The broadband bisection runs 30 halvings of the coverage interval [0, 80 %]; each tuned-panel bisection runs 25 halvings of [0, min(50 %, free space on the face)]. These are fixed iteration counts, not tolerance tests.
- The tuned-panel loop makes at most 3 × 4 attempts, stops when the worst band is within +10 % of target, and skips a band if no panel can be designed for it.

- Membranes are chosen from a list of 6 real boards with cavities 50–300 mm; Helmholtz panels use 12 mm board, 6 mm holes, open area 0.5–20 % (VERIFIED, SURFACES_IN_LOOP §4).
- "Why axis-balanced": the first version put 80 % on the ceiling and 90 % on the rear wall; the directional render then showed a double slope because the untreated side-wall axis kept ringing. Spreading panels over the axes removed it (VERIFIED, §4).
- **EBU preset (D-SL9):** target Tm = 0.25·(V/100)^(1/3) s (0.28 s for 144 m³), with pass/fail per band against EBU Tech 3276 tolerances (VERIFIED formula, §3).
- **"Make it a panel"**: on an EQ_PLAN resonance row below 300 Hz, design one tuned panel at that frequency (rows above about 500 Hz stay as decay cuts, because they are engine ringing, not room modes) (VERIFIED, §5; `make_it_a_panel`). This depends on the resonance rows (N5), which are not built.

IN PLAIN ENGLISH.

Auto-treat behaves like an acoustician with a tape measure and a catalogue: 1. **Broadband first.** It picks a thickness of absorber panel and covers the same share of the ceiling, back wall and two side walls, increasing the share until the mid-band decay matches your target (adding the front wall only if 80 % coverage is not enough). Spreading panels evenly matters: an early version piled them on the ceiling and back wall, and the untreated side walls kept ringing on their own, which you hear as a double decay. 2. **Then the bass.** It looks for the third-octave band between 40 and 500 Hz that rings longest compared with the target, designs a tuned panel for that pitch (a membrane below 180 Hz, a perforated Helmholtz board above), places it on the pair of walls whose resonance is closest to that pitch, and grows the panel until that band reaches the target. It repeats up to four times, or until every band is within 10 % of target. 3. **Pick the best.** It does this for each of four absorber thicknesses and keeps the one with the smallest overall error from 200 Hz to 8 kHz. 4. **Show, don't do.** It shows you the list, the before/after curves and a pass/fail check, and waits. You can accept all, accept some, edit or discard. "Bisect" means halving the search range each time, like guessing a number with "higher / lower"; the program does a fixed number of halvings rather than stopping at a tolerance.

6.7 The default-room plan and results

Default room 6 × 3 × 8 m, plaster walls and ceiling, parquet floor, target T_mid 0.8 s (VERIFIED, SURFACES_IN_LOOP §4):

#	Face	Kind	Build	f0	Area / coverage
1	F, D, A, C	broadband porous	4″ panel on 4″ air gap	—	21 % of each face
2	D (rear)	membrane	3 mm hardboard on 59 mm cavity with wool	125 Hz	9.0 m²
3	F (ceiling)	membrane	9 mm plywood on 119 mm cavity with wool	62.5 Hz	10.3 m²
4	D (rear)	Helmholtz	12 mm board, 6 mm holes, 2.3 % open, 100 mm cavity	198 Hz	3.4 m²
5	B (front)	Helmholtz	12 mm board, 6 mm holes, 1.4 % open, 100 mm cavity	157.5 Hz	8.5 m²

- Model: 20/24 ⅓-octaves within ±10 %. **Rendered** (engine + calibration, T20): **16/23 within ±10 %, 19/23 within ±20 %** (VERIFIED). The prebuild reproduces this plan exactly (VERIFIED, prebuild README).
- Remaining misses: 50–80 Hz +23…+65 % (below the lowest panel, modal; EBU allows +0.3 s at 63 Hz), 157 Hz +26 %, 250 Hz +17 %, 400 Hz and 8 kHz −12 %.
- Strict EBU ±0.05 s is not met (worst 0.136 s over 200 Hz–4 kHz), but EBU is a control-room standard (Tm 0.28 s here), while 0.8 s is a treated live-room target.

IN PLAIN ENGLISH.

For the default 6 × 3 × 8 m room with plaster walls and a parquet floor, aiming for 0.8 s, auto-treat suggests five items: 4-inch absorber panels on a 4-inch gap covering about a fifth of four surfaces, two membrane panels (tuned to 125 Hz and 62.5 Hz) and two perforated panels (157.5 Hz and 198 Hz). After rendering, 16 of 23 third-octave bands land within 10 % of target. What remains is mostly the deep bass between 50 and 80 Hz, below the lowest panel, which real rooms also struggle with. A "third-octave" is a band one third of an octave wide, so three of them make one octave.

6.8 Accuracy

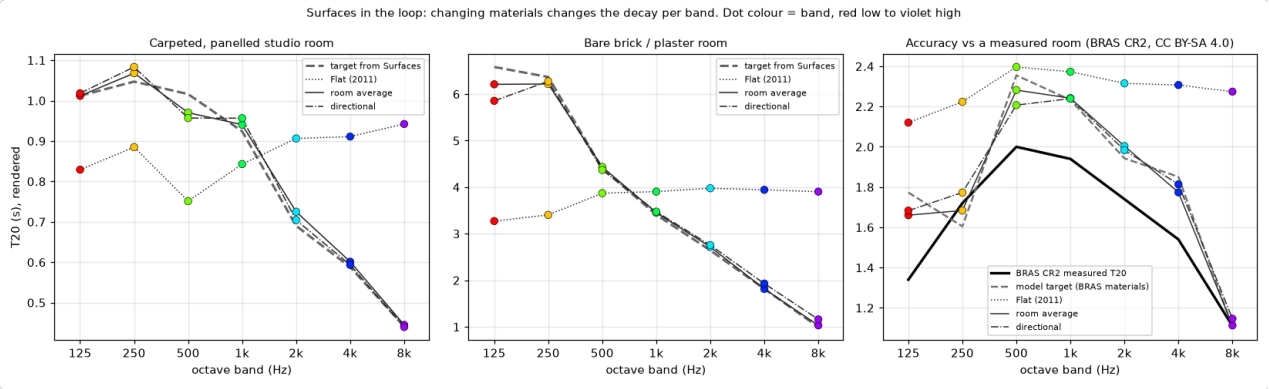


Figure 6.4 (copied from `roadmap/sound_tricks/surfaces_in_loop_rooms.png`). Changing materials changes decay per band; Flat (2011) does not. Right: accuracy against the measured BRAS CR2 room.

Against a measured room, BRAS CR2 (T20 125 Hz–8 kHz: 1.34 / 1.72 / 2.00 / 1.94 / 1.74 / 1.54 / 1.11 s) (VERIFIED, SURFACES_IN_LOOP §6):

Mode	Error per octave	8k/1k ratio (measured 0.57)
Flat (2011)	+58 / +29 / +20 / +22 / +33 / +50 / +105 %	0.96
Surfaces, room average + cal	+24 / −2 / +14 / +16 / +15 / +15 / 0 %	0.50
Surfaces, directional + cal	+26 / +3 / +10 / +15 / +14 / +18 / +3 %	0.51

The engine sits within −6…+10 % of its own target; the source attributes the remaining +10–18 % at mid to the material data and a humidity mismatch (app 0.35 against BRAS 41.7 % RH). Material contrast (T20, VERIFIED): studio 1.0 / 0.94 / 0.44 s at 125 Hz / 1 kHz / 8 kHz against bare brick 6.2 / 3.45 / 1.03 s.

IN PLAIN ENGLISH.

To check the design against reality, a real measured room (BRAS CR2, a published research room) was rebuilt in the program. The 2011 Flat mode rang 20–105 % too long depending on the octave and kept the treble almost as long as the mids (real: treble about half as long as mid; Flat: nearly equal). Surfaces mode came within about −2 to +26 % and got that treble-to-mid ratio right. The source puts most of the leftover error down to the material data and to the air humidity setting not matching the real room. And changing materials now really changes the decay: a carpeted, panelled studio rings 0.44 s at 8 kHz while a bare brick room rings 1.03 s there and 6.2 s at 125 Hz.

6.9 Filter choice and CPU

Of the structures studied offline, one shelf biquad (S1) was too coarse for directional lines (worst octave error 26–46 %), a ~9-band graphic EQ was most accurate (3–7 %) but estimated at ~0.42 ms, too much for Play; **S1P2** (3 biquads) gave 6–14 % room-average and 8–13 % directional (VERIFIED errors, ASSUMED costs, SURFACES**N_LOOP §7** The prebuild's measured Play cost: Flat 0.677 ms → Surfaces 0.708 ms per block, **+0.031 ms** (wasm, box; VERIFIED, prebuild README). The PC t19 gate decides. Render could use the graphic EQ later if listening asks (D-SL7).

> **IN PLAIN ENGLISH.**
> A tone control with more stages would be more accurate but costs more processing in Play, which has a strict time budget per slice of audio. One stage was too crude, nine stages too expensive for live use; three stages (S1P2) were the sweet spot. On the research machine, Surfaces mode added only 0.031 thousandths of a second per audio slice. The real test is on Idle's PC.

6.10 Open questions and findings (for Idle, before SL4–SL6)

From `roadmap/prebuild/sl/README.md` "Findings" and `NEXT_SESSION_PLAN.md` (VERIFIED as open):

1. **Helmholtz Q:** the physical model resonates 5–10 % below the formula f0 with Q ≈ 1–2, not the ASSUMED 3. Changing it moves auto-treat plan numbers, not goldens.
2. **D-SL3 reading:** keep cuts below 30 Hz (16/23 bands in the treated render) or drop all (15/23).
3. **SL4 double slope (xfail, not hidden):** untreated plaster room with directional lines, EDT/T30 0.62 / 0.46 / 0.57 at 250 Hz / 4 k / 8 k against Flat 1.04 / 0.85 / 0.89. Directional or room average to be decided by listening (A/B 3–4).
4. **Treatment also colours the early reflections** in Surfaces mode (treated α goes into the cards for early reflections and injection).
5. When the target is longer than the room can be, auto-treat suggests no porous rows and fits only the bass shape (`shape_only`); the UI must say so.
6. SL2 room average before calibration is −12 / −33 % at 125 / 250 Hz; calibration is what brings it in.

> **IN PLAIN ENGLISH.**
> These are the questions Idle will be asked before the later Surfaces sessions: how sharply the perforated panels should be tuned (the physics says less sharply than first assumed); whether to keep the old deepest dips; how strongly each corridor should follow its own walls (to be settled by listening, because the strong version gives a slight double decay in a bare plaster room); and the reminder that treatment will also change the early echoes, not just the tail. When you ask for a decay longer than the bare room can give, auto-treat suggests no absorbers and only shapes the bass, and the screen must say so.

6.11 What is not known yet

- **Fitter convergence is not specified.** The line fit runs a bounded least-squares solve capped at 600 evaluations; the prebuild does not test or log whether the solver converged, and no tolerance is set beyond the solver's defaults (VERIFIED by reading `design_line`). Quality is judged afterwards by the guard and the octave error. A convergence report per line is OPEN.
- **The calibration is one pass by design**, not an iteration; whether one pass suffices for rooms far from the test set is not proven (OPEN; tested on default, studio, brick and CR2).

- **Speed of sound:** the Surfaces model uses `C_AIR = 343.0` m/s (VERIFIED, `surf_target.py` line 20); the engine's locked value is `c = 340` (HANDOVER). Mode labels and air-per-path differ slightly; whether this matters audibly is not tested (OPEN).
- **Panel α values** (Q, αmax) and the room-shell values are ASSUMED until fitted to datasheets or measurements.
- The **Room result panel** UI (target vs rendered per band, Treatment list, Auto-treat button, Schroeder and mode markers) exists as a mockup and a prebuild proof only (DESIGNED, SURFACES_IN_LOOP §8; `prebuild/sl/shots/`).

> **IN PLAIN ENGLISH.**
> Some things are honestly unknown. Nobody has checked that the fitting routine always "finishes" properly for every corridor; the design relies on the safety check afterwards. The one-pass calibration has only been tested on four rooms. The Surfaces model uses a speed of sound of 343 m/s while the engine uses 340 m/s, a small mismatch still to be settled. The exact absorption of the tuned panels is an educated guess until it is fitted to real data. And the "Room result" screen exists only as a mock-up.

Chapter 7 — Microphones and the mix desk

> **Status:** DONE **(draft 2026-10-05)** for build 51. Thursday's desk and mic sessions (T1, T2, T3, T3b, T4b, T5, T7–T9) and the U87 default (SL7b) are LOCKED**, not merged**.

Chapter purpose. The room is heard through three virtual microphones and a small mixing desk. This chapter explains how a capsule "hears" a reflection (position, aim, pattern and measured colour), which stereo setups exist, how the desk folds L/C/R to stereo, and what Thursday will change. It also records the conventions that must never be flipped (left is left), because they took the longest to settle.

7.1 The stand and the capsules

Positions. Three capsules L, C, R sit on one stand. The stand X (`mcx`) and Z (`mcz`) are free in the modern UI (2011 locked `MCX = W × 0.5`); L and R sit at `MCX ∓ Space/2`; the centre's Z follows the locked rule `MCZ = L − MZ` (VERIFIED, HANDOVER hard locks; `/workspace/docs-publish/docs/HOW_IT_WORKS.md` §3). Today L and R are always symmetric about the stand: dragging either capsule sets `space = 2·|x − mcx|`; only C has its own offset (VERIFIED, THURSDAY_LIST U2, citing `mic_desk.js` 1056–1062).

Heights. Per-mic heights landed in S-AUDIT with the locked variant H-1: "FDN lines stay at stand height" (tail change ≤ 0.42 dB, against up to 4.6 dB for the rejected full version) (VERIFIED, BIG_JOBS §0.0).

Aim and the left/right convention (K1). One sign constant, `YAW_X_SIGN = +1` in `engine/early_field.py`, mirrored as `RoomMath.YAW_X_SIGN` in `room_math.js`, is used by every look/angle formula. It fixed "crossed capsules": with a source 32° left, the direct sound is now +3.5…+8.9 dB louder in L (it was −1.7…−8.7) and L arrives first or no later (VERIFIED, DECISIONS_LOCKED_2026-09-28 K1). Default ORTF is L −55° / R +55°, aiming outward: "the left capsule looks toward −X and the right capsule looks toward +X … Do not flip L/R to chase the picture." Floor plan, 3-D and speakers agree: room −X = left capsule = bus `l` = `output[0]` (VERIFIED, HANDOVER "Two-source image", "Still open" item 2).

> **IN PLAIN ENGLISH.**
> Think of a real mic stand with a bar holding three microphones: left, centre and right. You can move the stand around the room and set how far apart the left and right mics are ("Space"). Each mic can now have its own height, but the tail is still calculated from the stand's height (a locked compromise, H-1, that changes the tail by less than half a dB). Each mic points somewhere ("aim" or "yaw"). After a long hunt for "crossed" left/right images, one rule was fixed for good: the left capsule is on the room's left, points left, and feeds the left speaker. A **capsule** is the sensing part of a microphone.

7.2 How a capsule hears a reflection

"**Mic** is the listen tap off the diffuser output, in parallel with feedback. It is not on the inject, and the coloured signal is not written back." (VERIFIED, HANDOVER "Mic, air, and wall cards".) For each arrival — direct, each early tap, each FDN line:

1. **Angle.** "The angle is the 3D angle between capsule aim and the incoming ray (dot product), not the floor-plan angle." (VERIFIED, same.)

2. **Pattern gain.** The capsule's polar pattern at that angle scales the tap gain. Patterns never change which images are used (Chapter 5).
3. **Measured colour.** A measured capsule response at that angle, built as a minimum-phase FIR. In Play it is an 80 ms FIR at the AudioContext rate, applied on the way out of the direct read, the early left/right read and the FDN output; those copies are not written back (VERIFIED , HANDOVER "NOW — Play is one C block" and "Mic, air, and wall cards", Phase 3).
4. **Direct fold.** The direct sound is filtered per capsule *before* the fold, then × 0.55 (VERIFIED , HANDOVER; "do not retune … the 0.55 Direct fold").

Where the colour data comes from. Measured presets are fitted by `mic_fitter` into SOS tables; there are 20 tables per rate, at 44.1 and 48 kHz only, worst fit 0.408 dB at 44.1 kHz; other rates switch colour off with a notice (VERIFIED , DECISIONS_LOCKED_2026-09-28 item 7). The Neumann presets are U87 Ai Cardioid / Omni / Bidirect from the Franco et al. dataset (Zenodo 10.5281/zenodo.4633508); U87 Ai Cardioid is the only measured Neumann cardioid; other measured cardioids include AKG C414, Røde NT2-A, Schoeps CMC5-U (VERIFIED , THURSDAY_LIST U3).

Colour modes. Measured · User defined · Bypass (VERIFIED , BIG_JOBS §0.0 S-UI2 fix-up). In Measured the preset owns the pattern (D-UI21); Bypass = preset pattern, no measured colour (D-UI22). "Assumed toys" was renamed User defined (VERIFIED , BIG_JOBS §0.H).

> **IN PLAIN ENGLISH.**
> For every sound arriving at a mic (the direct sound, each early echo, each corridor of the tail) the program asks: from which direction does it arrive, compared with where the mic is pointing? The mic's **polar pattern** says how loud it hears sound from each direction. An **omni** hears all directions equally, a **cardioid** (heart-shaped) mainly the front, a **figure-8** front and back but not the sides. Real mics also change tone with direction, so each mic model's measured response is applied too (its **colour**). That uses a short filter worked out from lab measurements of real mics (the U87 Ai, AKG C414, Schoeps CMC5-U and others), and it only exists at 44.1 and 48 kHz. The mic only listens; what it hears is never fed back into the room. **FIR** (finite impulse response) is a filter made from a short recorded "fingerprint"; **minimum-phase** means built so that it adds as little delay as possible. **SOS** (second-order sections) is a compact chain of small filter stages.

7.3 The setups

The Mic setup menu in build 51 (VERIFIED , `ui_audit_1004/app/software/static/index.html` 547–558):

Setup	Notes
AB spaced pair (omni / cardioid)	spaced pair
XY coincident 90°, XY 120°	coincident
ORTF (default)	±55°; today includes the C mic at 0 dB (the 2011 477 fold)
NOS, DIN	near-coincident
MS — omni mid, cardioids inward	labelled MS
MS classic (Mid cardioid + Side fig-8)	M/S decode on the desk
Mono centre	C only; L/R locked (D-AUD-1)
Binaural (dummy head)	see below
Custom	free tweak

Binaural. Default head SADIE II KU100; ears 0.18 m apart (0.175 m for THK); `1/r²` from the head centre with ILD from the HRTF only; head yaw turns the ears as one unit; centre mic muted; K = 6 biquads per ear with a 60-entry mirror table; a 32-image set with 5 rear images (`BINAURAL_REAR32`); the diffuse FDN tail is matched to the head's interaural coherence in Render and Play (VERIFIED, BIG_JOBS §2). Still missing (UI): head drawing, HRTF polar/ILD panel, floor rotate handle, Play rate notice (VERIFIED, BIG_JOBS §2 and J12; session S6, not on Thursday).

Default mics today: SM57 × 3 (VERIFIED, THURSDAY_LIST U3, citing `app.js` 97, `app_server.py` 636 / 1028–1030 / 1317–1319, `render_pipeline.py` 1351–1353). Changing to U87 Ai Cardioid is D-UI30, merged into the single default switch SL7b and gated on Idle's written OK on the SL7a final A/B (LOCKED). T5 renders an early SM57 vs U87 A/B (with CMC5-U Cardioid as a third reference) for listening only.

> IN PLAIN ENGLISH.
> A **setup** is a recipe for arranging the mics, as recording engineers do. **AB**: two mics spaced apart (good sense of space). **XY**: two directional mics at the same point angled apart (sharp, mono-safe image). **ORTF**, **NOS**, **DIN**: in between, slightly spaced and angled (ORTF is the default; in the recording world it means two cardioids about 17 cm apart angled ±55°: the ±55° is VERIFIED in this program, the 17 cm is general background). **MS**: one mic facing forward (Mid) plus a sideways figure-8 (Side), decoded into left and right on the desk, which lets you change the stereo width afterwards. **Binaural**: a dummy head with ears, for headphones. **Mono centre**: just the centre mic. The default mics today are three Shure SM57s; the plan is to switch to Neumann U87 Ai cardioids in the final default switch, after Idle has listened.

7.4 The desk (S-MIX, build 51)

Idle locked D-UI3 (three independent mics and L/R link modes), D-UI4 (desk matrix, Width, route chips including "S → +L, −R" on any strip) and the 2011 linear pan law as default on 3 Oct 15:57 (VERIFIED, HANDOVER "S-MIX").

Strips and buses. Mic strips L, C, R and buses Direct, FDN, Early, Host dry. Each strip has trim, route, width (on route s), pan, phase and a fader in dB. There is no EQ on the desk (VERIFIED, THURSDAY_LIST U7; EQ_PLAN §5). Solo and Mute exist on the four buses only; "Solos and mutes zero the send and leave the fader" (VERIFIED, THURSDAY_LIST U7; HANDOVER older Mix note).

Routes (VERIFIED, THURSDAY_LIST U7, citing `app.js` `pair()` 442–457), with `a` the strip gain:

Route	Left, Right
centre	0.5a, 0.5a (equals pan at centre under the linear law)
both	a, a
l / r	a, 0 / 0, a
s	a, −a (MS side; Trim becomes Width)
pan	a(1 − t), a·t (locked 2011 linear law)

Phase (`phase_l/c/r` = ±1) flips the whole strip. MS classic sets C = both with `gain_exact_c = 1.5` (+3.5 dB) and S at 0 dB (VERIFIED, UI_AUDIT Q2).

Array and Room. "Array = this mic stays on the stand with the others. Room = this mic is unplugged from the stand and placed at the room-mic spot typed in the Mic tab (X, Z, height, yaw)." All Room mics share one spot today (VERIFIED, THURSDAY_LIST U7, U2).

Known desk bugs (VERIFIED, UI_AUDIT §A): B1 pan/trim knobs nearly uncontrollable (whole range in about 16 px; trim 1.5 dB per px); B2 width needle never moves; B5 Host strip hidden at 1280 px; B8 Pan live but ignored off the Pan route; B3/B4 MS classic flips to "Custom" and has a cardioid S in Measured.

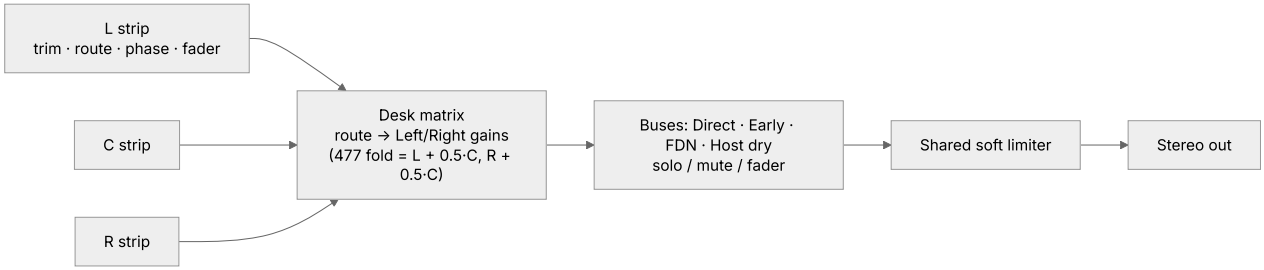


Figure 7.1. Desk, simplified. In the engine the per-bus split happens before the matrix; the order here is schematic (ASSUMED simplification).

IN PLAIN ENGLISH.

Each mic has a channel strip on the desk: **trim** (input level), **route** (where it goes: left only, right only, both, panned, or "side" which goes +left and −right for MS), **width** (for the side signal), **pan**, **phase** (flip upside down) and a **fader**. The finished mix is made of four **buses** (groups): direct sound, early echoes, the tail (FDN) and the untouched dry signal, each with its own solo, mute and fader. A mic can stay "on the stand" with the others (**Array**) or be taken off and put somewhere else in the room as a room mic (**Room**). Some knobs are currently very hard to use (the whole range in a few pixels); that is the first Thursday fix. The **pan law** says how level is shared between left and right as you pan: in the 2011 linear law, the two sides always add up to the same total.

7.5 What Thursday changes (LOCKED, not merged)

From `NEXT_SESSION_PLAN.md` and THURSDAY_LIST §5 (VERIFIED as plan):

Session	Change	Sound / goldens
T1	Knobs: relative drag over 200 px (trim 0.12 dB/px, pan 0.01/px), Shift = fine, double-click reset, wheel step, ±6 px detent, value readout, click-to-type everywhere (37 static + 14 per-face sliders)	none
T2	D-UI31: L and R buttons with phase invert on **every** strip (truth table below); L+R = "both" when the strip is the Mid of an MS pair, otherwise pan; Ø buttons removed; Trim as text; mic-strip **solo-in-place**; "On stand" / "Room mic" names (D-UI36)	none (old scenes map 1:1)
T3	D-UI29: MS presets (a) fig-8 S + omni M, (b) fig-8 S + cardioid M — **default (b) = U87 Ai Bidirect + U87 Ai Cardioid**, (c) back-to-back cardioids as S; (d) dropped; fixes B3/B4	new MS tests
T3b	D18: unity MS decode, −3 dB on M and S (gain 0.708); D19: with route s, the direct sound uses the pattern spikes in Render and Play	MS references only
T4b	D20: ORTF = C off by default; "ORTF + C" preset = today's sound	**main golden YES**
T5	SM57 vs U87 A/B renders (box job), early listen	none
T7 / T8	D-UI37: one **Mirror / Linked / Split** control; in Split, R gets its own preset, pattern, HPF, position and yaw, also in MS; one room spot per strip; drag R and room mics; wall drag with "stop"	none (Mirror bit-exact)
T9	Mic tab clean-up (binaural and mono-centre dead controls, overlapping drag targets)	none

T2 truth table (proposed buttons → existing routes; VERIFIED as specified in THURSDAY_LIST U7):

L button	R button	Route	Pot
on	off	l	greyed
off	on	r	greyed
on	on	pan (centre detent) — or *both* for an MS Mid	balance
on	inverted	s (+L, −R)	width
inverted	on	s with Ø (−L, +R)	width
inverted	inverted	pan with Ø	balance
off	off	muted	—

Why D19 matters: with one source, the direct sound ignored the mic pattern, so an MS direct sat 11.2 dB to the left (VERIFIED as finding R18, THURSDAY_LIST §5 note).

IN PLAIN ENGLISH.

Thursday's mic and desk sessions, none of which changes today's default sound except T4b: knobs you can actually control and values you can type (T1); simple L and R buttons on every strip, where pressing both or inverting one picks the right route automatically (T2); three proper MS presets, defaulting to the classic U87 figure-8 plus cardioid (T3); two MS sound fixes, so MS decodes at the right level and the direct sound respects the mic patterns (T3b, which changes MS reference recordings only); ORTF without the centre mic by default (T4b, which does change the main reference); an early U87-versus-SM57 listening test (T5); and Mirror / Linked / Split so the right mic and the room mics can be moved freely (T7, T8).

Open / gaps

- Q3 (should AB omni / MS omni-mid stay Measured with an omni preset?), Q4 (keep "Heavy curtain"), Q6 (yaw boxes under the polar plots) are open (UI_AUDIT §C).
- A unity-at-centre balance law would change locked D-UI4; it is a question for Idle, not a default (THURSDAY_LIST U7).
- The exact per-bus order of the desk matrix in the C engine is not described in the sources read; Figure 7.1 is schematic.

Chapter 8 — Testing, goldens and how nothing changes by accident

> **Status:** DONE **(draft 2026-10-05)** for the test system as it stands at build 51. The Flat (2011) parity suite (SL7a) and the SL7b re-recording are LOCKED**, not built**.

Chapter purpose. This project's promise to Idle is "no sound changes without your decision". That promise is kept by tests, not by good intentions. This chapter explains the four layers of checking (golden files, the run_all harness, the phase acceptance suite and the software unit tests), what each status word means, how a golden is allowed to change, how CPU is measured fairly, and which gates are still open. It is the chapter to read before trusting any "bit-exact" claim elsewhere in the book.

Sources: `3_phase_acceptance_tests/README.md`; `3_phase_acceptance_tests/harness/run_all.py` (vendored copy of `engine/tests/run_all.py`); `roadmap/eq_plan/src/HANDOVER.md`; `roadmap/THURSDAY_LIST.md` §5 and session rules; `roadmap/DECISIONS_LOCKED_2026-10-04.md` (D-SL10); `roadmap/prebuild/sl/README.md`; `BIG_JOBS_ROADMAP_2026-10-01.md`.

8.1 Why goldens

A **golden** is a stored impulse response for a fixed scene. After any change the engine renders the scene again and compares sample by sample. If the null is perfect (max-abs 0, or ≤ −100 dB for the golden tests), nothing audible could have changed (VERIFIED, `run_all.py` t03/t04 docstrings).

There are five golden files (VERIFIED, `run_all.py` t03/t04; HANDOVER):

- four **Play** goldens: `play_std_unlocked_byp`, `play_std_unlocked_col`, `play_std_lock18_byp`, `play_std_lock18_col` (decay unlocked and locked at 1.8 s, colour bypassed and on);
- one **Render** golden: `render_std_unlocked_byp.f32` (Render, BYPASS colour, 0.5 s).

Next to them, `GOLDEN_META.json` records the SHA1 of every file and a **reason** string. A golden may only be rewritten on purpose, with a reason, and that change is the only sound change in its session (VERIFIED, THURSDAY_LIST session rules: "**NO** = the main Render golden `ebd77c3b…` must stay bit-exact. **YES** = the session regenerates goldens on purpose, and it is the only change in that session").

The recent chain of the Render golden (VERIFIED unless noted):

When	SHA1	Reason (GOLDEN_META)	Source
2 Oct 19:22	—	"S7 z^-1 in both + equal F3 trim (D3/D3b)"	eq_plan HANDOVER
3 Oct (S5 CPU fix-up)	`7894bab0da4526c975e78084ecdf62163fada1bc`	Render inject uses Play's fitted surface filters (D-S5c-3); max-abs 9.894e-04	eq_plan HANDOVER
3 Oct (D-MIX-J)	`3e992bf0f3b29d7cbf2614b93af5e529ae7a4235`	"D-MIX-J mirror-invariant early jitter"	eq_plan HANDOVER
after S-AUDIT	`ebd77c3b599e80a037f4a3c552c93f6e37343d1b`	not on the box	THURSDAY_LIST (it cites HANDOVER and GOLDEN_META on the PC)

In each of those steps the four Play goldens stayed byte-identical (t03 max-abs 0) (VERIFIED for S5 fix-up and D-MIX-J).

Which sessions may change goldens (VERIFIED as plan, `NEXT_SESSION_PLAN.md`): T4 Smooth and T4b ORTF C off (main golden YES), T3b (MS references only), and the SL7b default switch (all defaults, once). Everything else is NO; T11 Movement and SL2–SL6 add new references for new modes but leave the default bit-exact.

> **IN PLAIN ENGLISH.**
> A **golden** is a reference recording of a fixed test scene, stored as a file. After any change, the program records the same scene again and compares it sample by sample with the golden. If not a single sample differs ("bit-exact"), nothing you could possibly hear has changed. Each golden also has a **SHA1**, a 40-character fingerprint of the file: change one bit of the file and the fingerprint changes completely, so comparing fingerprints is a quick way to prove two files are identical. A golden may only be replaced on purpose, in a session allowed to do so, with the reason written down (in `GOLDEN_META.json`).

8.2 The run_all harness (t01–t19)

`run_all.py` drives Play through node + wasm and Render through the Python app, on fixed scenes (`std_unlocked`, `std_lock18` and others). Descriptions are from the vendored copy's docstrings (VERIFIED; thresholds in the PC copy may have moved since, e.g. t19 was re-based in the S5 fix-up):

Test	Checks
t01	Packed parameter blob matches the worklet layout
t02	wasm output equals a native build of `live_stereo.c` (skips without gcc/clang)
t03 / t04	Play / Render goldens, null ≤ −100 dB
t05	Omni source radiation reproduces the golden
t06	Scattering 0 reproduces the golden (pending: scattering not built)
t07	Tail energy (80 ms–1 s) within 0.5 dB over source yaw 0/90/180
t08	FDN stability: no NaN/Inf/growth over the full Dffs range, 8 s decay, 15 s noise then silence
t09	Silence costs ≤ 1.3 × playing (no denormal blow-up)
t10 / t12	Play / Render T30 per octave 500 Hz–8 kHz within ±15 % of the dial
t11	HF residue in Play
t13	Play vs Render parity (T30 within 10 %, tail level, early level)
t14	Level sanity: peak < 0.95 and FDN bus within ±0.5 dB of the pre-F2 level
t15	Active CPU: 1 source + colour ≤ 60 % budget, 2 sources ≤ 85 %
t16	Surface jitter and Structure Dffs are separate keys
t17	Play follows `hadamard_n`
t18	FDN-only Play tail nulls Render ≤ −30 dB (slow)
t19	Two directional sources + colour within the CPU ceiling

Latest full run on the PC (S5 fix-up, VERIFIED, eq_plan HANDOVER): **16 pass, 0 fail, 1 skip (t02), 1 pending (t06), 1 xfail (t13)**; t19 shared-off median 2.238 ms (limit 2.40), shared-on 1.760 ms (limit 2.133). t18 measured −71.9 / −71.8 dB on quiet runs (VERIFIED, same).

IN PLAIN ENGLISH.

The **harness** is a set of 19 automatic checks run after every change. In everyday terms they ask: do the live engine and its source code agree? Is the reference sound unchanged? Does the tail die away instead of growing? Is the decay time what the dial says? Do Play and Render match? Does it stay within the processing budget? Each check prints PASS, FAIL, SKIP (could not run here) or XFAIL (expected to fail because the feature it tests is not built yet). On the PC, the latest full run had 16 passes, no failures, one skip, one pending and one expected failure (the known early-echo difference).

8.3 The phase acceptance suite (package 3)

`3_phase_acceptance_tests/` is a pytest package that encodes the roadmap's success criteria (P0 engine, P1 blob, P2 air, P3/P4 surfaces, P5/P6 scattering and radiation), the product locks (unscaled ±1 Hadamard, $1/r^2$ level, all 32 early images inject into the FDN, no scattering inside the loop, no air on the FDN return), the 28 Sep decisions (K1 left-is-louder-and-earlier for XY/ORTF/NOS/DIN/MS, K3 29 distinct delays per bus), mic colour (20 SOS tables, ≤ 0.5 dB at 48k, ≤ 0.6 dB at 44.1k, unsupported rates off with a notice) and mirror symmetry (≤ 0.1 dB per third-octave, null ≤ −60 dB) (VERIFIED , package README).

Status words (VERIFIED , package README):

- **PASS** — criterion met.
- **FAIL** — the feature *is* in the tree and misses: a real bug.
- **XFAIL** — a probe says the feature is not in the tree yet; when it lands the probe flips and the real assertion runs. This is how planned work is tested before it exists, without lying about it.
- **SKIP** — node, zig or `--slow` missing.

The last recorded box results are from 28 Sep and 1 Oct (e.g. 60 / 2 / 22 / 0 on the patched tree); they predate builds 46–51 and are history, not current status (VERIFIED as dated).

IN PLAIN ENGLISH.

This is a second, larger set of tests written from the roadmap itself: every promise in the plan ("the direct sound has no air filter", "left source means left mic louder and earlier", "20 mic colour tables within 0.5 dB"...) becomes a test. The clever part is **XFAIL**: a test for a feature that does not exist yet first checks whether the feature is there; if not, it reports "expected to fail" instead of a false alarm. The day the feature lands, the real test switches on automatically.

8.4 Software tests, selfcheck and prebuild tests

- `software/tests` (pytest): **55 passed, 0 failed** after D-MIX-J (VERIFIED , eq_plan HANDOVER).
- `engine/selfcheck.py` : **45/45** at the S5 fix-up (VERIFIED , same). It asserts structural facts such as the direct image at index 2 and the 29-unique delay count (VERIFIED , MIRROR_ASYMMETRY_BRIEF).
- Surfaces prebuild (`roadmap/prebuild/sl/tests`): fast **36/36**; slow SL2 4/4, SL5/SL6 render 2/2, SL4 4 pass + 1 xfail; `hash_scenes.py` checks the PC golden hash (VERIFIED on the box, prebuild README). These are box results on a copy, not the PC.

Suggested order after any engine change: `run_all.py` , then `run_phase_tests.py --slow` , then `pytest software/tests` (VERIFIED , package README). Every session ends with the full suite green and the Render golden hash recorded in HANDOVER (VERIFIED , THURSDAY_LIST session rules).

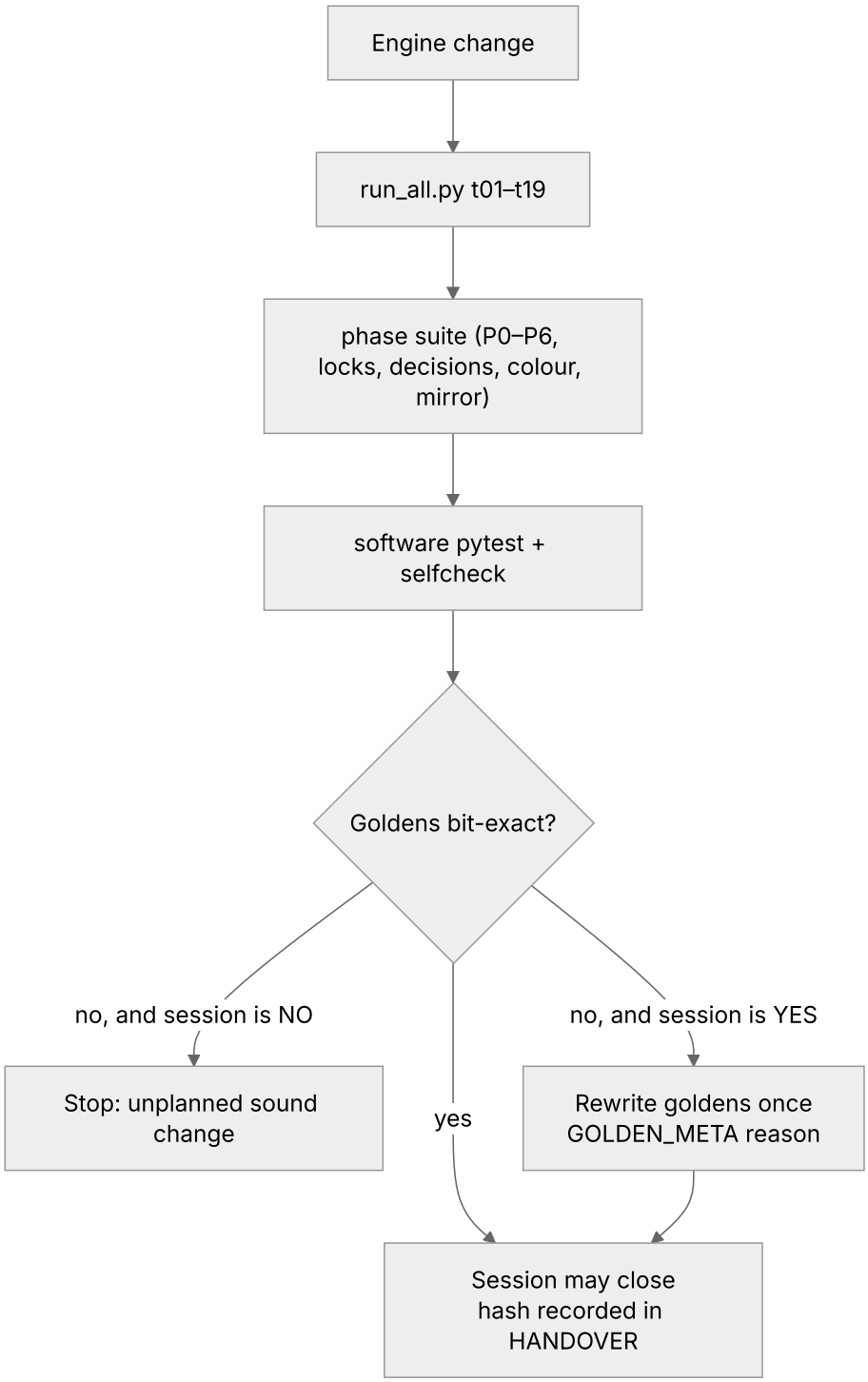


Figure 8.1. The gate every session passes (VERIFIED as described in THURSDAY_LIST session rules; drawn here).

IN PLAIN ENGLISH.

Two more layers: the program's own unit tests (55 passing) and a built-in self-check (45 of 45) that confirms basic facts such as "the direct sound is image number 3". The Surfaces code waiting on the research machine has its own tests too, all passing there apart from one honest expected failure. The order to run them after any engine change is harness, then phase tests, then unit tests.

8.5 Measuring CPU fairly

Single CPU numbers drift with machine load, so the locked method is a **paired ABBA run**: old and new wasm alternate in the same session, 12 rounds, `bench 4 10` on the t19 scene, and the result is the median of per-round paired differences. At the S5 fix-up this gave a median 2.2355 ms (ceiling 2.40) and a paired +5.88 % (VERIFIED, eq_plan HANDOVER D-S5c-2; locked as D-S5c-2 in BIG_JOBS). The box can disagree with the PC (box +8–9 % in the same comparison); **the PC is the gate** (VERIFIED, same).

> **IN PLAIN ENGLISH.**
> Computer timings wobble with whatever else the machine is doing. To compare an old and a new version fairly, they are run alternately (old, new, new, old: "ABBA") twelve times in the same sitting, and the middle result is used. Idle's PC is the judge, because the research machine can give different numbers.

8.6 Flat (2011) parity suite (LOCKED, not built)

D-SL10: when the defaults switch to Surfaces + natural decay + U87, "Flat (2011) stays a selectable mode, and its 2011 references are kept as a **separate frozen parity test** (`tests/refs_flat2011/`, bit-exact with a legacy profile)." SL7a builds it together with `legacy_2011_profile()` and leaves the `DEFAULT_SET` flag at 2011; SL7b flips it, regenerates every reference once with GOLDEN_META "SL7b default switch", and Flat must stay bit-exact (VERIFIED as plan, DECISIONS_LOCKED_2026-10-04; THURSDAY_LIST SL7a/SL7b). The number of references the suite will hold is not fixed in the sources read (Open).

> **IN PLAIN ENGLISH.**
> When the new sound becomes the default, the 2011 sound must not be lost or quietly altered. So, before the switch, its reference recordings will be frozen into a separate test set with a "2011 profile" that recreates the old settings. From then on, any change that alters the 2011 sound by even one sample fails that test.

8.7 Open gates

- **t13** early parity XFAIL: Play early about +3 dB vs Render (32 taps vs the full set); last measured L +3.1 dB, R +3.25 dB (VERIFIED, THURSDAY_LIST L11; eq_plan HANDOVER).
- **P5 coherence** miss (VERIFIED as open, THURSDAY_LIST L11).
- **Small-room decay**: the figures differ between sources (−8.5 % vs −3.5 % after the S4b signs); the P-D7 step re-measures it in Play on the PC (Open; see Chapter 5).
- **Movement** (EQ-1): modulated output cannot null against a golden; the plan uses statistical gates and keeps goldens bit-exact with modulation off (VERIFIED as plan, EQ_PLAN decision 4). The exact statistical thresholds are for the session to write.
- **Surfaces accuracy** gate ST-7 is open (Chapter 6).
- The latest golden's GOLDEN_META and the S-AUDIT HANDOVER section are on the PC only; this chapter's last row is second-hand until they are mirrored.

> **IN PLAIN ENGLISH.**
> The known open items: Play's early echoes are about 3 dB louder than Render's; one coherence test misses; the small room's decay figure needs re-measuring on the PC; Movement will need statistical tests (it can never match a fixed recording because it changes over time); and the Surfaces accuracy gate is still open.

Chapter 9 — Non-rectangular rooms

> **Status: STUB, expanded 2026-10-05 — PLANNED (research only, no engine code).** Nothing here is in the app. The Box stays the default.

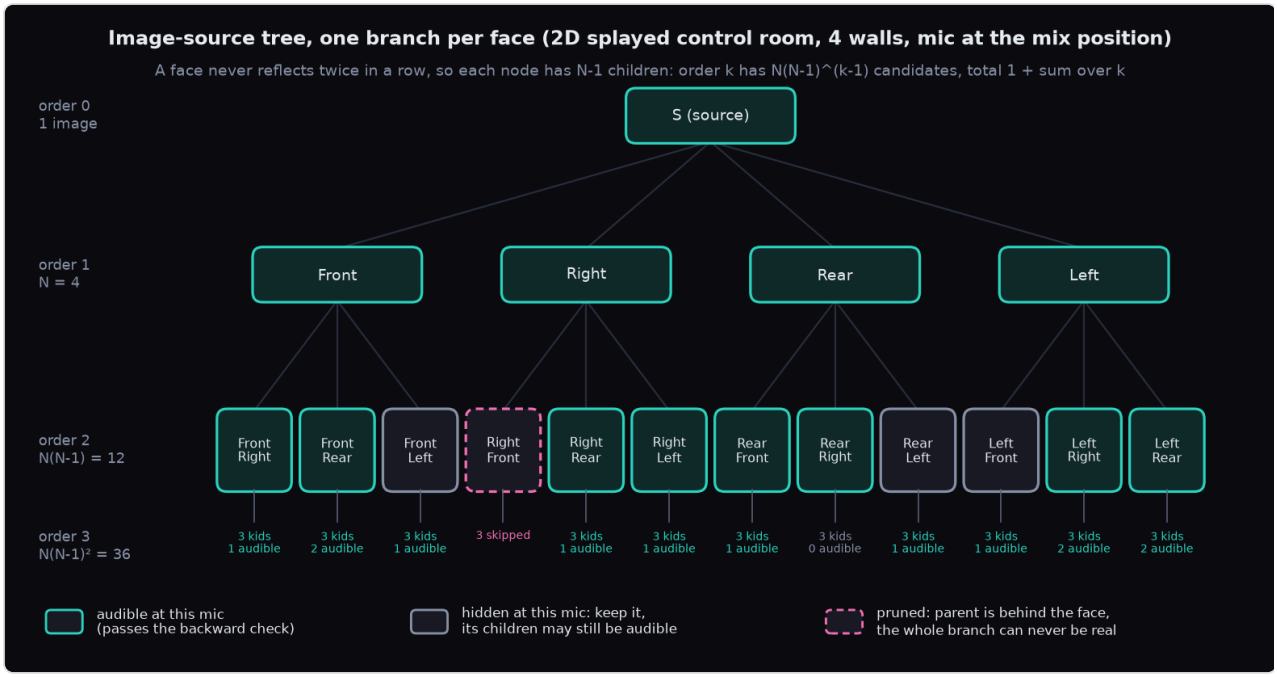
Chapter purpose. Idle proposed studio-shaped rooms (splayed walls, booths, L-shapes) that scale with the W/L/H sliders, with image sources found as a tree, one branch per face. This page records what the research found, so the idea is not lost; the full chapter is written only if the work is scheduled after SL7b.

Sources: `roadmap/nonrect/NONRECT_RESEARCH.md` (4 Oct 2026); figures `figures/D1_branch_tree.png`, `figures/D3_counts.png`.

> **IN PLAIN ENGLISH.**
> Today every room in the program is a shoebox. Real studios often are not: control rooms have angled side walls, vocal booths can be six-sided, ceilings slope. The idea is to offer a few such shapes as presets that still grow and shrink with the usual width/length/height sliders. The mirror trick (Chapter 5) still works for any flat-walled room, but with a catch: in an odd-shaped room many "mirror images" are fake, because the wall you would have to bounce off isn't actually in the way of the path. So each image has to be checked, and most are thrown away. The research shows there are still plenty of real echoes to fill the 32 slots.

What the research says (VERIFIED in the source unless marked)

- **The tree is right.** This is Borish's (1984) extension of the image model to polyhedra. A face cannot reflect twice in a row, so each node has N−1 children: total candidates up to order K = **1 + N[(N−1)^K − 1]/(N−2)** (a box at order 4: 937).
- **Most candidates are not real.** Each image is checked backwards (mic → deepest image → through each face outline in turn). A **parent-side test** (the parent image must be in front of the face) is independent of the mic and prunes whole branches. In the prototype only 2–9 % of order-4 candidates are valid, yet every shape tested has more than 32 valid images by order 3, enough for today's 32 taps.
- **Presets proposed:** Splayed control room (RFZ-style), LEDE-style and Non-parallel live room first (convex, no blocking tests); then Hexagon and Gable booths; the non-convex L-shape last.
- **Staged plan:** 1 Render-only `room_shape` with 3 convex presets (Box bit-exact) → 2 A/B renders → 3 Play support → 4 UI with N-face Surfaces → 5 optional non-convex/beam tree/diffraction.
- **Decisions for Idle:** D-NR1–D-NR8 (presets first, tap picking, frozen FDN delays per preset, Render-first, 15 ms tap fade ASSUMED, face mapping to A–F, whether to do the L-shape, park SDN).



What it could feel like (PLANNED , plain English)

A description of the intended experience, not of built behaviour.

You pick "Splayed control room" from a shape menu. The floor plan redraws with the front side walls angled out; the Surfaces tab shows the same faces, now not all parallel. The early reflections change, because the angled walls throw the first echoes away from the listening position, which is why studios build rooms this way. In the first stage this would be heard in Render only, compared side by side with a shoebox of the same volume, before anyone decides whether Play should support it (D-NR4).

Open / not yet built

No `room_shape` parameter exists. How Surfaces-in-loop (Chapter 6) maps onto N faces is undecided (D-NR6). Not scheduled in T1→SL7b.

Chapter 10 — One shared room in Studio One

> **Status: STUB, expanded 2026-10-05 — PLANNED (design research only, no plug-in code).**

Chapter purpose. Idle's vision: the reverb as a plug-in on many Studio One channels, all instances living in the **same virtual room**, moved and controlled from one place in 3-D. This page summarises the existing design study so the chapter can grow when a prototype exists. It is after the roadmap, not part of T1→SL7b.

Sources: `/workspace/build_support/research/studio_one_plugin/DESIGN.md`, `GROK_BUILD_BRIEF_Plugin_prototype.md`, `SOURCES.md`; `roadmap/THURSDAY_LIST.md` L4, L6.

> **IN PLAIN ENGLISH.**
> Normally, if you put a reverb on five tracks in a song, you get five separate reverbs that know nothing about each other, each pretending to be its own room. Idle's idea is the opposite: every track's plug-in is a *window onto the same room*. Drums, bass, guitar and vocal become dots in one 3-D room, and from one screen you drag them around: the vocal forward, the drums back left. Each instrument then gets the echoes it would really get from that spot, and they all share one tail, the way instruments in a real live room all excite the same air.

The proposed shape (from DESIGN.md; tags as given there)

- **Architecture A (recommended):** a *Source* insert on each track renders direct + early for that instrument and sends to one *Room* instance that runs **one** Hadamard FDN per L/C/R bus. The host's own send graph keeps latency and bounce/freeze behaviour normal.
- **Registry:** an in-process singleton keyed by `(documentID, roomId)` with versioned RoomState/SourceState; a bridge thread links to the web tweak app as the central editor.
- **A deliberate change from 2011:** with many sources and one FDN, the line delays cannot follow every source, so the shared FDN uses **room-fixed delays**. This must be flagged to Idle and A/B-tested against per-instance engines (P2).
- **Formats:** VST3 SDK (MIT since 3.8) first, CLAP (MIT) second.
- **Prototype phases:** P0 singleton proof → P1 registry + bridge → P2 audio A/B → P3 Studio One hardening → P4 CLAP + heavy offline export.
- **Prerequisite:** one C core for Play, Render and plug-in (J2 / L4, C migration).
- **Prior art checked** includes linked-instance designs (Fiedler Beam/Composer, FLUX SPAT LAP, Audio Ease 360pan, 2CAudio Precedence + Breeze). So "one shared room" is not entirely new; this design's twist is the 2011 geometry engine behind it (ASSUMED judgement).

A session in Studio One, as it might look (PLANNED, plain English)

An illustration of the intended workflow from DESIGN.md's architecture, not built behaviour.

1. You put the **Source** version of the plug-in on each instrument track, and one **Room** version on an FX channel.

2. Each Source works out its own instrument's direct sound and early echoes from where that instrument stands, and sends the sound on to the Room.
3. The Room runs one shared tail for everyone.
4. You open the room editor (the familiar web screen) and see every track as a named dot. Dragging a dot updates that track's echoes live.
5. Saving the song saves the room and all positions with it. Freezing or bouncing behaves like any normal plug-in.

The trade-off Idle should know about. In the 2011 design each tail corridor is exactly as long as one echo from *the* source. With many sources sharing one tail, the corridors can't match every source at once, so the shared tail would use corridor lengths fixed by the room, not by each source. That is a deliberate change from 2011, and the plan is to A/B it against giving every instrument its own full engine (DESIGN.md §3.2, prototype step P2).

Has it been done? Linked plug-ins that share one space exist (DESIGN.md lists Fiedler Beam/Composer, FLUX SPAT, Audio Ease 360pan, 2CAudio Precedence + Breeze). What would be distinctive here is the engine behind it: a geometric room with mirror-image echoes, real wall materials and modelled microphones (ASSUMED judgement).

Partitioned convolution (Idle's and Ben's idea, L6)

In plain words: instead of running the whole room engine live, each plug-in could just apply a pre-rendered echo fingerprint (convolution). Long fingerprints are expensive to apply in one go and would add delay, so the usual technique is to cut the fingerprint into short slices and process them in parallel, so that the sound starts immediately and the cost is spread out (this is a general description of the well-known technique, ASSUMED background; Ben mentioned it to Idle). Each instance could then be very cheap, with an offline full-quality render available when needed. Being one IR snapshot, it **cannot carry Movement exactly**; options are no modulation or a few cross-faded IRs (VERIFIED as noted in THURSDAY_LIST L6). No code or benchmark exists.

Open / not yet built

Everything. Studio One multi-Session behaviour (H9) and effect multi-outs (H10) are ASSUMED and need testing.

Chapter 11 — Variants and a reverb designer

> **Status: STUB, expanded 2026-10-05 — PLANNED (ideas only; no design document, no code).**

Chapter purpose. Idle sees the engine as a source for other products, and the app as a *designer*: once a room sounds right, it exports something another product can use. This page lists the ideas as Idle stated them (parent conversation, 5 Oct 2026) so they are kept with the book. None has a source file or a decision yet.

Sources: Idle's brainstorm (5 Oct 2026, relayed in the book brief); Chapter 10 for the shared-room plug-in. No research files exist for these items.

The ideas (all ASSUMED / unscheduled)

- **Reverb designer / code export.** Tune a room, then export code, IR packs or presets for a fixed product. What "code" means (C core plus a frozen parameter set? IRs?) is not decided.
- **Source files for variants.** The documented engine (this book) as the starting point for spin-offs.
- **3-D delay.** Echoes that ping-pong around the room in three dimensions, reusing the image geometry.
- **A nice room with cut-down controls.** A few knobs over a fixed, well-tuned room.
- **Game / film binaural walkthrough.** A listener moving through a virtual environment, rendered binaurally (the binaural mic path exists today, Chapter 7; movement through a scene does not).

The ideas as pictures (PLANNED, plain English)

How each idea might feel to a user. No designs or algorithms exist; these are Idle's ideas restated.

- **The designer.** You use the full program as a workbench: shape the room, choose walls and panels, place mics, listen. When it sounds right you press "export", and out comes something a simpler product can load: a fixed preset, a set of echo fingerprints, or code with the settings baked in. The workbench stays complicated so the product can be simple.
- **The "nice room" plug-in.** One carefully tuned room and three or four knobs (perhaps size, distance, decay, tone). Under the hood it is the same engine with almost everything fixed by the designer.
- **3-D delay.** Instead of echoes bouncing between two speakers, the repeats travel around a virtual room in three dimensions, coming back from the walls and ceiling with the right timing and colour for each surface, and through the right mic pattern or binaural ears.
- **A walk-through for games and film.** A listener (the camera or the player) moves through a virtual building. Each room has its own surfaces; the echoes and tail change smoothly as you walk from a tiled corridor into a carpeted office, rendered binaurally for headphones. The binaural mic exists today (Chapter 7); moving through a scene, several connected rooms, and smooth changes are all unbuilt.
- **Source files for variants.** Because this book documents the engine, a developer (or Ben) could start a new product from the engine and the book rather than from scratch.

What would have to be true first

The C migration (J2) for a portable core, a stable parameter set after SL7b, and this book's Surfaces and auto-treat chapters complete enough that someone else can rebuild the behaviour (ASSUMED).

Open / not yet built

All of it. No item is in T1→SL7b.

Virtual speakers as virtual microphones (PLANNED , idea of 2026-10-07)

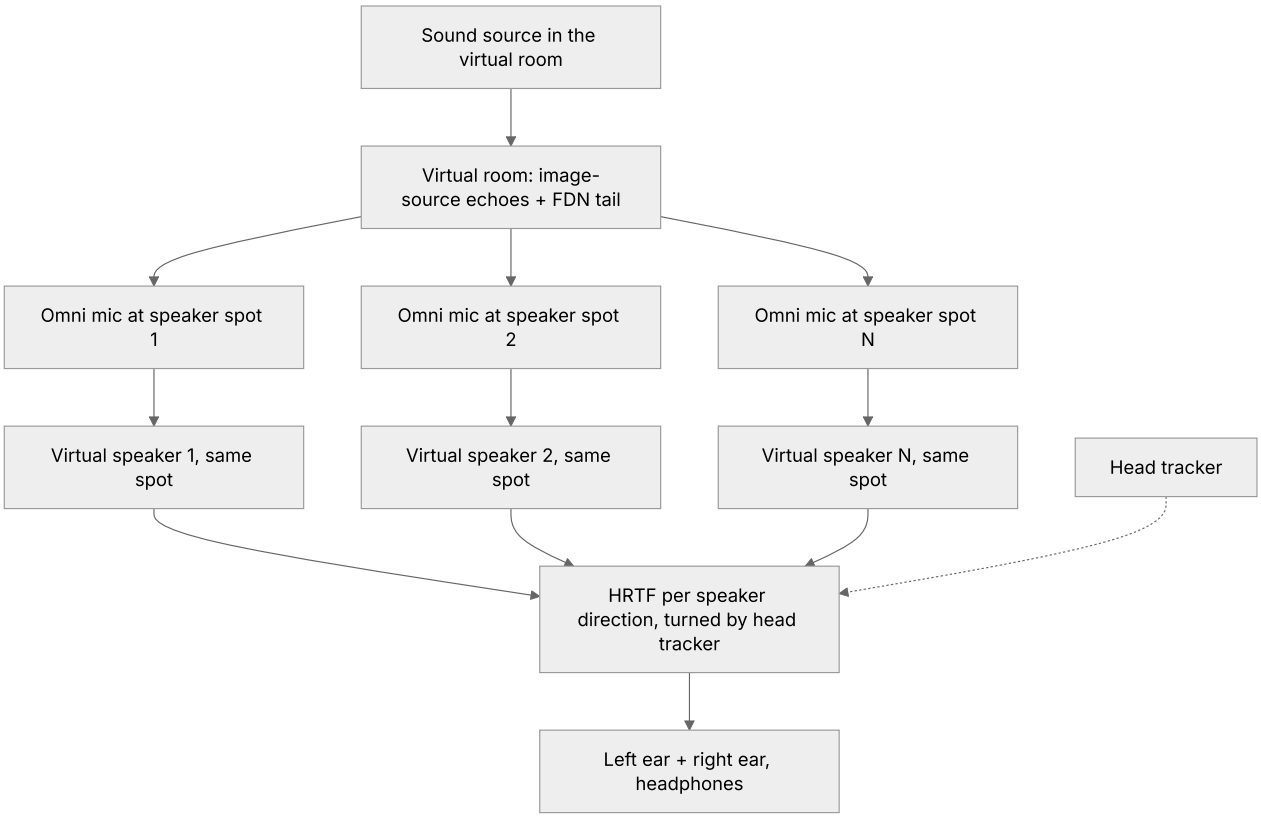
Status: PLANNED . An idea from Idle's talk with Ben Supper, researched on 2026-10-07. Nothing here is designed or built. Most likely a separate future app on the shared engine, not a mode of the reverb, and not part of the T1 to SL7b roadmap. Full research notes: `research/virtual_speaker_mics_RESEARCH.md` .

11.V.0 The idea

IN PLAIN ENGLISH.

Put a virtual omni microphone wherever a loudspeaker would stand in a virtual room. Each mic hears the room as it sounds at that exact spot. Then play each mic's signal out of a virtual speaker in the same place, and turn all of those speakers into headphone sound with HRTFs (the "head filters" that tell your ears which direction a sound comes from). If it works, you hear the room's real timing, loudness and echoes around you, and you can move your head (with a head tracker such as Ben's Supperware tracker) and the room stays still. Because the room, the mics and the speakers are all virtual, a layout can be changed and re-rendered in seconds instead of re-rigging a real room.

The loop, end to end:



The rest of this section asks four questions: why should this work (theory), what do real recording engineers already do (practice), how do you turn many speakers into headphone sound (virtual speakers), and what would the engine need (simulation). It ends with a first experiment.

11.V.1 Why it should work: the theory

> **IN PLAIN ENGLISH.**
> Imagine a pond. If you could copy exactly how the water moves all along a line across the pond, and then make new ripples from that line, the waves beyond it would look the same as the original. Sound works the same way. If you record the sound everywhere on a surface around the listener and re-create it at those points, the sound inside is re-created too. Real systems can only use a few points, so they approximate this. An omni mic at each speaker spot is the crudest, cheapest form of the idea.

Huygens and Kirchhoff-Helmholtz. The Kirchhoff-Helmholtz integral says that inside a source-free region, the sound pressure at any point can be calculated if both the pressure and the normal particle velocity are known on the surface enclosing that region (VERIFIED, Wittek's literature review of WFS [W1]). This is the precise form of Huygens' principle: every point on a wavefront acts like a new small source.

The acoustic curtain (Bell Labs, 1934). Steinberg and Snow, with Fletcher, described a "curtain" of microphones in front of a stage, each feeding a loudspeaker at the matching position in the listening hall, so the wavefront would be carried across (VERIFIED [W1], [B1]). They found it impractical and settled on two or three channels, accepting that the real sound field would no longer be recreated (VERIFIED [W1]). That compromise is the origin of spaced-microphone stereo.

Wave field synthesis (Berkhout, Delft). Berkhout and colleagues turned the idea into engineering in 1988 to 1993: measure the source field, extrapolate it electronically, and re-emit it from loudspeaker arrays; reflections can be generated with the image-source concept in the same way as direct sound (VERIFIED, Berkhout, de Vries and Vogel 1993 abstract [B2]; Berkhout 1992 [B3]). WFS needs many closely spaced speakers to work above a limiting ("aliasing") frequency (ASSUMED here; standard WFS knowledge, not quantified in the sources read).

How "an omni at each speaker" relates. It is a sparse acoustic curtain wrapped around the listener. Two simplifications separate it from true WFS (ASSUMED, reasoning from [W1]):

1. Only pressure is captured (omnis), not velocity, so the "inside/outside" information that the full integral needs is missing.
2. There are 5 to 12 points, not hundreds, so above a few hundred hertz the field is not physically rebuilt. What remains is *perceptual*: each speaker carries the right arrival times, levels and echo pattern for its direction, and the ear-brain does the rest.

How it differs from Ambisonics / HOA. Ambisonics records the sound field at *one point* as a set of spherical components (order 1 = 4 channels, higher orders more), then *decodes* it to whatever speakers exist. A spaced-omni array records at *many points* and each channel is tied to one speaker. Trade-offs (ASSUMED, general knowledge, consistent with Lee's review [L1]):

- Ambisonics: layout-independent, rotates easily for head tracking, sharp-ish imaging at high order; but low orders sound less spacious because the channels are highly correlated.
- Spaced omnis: very spacious and natural room sound because the channels are decorrelated by distance; but tied to one layout and with weaker, less stable phantom images between speakers.

For this engine the decisive point is that a virtual room can produce *both*: N spaced omni feeds, or an Ambisonic feed at the listener. That is exactly what makes an A/B possible.

11.V.2 What engineers already do: spaced-omni surround and 3D arrays

> **IN PLAIN ENGLISH.**
> Recording engineers have tried this many ways for decades. The rule they all found is: microphones close together give a sharp but "small" picture; microphones far apart give a big, airy room but blurrier images. Most real setups mix the two: a front group for sharp images, and widely spaced mics for the room.

The 2L-cube (Morten Lindberg) — almost exactly the idea from the talk. Lindberg writes that his main array "is really a direct consequence of the speaker configuration": one omni per speaker in 5.1.4 or 7.1.4, so that "time of arrival, SPL and on-axis HF texture is directly preserved". The cube is *scaled down*: about 120 cm for a large orchestra and 40 cm for chamber music. He always uses omnis (DPA 4003 or 4041) (all VERIFIED [2L1]). This matters: he does **not** put mics at the real speaker positions (several metres apart); he shrinks the layout. So the talk's claim is supported in spirit, with that caveat (VERIFIED quote; interpretation ASSUMED).

Hamasaki square (NHK). Four sideways-facing figure-8 mics for room sound, nulls toward the stage. Spacing was first 1 m, then raised to 2 to 3 m; the omni coherence calculation showed 2 m gives decorrelation above about 100 Hz, and listeners preferred 3 m over 2 m and 2 m over 1 m (VERIFIED [H1], [L1]). The Hamasaki cube adds an upward-facing layer at the same 2 to 3 m spacing (VERIFIED [L1]).

OCT (Theile) and Williams MMAD. OCT is a front triangle of directional mics designed for low crosstalk and stable front images; base widths of about 50 to 100 cm are quoted; it is often combined with a Hamasaki square for ambience (VERIFIED, Wittek [WI1]). Williams' MMAD method calculates spacing and angles so neighbouring pairs' recording angles join up without gaps or overlaps (ASSUMED; mentioned in [WI1] but not read in detail).

Decca tree, Fukada tree, ORTF-3D. Decca tree: three spaced omnis in front of an orchestra (classic, ASSUMED general knowledge). Fukada tree (NHK) and ORTF-3D (Schoeps) are 5-channel and 8-channel arrays reviewed in Lee 2021 [L1] (VERIFIED that they are covered there; exact dimensions not repeated here, ASSUMED/omitted).

Hyunkook Lee (Huddersfield, APL) on spacing and height.

- Horizontal spacing gives the interchannel decorrelation that creates spatial impression (VERIFIED [L1], [L2]).
- Vertical spacing between the main and height layers (0, 0.5, 1, 1.5 m tested) made no significant difference to spatial impression; 0 m was rated slightly higher (VERIFIED, Lee and Gribben 2014 [L3]).
- In PCMA-3D the height mic's direct sound should be at least 7 dB below the main mic's to avoid the source image lifting upward (VERIFIED [L4]); the review quotes 9.5 dB for another configuration (VERIFIED [L1]).

What fails (all ASSUMED unless cited, standard spaced-array knowledge):

- **Comb filtering:** if two spaced channels carry the same direct sound at different times and are summed (downmix, or both reaching one ear), notches appear. Wide spacing makes it less audible because the notches are dense, but a stereo or mono fold-down suffers.
- **Weak phantom images:** time differences between widely spaced omnis are too large for clean panning between speakers. Sounds tend to "jump" to the nearest speaker.

- **Precedence effect:** the ear locks onto the first-arriving speaker. In a spaced-omni array the speaker nearest the source fires first, which is *helpful* (it points the right way), but a listener off the centre seat hears a different first speaker.
- **Height crosstalk:** too much direct sound in the height channels pulls images upward (VERIFIED [L4]).

> **IN PLAIN ENGLISH: WHAT THIS MEANS FOR US.**
> Omnis at the speaker spots should give a lovely, natural room around you. Direct sound images will be roughly in the right place but not pin-sharp. Height omnis must carry little direct sound, or voices will float upward. In a virtual room we can cheat in ways a real engineer cannot: shrink the array like Lindberg, or keep wide spacing for the tail but feed sharper directional cues for the direct sound.

11.V.3 Turning many speakers into headphone sound

> **IN PLAIN ENGLISH.**
> Each virtual speaker is just a sound coming from one direction. To put it on headphones, run it through the pair of head filters (HRTFs) measured for that direction, add up all the speakers, and you have the left and right ear signals. With a head tracker, when you turn your head the filters change so the speakers stay put in the room.

VRM as a precedent. Ben Supper's Focusrite VRM work had four parts: a loudspeaker radiation model, a room reverberation simulator, a database of directional head cues, and a real-time engine (VERIFIED, AES paper 8267 [S2]). Paper 7994 measured studio monitors' impulse response and directivity in an ordinary room and removed the room reflections (VERIFIED [S1]). Paper 8267 processed the KEMAR HRIR set: correction, equalisation (removing colouration) and spatial interpolation to a finer grid, aiming for angular resolution at least as fine as human hearing and working for most listeners on unknown headphones without personalisation (VERIFIED [S2]). VRM simulated *stereo* loudspeakers; the virtual-speaker-mic idea extends the same structure to N speakers in a simulated room.

How many virtual speakers? Each virtual speaker is a fixed direction, so N speakers need N HRTF pairs (N two-channel convolutions, or 2N FIR filters). A 5.0 layout means 5 pairs; 7.1.4 means 11 (LFE usually not binauralised) (ASSUMED arithmetic). The binaural result can only be as good as the speaker layout: gaps between speakers become gaps in the image. More speakers help imaging but cost CPU and, for spaced omnis, need more mics.

HRTF interpolation. Measured sets have a finite grid; directions in between need interpolation, which Supper did offline for KEMAR [S2] (VERIFIED). For fixed virtual speakers this only matters when the head moves: each speaker's direction relative to the head then changes continuously.

Head-tracking latency. Lindau (TU Berlin, 2009) found a lowest individual detection threshold of about 52 to 53 ms and a mean of about 108 ms for total system latency in dynamic binaural synthesis (VERIFIED as reported in [LI1]); a 2023 study found higher thresholds under its method but recommends under 40 ms for robust use (VERIFIED as summarised from [LI2]). Treat 40 ms total (tracker + block + convolution) as the target (ASSUMED design target based on those).

Room doubling. If the virtual room's sound is played through virtual speakers that themselves sit in a *second* simulated room (as VRM did, by design), the listener hears two rooms stacked: more reverb, muddier time cues. Options (ASSUMED, design choices):

1. Anechoic virtual speakers (HRTF only, no listening room). Cleanest test of the capture idea.
2. A short, well-damped "control room" on purpose, to make headphones feel like speakers (VRM-style externalisation).

3. Measure how much the listening room adds and reduce the capture tail to compensate. Option 1 should come first.

11.V.4 The simulation side: what the engine would need

> **IN PLAIN ENGLISH.**
> The early echoes are easy: the engine already works out each echo's path to a mic, so more mics just means doing the sums more times. The tail is the hard part. In a real room the tail at two spots far apart is different (that is why spaced mics sound spacious). If our N mics all got the same tail, the room would collapse into one blob in the middle of your head. Each mic needs its own, different-but-matching tail.

Early field to N receivers. Image-source computation for many receivers is well established: MCRoomSim (Wabnitz, Epain, Jin, van Schaik, 2010) simulates any number of sources and receivers in a shoebox room, with directivities, HRTFs and spherical-harmonic receivers, specular and diffuse reflections, and realistic inter-sensor delays (VERIFIED [M1], [M2]). In this engine, the 2011 design uses a fixed set of 32 image "voices" and three mics (Ch. 1, Ch. 5). Whether that fixed set is adequate for mics placed several metres apart (each needs its own delays and levels per image, which the engine already computes per mic) is ASSUMED plausible and must be checked.

Decorrelated tail per output. Two routes from the literature:

- **FDN with several outputs.** Research at Aalto found that correlation between FDN outputs is set mainly by the feedforward (input/output) paths rather than the long recursive tail; larger FDNs and filtered feedback matrices, especially velvet scattering, decorrelate best (VERIFIED, as summarised from [D1]). The engine's 32-line FDN already has many taps, so N different output mixes are cheap (ASSUMED, check the code).
- **Velvet noise.** Sparse random ±1 sequences used as decorrelators or tails; interleaved velvet noise can produce many decorrelated channels from one reverberator by permuting sequences and mixing (VERIFIED [D2], [D3], [D4]). The tail's correlation between two mics should ideally follow the diffuse-field rule for their distance (strongly correlated at low frequencies, decorrelated above roughly c/(2d)), so that a 2 m pair behaves like Hamasaki's measurement [H1] (ASSUMED target; the 2 m / 100 Hz figure is VERIFIED from [H1]).

CPU and quality trade-offs (ASSUMED estimates, to measure). Offline Render: N times the early work plus N tail taps, then N HRTF pairs; trivial offline. Real-time Play: the HRTF stage dominates; partitioned convolution (Ch. 10) keeps it manageable; 11 speakers × 2 ears is 22 short convolutions.

Existing tools and how the idea differs.

- **MCRoomSim** — offline MATLAB research simulator for multichannel arrays [M1]. Closest technically; not a production or listening tool.
- **SPARTA** (Aalto, McCormack) — open-source plug-ins; AmbiRoomSim simulates shoebox image sources into Ambisonics; Binauraliser renders up to 128 object channels to binaural with SOFA HRTFs and OSC head tracking (VERIFIED [SP1]). One could chain a multi-receiver simulation into Binauraliser, but the omni-at-speaker capture is not a built-in workflow (ASSUMED).
- **IEM Plug-in Suite** — RoomEncoder places source and listener in a shoebox and encodes 200+ reflections into Ambisonics (VERIFIED [I1]). Single-point Ambisonic, not spaced.
- **Spat Revolution** (FLUX::) — commercial spatial mixer with virtual rooms, binaural monitoring, HOA, custom layouts (VERIFIED [F1]).
- **dearVR, Steam Audio, Anaglyph** — object/game binaural renderers with room models (ASSUMED from general knowledge; not checked in this pass). They render the room directly at the listener's

head, not via virtual omnis at speaker spots.

11.V.5 Is "each virtual speaker is also a virtual microphone" new?

> **IN PLAIN ENGLISH.**
> The idea itself is old: it is the 1934 acoustic curtain, and Lindberg records real music this way today. Simulating rooms for many mics is also done, and virtual speakers on headphones are done (Ben did it). What we did not find is a ready-made tool that does all three together as one design: simulate a room, capture it with omnis at the speaker spots, and play it back on head-tracked virtual speakers so you can audition and move the layout. That may be new as a product. It is not new science.

Honest verdict: no prior art for the exact combined workflow was found in this short search, but the search was not exhaustive and similar research setups very likely exist in university labs (ASSUMED). Do not claim novelty publicly without a proper search. The useful framing is "a virtual 2L-cube / acoustic curtain designer with headphone monitoring".

11.V.6 Suggested first experiment (PLANNED)

> **IN PLAIN ENGLISH.**
> Render the same short sound three ways in the same virtual room, put them on headphones, and compare. If the "omnis at speakers" version sounds more like being in the room than our current binaural, the idea is worth an app.

Setup (offline Render only, no Play, no changes to the reverb's defaults or goldens):

1. One room the engine already handles well (for example a medium hall), one source front-left, listener in the middle.
2. **A — current binaural render** of the engine (reference).
3. **B — 5.0 omni set:** omnis at ITU 5.0 positions (0°, ±30°, ±110°) on a circle of about 2 m radius (ASSUMED value) around the listener, each with its own decorrelated tail output; each played from an anechoic virtual speaker at its angle through the same HRTF set; summed to two ears.
4. **C — same as B, but a Lindberg-style scaled array** (mics on a 0.5 to 1 m cube, speakers still at full layout).
5. Later: **D — 7.1.4** to test height, keeping height direct sound at least 7 dB down [L4].
6. Loudness-match all versions; blind A/B/X; with and without head tracking if available.

Listening criteria (score 1 to 5 each):

- Externalisation: does it sit outside the head?
- Envelopment: does the room surround you?
- Source direction and sharpness: is front-left in the right place, and how wide?
- Distance and depth.
- Tone colour and comb-filter artefacts (listen for "phasey" or hollow sound).
- Tail smoothness: no flutter, no "blob in the middle".
- Stability when turning the head (if tracked).
- Overall naturalness and preference.

Measurements alongside: interaural cross-correlation (IACC) of early and late parts of A, B and C (ASSUMED useful metric); per-mic tail coherence vs frequency compared with the diffuse-field curve.

Gate: if B or C does not clearly beat A on envelopment without losing badly on direction, the idea stays a book note. Decided by Idle (and ideally Ben) by ear.

11.V.7 Open questions

- Does the 32-voice early field hold up for mics several metres apart? (Check in code.)
- How should the FDN produce N decorrelated outputs while keeping Flat (2011) bit-exact for the reverb itself?
- Real speaker spacing vs Lindberg-style scaled array: which sounds better virtually?
- Which listening room: anechoic, or a deliberate control room?
- Which HRTF set, and does Ben's processed KEMAR set fit?
- Head-tracker integration and the 40 ms latency budget.

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Chapter 12 — Handing the technology to Ben Supper

> **Status: STUB, expanded 2026-10-05** — PLANNED. No handoff has been arranged in any source file; this is Idle's stated wish (5 Oct 2026).

Chapter purpose. Idle would like to hand this technology to his friend Ben Supper, who now makes synths and might build something with it. This page lists what a handoff pack would need, so the book grows toward it. It does not speak for Ben.

Sources: the book brief (`/tmp/book_brief.md`: "Ben Supper (supperware.net, Focusrite VRM inventor)"); the parent agent's web look-up (Focusrite VRM around 2010, later ROLI, his own head tracker and synths — ASSUMED, not re-checked here); licence files listed below.

> **IN PLAIN ENGLISH.**
> Idle wants this technology to have a life beyond his own project, and Ben, who built Focusrite's headphone room simulation (VRM) and now makes synths, is the person he has in mind. A handover only works if Ben can understand the system without the hundreds of chat messages behind it. That is the main reason this book exists: it is the handover document in the making. The more of it that is written from built, tested code, the easier the handover.

Why Ben (ASSUMED, from the brief)

VRM already combined measured monitors, an artificial room and HRTF convolution for headphones. This engine speaks the same language: sources, surfaces, mics and binaural.

What a handoff pack needs (proposal, ASSUMED)

1. **This book**, with Chapters 6 (Surfaces and auto-treat) and 8 (tests) written from shipped code, not plans.
2. **One C core** (J2) with build instructions, goldens and the Flat (2011) parity suite.
3. **The locked-decision record** (DECISIONS_LOCKED files) so he knows what was a choice and what was physics.
4. **Licences.** Data in the app carries its own terms (VERIFIED, `BIG_JOBS_ROADMAP_2026-10-01.md` and the vendored `LICENSES/` folder): SADIE II HRTFs Apache-2.0; TH Köln KU100 data CC BY-SA 3.0 (share-alike for derived data files); TU Berlin source-radiation sets are excluded by the loader (`_reject_tu()`). The mic dataset (Franco et al., Zenodo 10.5281/zenodo.4633508) and BRAS material terms need checking before redistribution (Open).
5. **Ownership and licence of Idle's own code** — not stated anywhere (Open; Idle's decision).

A possible order of events (PLANNED, plain English)

A suggestion, not a decision; Idle and Ben decide.

1. **Finish the roadmap** to SL7b, so the default sound is the one Idle wants to hand over.

2. **Bring the book up to date** with Surfaces, auto-treat and the tests written from shipped code.
3. **An informal look:** send Ben this book plus a few A/B renders and ask what interests him (a synth's built-in room? a binaural tool? the shared-room plug-in?).
4. **Decide what is handed over:** the whole engine, a single C core, or the knowledge only; and under what licence (Idle's choice).
5. **Make the pack:** code, tests, goldens, decision records, data licences and the book.

What Ben would find most useful (ASSUMED)

The parts that are new rather than textbook: how the 2011 mirror echoes feed the corridors one-to-one; Surfaces-in-the-loop with directional lines, calibration and auto-treat; the Play/Render parity method; and the test discipline that keeps the 2011 sound reproducible.

Open / not yet built

No date, scope or licence choice. Revisit after SL7b.

Glossary

> **Status:** DONE **(draft 2026-10-05)** for every term used in Chapters 1–8. It grows with the book.

How to use it. Each entry gives an everyday sentence first, then a *Precise* note for engineers (and for Ben), then the chapter where the term matters most. Tags (VERIFIED / ASSUMED / LOCKED / PLANNED) live in the chapters, and the glossary follows them. Definitions marked *(general acoustics)* are standard textbook meanings, not project-specific facts.

Numbers and symbols

- **1/r²** — Further away is quieter: twice the distance gives a quarter of the level. *Precise:* the locked geometric level law for each image in this engine ("Air Abs geometry is 1/r² VERIFIED "). An older public doc says 1/r and is out of date. (Ch. 1, 5)
- **2011 / Flat (2011)** — The original sound of Idle's Reaktor reverb, kept forever as a mode. *Precise:* the decay mode where every FDN line loses the same dB per second at every frequency; it will be protected by a frozen parity suite (D-SL10). (Ch. 1, 2, 8)
- **477 fold** — The rule for mixing three mics to two speakers: left + half the centre, right + half the centre. *Precise:* `L + 0.5·C`, `R + 0.5·C`; "477" is the Structure reference cited for it (that reading of the number is ASSUMED). (Ch. 1, 7)
- **z⁻¹** — "One sample later." *Precise:* unit delay; "z⁻¹ in both" means Play and Render both apply the Hadamard feedback on the next sample (S7). (Ch. 3, 5)
- **α (alpha), absorption coefficient** — How much of the sound a surface swallows per bounce, from 0 (mirror) to 1 (open window). *Precise:* random-incidence absorption per octave on a material card; converted to per-bounce α with 1 − e^(−α) in Surfaces mode. (Ch. 6)
- **κ (kappa)** — A scale factor in the Surfaces design. *Precise:* normalises the directional part of each line's rate so the delay-weighted mean equals the room target. (Ch. 6)

A

- **A–F faces** — The six room surfaces. *Precise:* A Left (−X), B Front (+Z), C Right (+X), D Back (−Z), E Floor (−Y), F Ceiling (+Y); orientation ASSUMED from the older doc, D = back VERIFIED . (Ch. 6)
- **A/B** — Listening to two versions back to back to choose. *Precise:* level-matched render pairs with a checklist; every sound change is decided this way. (Ch. 4)
- **AB (mic setup)** — Two mics spaced apart. *Precise:* `ab_omni` / `ab_card` setups. (Ch. 7)
- **ABBA** — A fair way to time old against new. *Precise:* paired runs in the order old/new/new/old, 12 rounds, median of per-round differences; the PC result is the gate. (Ch. 3, 8)
- **Air absorption** — Air soaks up treble over distance. *Precise:* ISO 9613-1 by path length on early images and the FDN inject; the direct sound is air-free; never on the FDN return today. (Ch. 2, 5)
- **Allpass (diffuser)** — A filter that smears a click in time without changing its tone. *Precise:* a Schroeder allpass around each delay line with coefficient d = Dffs; |T| = 1 when lossless. (Ch. 5)
- **Arau-Puchades** — A reverb-time formula for rooms that are deader in one direction. *Precise:* area-weighted geometric mean of per-axis Eyring times; blended with Eyring by scattering in Surfaces

(blend ASSUMED). (Ch. 6)

- **Array / Room (mic)** — On the stand, or moved off as a room mic. *Precise:* Array = capsule stays on the stand; Room = placed at the room-mic spot (one shared spot today; one per strip from T7/T8). (Ch. 7)
- ASSUMED — "This is a judgement, or not re-checked." *Precise:* tag for any claim not measured or read in a cited file. (all)
- **AudioWorklet** — The browser's slot for running live audio code. *Precise:* hosts `live_stereo.wasm`, called once per 128-sample block. (Ch. 2, 3)
- **Auto-treat** — A helper that suggests panels to hit a target decay. *Precise:* PLANNED (SL6, D-SL5): broadband porous by bisection, then up to 4 tuned panels at the worst 40–500 Hz excess vs the Mäkivirta target; suggestions only. (Ch. 6)
- **Axial mode** — The lowest note a room "hums" along one dimension. *Precise:* f = c/(2·length); default room ≈ 21.3 / 28.3 / 56.7 Hz at c = 340 (computed in the book). (Ch. 5, 6)

B

- **Binaural** — Recording with a dummy head so headphones sound 3-D. *Precise:* SADIE II KU100 default, ears 0.18 m, K = 6 biquads per ear, `BINAURAL_REAR32`, IC-matched tail, centre mic muted. (Ch. 7)
- **Biquad** — One standard EQ building block. *Precise:* a second-order IIR section (two poles, two zeros); S1P2 uses three per line. (Ch. 6)
- **Bisection** — Finding a value by halving the range each time, like "higher / lower". *Precise:* auto-treat uses 30 halvings for coverage and 25 for panel area, fixed counts not tolerances. (Ch. 6)
- **Bit-exact** — Not one sample different. *Precise:* max-abs difference 0; required for any feature when switched off and for the main golden in NO sessions. (Ch. 8)
- **Block / block budget** — The small slice of audio Play must deliver on time. *Precise:* 128 samples at 48 kHz = 2.667 ms (`BUDGET_MS`). (Ch. 3)
- **The box** — The research computer. *Precise:* Linux machine that works on copies, prototypes, measures and writes prompts; never touches `software\` on the PC. (Ch. 1)
- **BRAS CR2** — A real, published, measured room used as a check. *Precise:* T20 125 Hz–8 kHz 1.34 / 1.72 / 2.00 / 1.94 / 1.74 / 1.54 / 1.11 s. (Ch. 6)
- **Build session** — One scoped job that changes the real program. *Precise:* one prompt per new Build window; ends with tests green and the golden hash recorded. (Ch. 1, 4)
- **Bus** — A group channel on the desk. *Precise:* Direct, Early, FDN, Host dry; with solo, mute and fader. (Ch. 7)

C

- **c** — The speed of sound. *Precise:* locked at 340 m/s in the engine; the Surfaces prebuild uses 343 m/s (OPEN mismatch). (Ch. 5, 6)
- **Calibration (Surfaces)** — Render once, measure, nudge once. *Precise:* D-SL4: T20 per octave, scale factor clamped 0.6–1.6, redesign once; Play uses the Render-designed coefficients. (Ch. 6)
- **Capsule** — The sensing part of a microphone. (Ch. 7)
- **Cardioid** — A mic that mainly hears what is in front. *Precise:* heart-shaped polar pattern. (Ch. 7)
- **C migration (J2)** — Moving Play and Render onto one C engine. *Precise:* PLANNED; makes drift impossible; prerequisite for the plug-in. (Ch. 3)

- **Colour (mic)** — How a real mic changes tone, including with direction. *Precise:* measured minimum-phase FIR (Play, 80 ms) from SOS tables; modes Measured / User defined / Bypass. (Ch. 7)
- **Comb (filtering)** — A hollow, "flanged" colouring from regularly spaced peaks and dips. *Precise:* removed in the default room by the S4b H·D column signs. (Ch. 5)
- **Convolution** — Applying a room's echo fingerprint to a recording. *Precise:* Render's `fft_convolve` of the dry file with the IR; only possible for time-invariant processing. (Ch. 2, 3)
- **Cut line** — In a session prompt, the line above which work is essential. (Ch. 4)

D

- **dB (decibel)** — The unit for loudness differences; −6 dB is about half the amplitude, −60 dB is a thousandth. *(general acoustics)* (all)
- **Decay dial / T60** — How long the tail rings. *Precise:* the dialled time for a 60 dB fall; with Decay lock on, it sets each line's loss `lg`. (Ch. 2, 5)
- **Decay lock** — The desktop default where the dial directly sets the decay. *Precise:* g = 1/√N plus per-line loss lg = 10^(−3·d_i/(sr·T)); off = the 2011 law g = Gain/N. (Ch. 5)
- **Decay mode** — How each line loses energy per trip. *Precise:* Flat (2011) shipped; Natural (air) LOCKED for SL7a; Surfaces LOCKED for SL1–SL6. (Ch. 2)
- **Denormal** — An extremely tiny number some processors handle very slowly. *Precise:* flushed below about 1e-20, plus a tiny DC offset in the loop. (Ch. 5)
- DESIGNED — Worked out and prototyped on the box, not in the program. (Ch. 6)
- **Desk** — The mixer page. *Precise:* three mic strips (L, C, R) and four buses; no EQ. (Ch. 7)
- **Dffs** — The strength of each line's smearing filter. *Precise:* allpass coefficient per line = surface diffusion sum × Diffusion Master, clipped to ±1; Smooth caps |Dffs| at 0.7 (LOCKED, T4). (Ch. 5)
- **Diffusion Master** — One knob scaling all the smearing. *Precise:* default −0.25, clamped [−0.5, 0]. (Ch. 5)
- **Direct sound** — The sound that travels straight from source to mic. *Precise:* image (0,0,0), voice 3, line index 2; filtered per capsule then × 0.55. (Ch. 2, 7)
- **Directional lines** — Letting each tail corridor follow its own walls. *Precise:* D-SL2, rate = air + s′·room + (1 − s′)·κ·own faces, s′ ≥ 0.5. (Ch. 6)
- **Double slope** — A decay that sounds like two fades one after the other. *Precise:* seen with raw directional rates (s′ = 0.15) and in the SL4 untreated-plaster finding. (Ch. 6)

E

- **Early reflections / early field** — The first distinct echoes. *Precise:* the 32 image taps (`LIVE_32`) in Play; the full image set in Render. (Ch. 2, 5)
- **EBU Tech 3276** — A broadcast standard for control-room decay. *Precise:* D-SL9 preset, Tm = 0.25·(V/100)^(1/3) s with per-band tolerances. (Ch. 6)
- **EDT** — Early decay time: how fast the first part of the fade goes. *(general acoustics: from 0 to -10 dB, scaled to 60 dB)* (Ch. 6)
- **Eyring** — The classic formula linking room size and absorption to reverb time. *Precise:* T = 0.161·V / (−S·ln(1 − $\bar{\alpha}$) + 4mV). (Ch. 6)

F

- **Face EQ** — A per-wall tone control. *Precise:* low shelf, peak, high shelf; in Surfaces only cuts count in the loop. (Ch. 6)
- **FDN (feedback delay network)** — The recirculating "corridors" that make the tail. *Precise:* 32 delay lines (per L/C/R bus), unscaled ±1 Hadamard H·D, one-sample feedback, per-line allpass and loss. (Ch. 2, 5)
- **FDN-first** — The default mix: tail up, direct and early down. *Precise:* K4. (Ch. 2)
- **Figure-8 (bidirectional)** — A mic that hears front and back but not the sides. (Ch. 7)
- **FIR** — A filter made from a short fingerprint of samples. *Precise:* finite impulse response. (Ch. 7)
- **Fitter** — The routine that sets each line's tone control. *Precise:* bounded least squares, 56 log-spaced points, at most 600 evaluations, stability penalty; convergence not checked (OPEN). (Ch. 6)

G–H

- **Gate** — A pass/fail condition a session must meet. *Precise:* with a stop rule: ship the safe variant or revert, never half-merge. (Ch. 4, 8)
- **Golden** — A stored reference recording that must not change by accident. *Precise:* four Play goldens + `render_std_unlocked_byp.f32`. (Ch. 8)
- **GOLDEN_META** — The file that records golden fingerprints and why they changed. (Ch. 8)
- **Guard** — A safety check on each line filter. *Precise:* reject if loop gain implies T > 3× the longest target or octave error > 50 %; refit with S1, else broadband. (Ch. 6)
- **H·D (column signs)** — The shuffle pattern with some columns flipped. *Precise:* Hadamard times a fixed mirror-safe sign diagonal (S4b); `reaktor_parity` keeps plain H. (Ch. 5)
- **Hadamard matrix** — A fixed pattern of +1 and −1 that blends every line into every other. *Precise:* unscaled ±1 Sylvester construction (ASSUMED), applied by `fwht32`. (Ch. 5)
- **HANDOVER** — The project's read-first notes file, kept with the code. (Ch. 1)
- **Harness** — The 19 automatic engine checks (`run_all.py`, t01–t19). (Ch. 8)
- **Helmholtz panel** — A perforated board over a cavity that soaks up one bass band, like blowing over a bottle. *Precise:* f0 = (c/2π)·√(ε/(t′·d)); Q 3 ASSUMED (physics suggests 1–2, open). (Ch. 6)
- **H-1** — The locked mic-height compromise. *Precise:* direct, early and inject follow capsule heights; FDN lines stay at stand height (tail change ≤ 0.42 dB). (Ch. 7)
- **HRTF** — How your head and ears filter sound from each direction. *Precise:* head-related transfer function; SADIE II data in this program. (Ch. 7)

I–L

- **IC (interaural coherence)** — How similar the two ears' signals are. *Precise:* the binaural tail is matched to the head's IC. (Ch. 7)
- **Image source** — A mirror copy of the source behind a wall, standing for one echo. *Precise:* 2011 mirror macros, full-R; delay d/340 s, level $1/r^2$. (Ch. 5)
- **Impulse response (IR)** — A room's echo fingerprint: what you'd record after one click. (Ch. 2, 3)
- **Inject** — Where the echoes enter the tail. *Precise:* all 32 images inject into the FDN; filters here change tone and level, not decay. (Ch. 2, 5)
- **Jitter (Δd)** — Tiny random changes in echo path length so the pattern sounds less rigid. *Precise:* seeded, mirror-invariant since D-MIX-J; separate from Dffs. (Ch. 5)

- **K1 / YAW_X_SIGN** — The one rule for left and right. *Precise:* `YAW_X_SIGN = +1` in Python and JS; source left ⇒ L louder and earlier. (Ch. 7)
- **Least squares** — Making the total of squared errors as small as possible. (Ch. 6)
- **LFO** — A slow wobble generator. *Precise:* Movement uses sines at 0.7 Hz × 0.6–1.4 per line. (Ch. 5)
- **lg / line_t60** — Each corridor's loss per trip. *Precise:* lg = $10^{(-3 \cdot d_i/(sr \cdot T))}$, i.e. $-60 \cdot \tau_i/T$ dB per pass. (Ch. 5)
- **Limiter** — A safety stop against clipping. *Precise:* one shared soft limiter in Play and Render (S-AUDIT). (Ch. 2)
- **LIVE_32** — The fixed list of 32 images. *Precise:* orders 0–4 as 1/6/11/10/4; `engine/early_field.py`. (Ch. 5)
- LOCKED — Approved by Idle, not yet in the program. (all)
- **Loop gain (g)** — How much the mixer passes back each trip. *Precise:* g = Gain/N (2011) or 1/√N (Decay lock). (Ch. 5)
- **LTI** — A process that treats every moment the same and just adds signals. *Precise:* linear time-invariant; any LTI chain can be replaced by one fixed filter, so it can live in an IR. (Ch. 2, 3)

M

- **Mäkivirta target** — The decay shape studios aim for. *Precise:* flat at T_mid (500 Hz–2 kHz mean), +0.2 s on log f from 300 Hz down to 50 Hz, air alone above 2 kHz; the auto-treat yardstick. (Ch. 6)
- **Material card** — A wall's absorption per octave. *Precise:* 12 real materials with role bans plus porous presets; the default card is not a real material (ASSUMED plaster-ish). (Ch. 6)
- **Median** — The middle value of a sorted list; used for CPU gates. (Ch. 3, 8)
- **Membrane panel** — A board over an air gap that vibrates like a drum skin and soaks up one bass band. *Precise:* f0 ≈ 60/√(m·d), ≈ 50/√(m·d) with wool; Q 2, αmax 0.6 ASSUMED. (Ch. 6)
- **Mic tap** — Where a mic listens. *Precise:* off the diffuser output, in parallel with feedback; never written back. (Ch. 2, 7)
- **Minimum-phase** — A filter built to add as little delay as possible. (Ch. 7)
- **Mirror (two sources)** — Putting the second source at the mirror spot across the room. *Precise:* B at (W − sx, sy, sz); `test_mirror_exact` ≤ −120 dB. (Ch. 2, 8)
- **Mirror / Linked / Split** — How the right mic follows the left. *Precise:* D-UI37, LOCKED for T7/T8. (Ch. 7)
- **Movement** — A gentle wobble of corridor lengths that stops metallic rings. *Precise:* LOCKED (T11): 16 of 32 lines, 0.10–0.15 ms peak, allpass reads, default 0; Render runs the loop when > 0. (Ch. 5)
- **MS** — Mid plus Side: a forward mic and a sideways figure-8 decoded to stereo. *Precise:* side route s = (+a, −a); D18/D19 fixes LOCKED for T3b. (Ch. 7)

N–O

- **Natural (air) decay** — Treble dies a bit faster, as in real air. *Precise:* LOCKED (SL7a, ST-1/ST-2): per-line air loss for the line's delay, per-axis cut loss folded in. (Ch. 2)
- **Null / null test** — Flip one version upside down, add it to the other, measure what's left. *Precise:* reported in dB; t18 −71.8 dB. (Ch. 3, 8)
- **Nyquist** — The highest pitch a sample rate can carry: half the rate. *(general)* (Ch. 6)
- **Octave / third-octave** — A band whose top is twice its bottom / a third of that width. *(general acoustics)* (Ch. 6)
- **Omni** — A mic that hears all directions equally. (Ch. 7)

- OPEN — Undecided or unknown. (all)
- **ORTF** — Two cardioids slightly apart and angled out; the default setup. *Precise:* ±55° outward; today includes C at 0 dB; C off by default from T4b (D20). (Ch. 7)

P

- **Pan law** — How level is shared as you pan. *Precise:* the 2011 linear law, pan = (a(1 − t), a·t). (Ch. 7)
- **Parity** — Two versions sounding identical. *Precise:* Play vs Render, or today vs 2011 (`reaktor_parity`). (Ch. 3, 8)
- **Peakiness** — How spiky the fine detail of the tail's spectrum is; a measure of metallic ringing. *Precise:* std of fine-bin dB over 0.3–1.3 s; engine 7.20 dB vs noise 5.53. (Ch. 5)
- **Peaking cut (per-axis)** — A small fixed dip inside the loop. *Precise:* one RBJ cut per line at the first axial mode of X, Y or Z (line i % 3), −1.5 dB per pass, Q 1.5. (Ch. 5)
- **Phase (invert)** — Flipping a signal upside down. *Precise:* `phase_l/c/r` = ±1. (Ch. 7)
- **Play** — The live engine in the browser. *Precise:* `live_stereo` C compiled to wasm in an AudioWorklet. (Ch. 2, 3)
- PLANNED — An idea or later design, not decided in detail. (all)
- **Polar pattern** — Which directions a mic hears. (Ch. 7)
- **Porous absorber** — A soft panel (mineral wool, foam) that soaks up mids and treble. *Precise:* library presets 1″+1″ gap, 2″, 2″+2″, 4″+4″. (Ch. 6)
- **Prebuild** — Code pre-written and tested on the box, ready for a Build session. (Ch. 6, 8)

Q–R

- **Q** — How narrow a filter or resonance is; higher = narrower. (Ch. 5, 6)
- **reaktor_parity ("Hall")** — The switch that recreates the 2011 behaviour. *Precise:* g = Gain/N, plain H, `LIVE_32_REAKTOR` ; Smooth and Movement forced off. (Ch. 1, 5)
- **Reflection order** — How many bounces an echo has made. (Ch. 5)
- **Render** — The offline engine that writes a file. *Precise:* `render_pipeline.py` , IR + FFT convolution; 24-bit WAV. (Ch. 2, 3)
- **Room result** — A planned panel showing predicted vs rendered decay per band. *Precise:* DESIGNED (SL1), mock-up only. (Ch. 6)
- **Room shell** — An allowance for doors, windows and stud walls. *Precise:* D-SL6: 90 Hz membrane, Q 1, 30 % of each wall, αmax 0.35 (ASSUMED values). (Ch. 6)
- **Route** — Where a mic strip goes. *Precise:* centre, both, l, r, s, pan. (Ch. 7)

S

- **S1P2** — The three-stage line filter. *Precise:* one biquad packing a first-order low and high shelf, plus two peaks (D-SL7). (Ch. 6)
- **Sample / sample rate** — One measurement of the waveform / how many per second (44,100 or 48,000). (Ch. 3)
- **Scattering** — How much a surface spreads sound in all directions instead of mirroring it. *Precise:* sets the Eyring/Arau blend and s′ = max(mean scattering, 0.5). (Ch. 6)
- **Schroeder curve** — A smoothed decay curve used to measure reverb time. *(general acoustics: backward integration of the IR)* (Ch. 6)

- **SHA1** — A 40-character fingerprint of a file. (Ch. 8)
- **Shared tail** — One tail for all mics in Play to save CPU. *Precise:* off by default; 24.7 % saving. (Ch. 2)
- **Shelf filter** — A tone control that tilts everything above or below a pitch. (Ch. 6)
- **Smooth** — A cap on the smearing filters to reduce metallic ringing. *Precise:* LOCKED (T4): $|\text{Dffs}| \leq 0.7$, default ON, decay-corrected; off = bit-exact. (Ch. 5)
- **Solo / Mute (face)** — Hear only, or drop, the echoes touching one wall. *Precise:* early field only; the inject is unaffected. (Ch. 5)
- **SOS** — Second-order sections: a chain of small filter stages. (Ch. 7)
- **Source** — The instrument or voice in the room. *Precise:* position, height, yaw, radiation pattern. (Ch. 2)
- **Structure mining** — Reading Idle's 2011 Reaktor patch screen by screen to recover the exact design. (Ch. 1)
- **Surfaces mode** — Walls and panels decide how long each pitch rings. *Precise:* LOCKED (D-SL1–9), DESIGNED, not merged; per-line S1P2 filters inside the loop. (Ch. 2, 6)

T–Z

- **T20 / T30 / T60** — Ways of stating decay time. *Precise:* T60 = time for a 60 dB fall; T20 and T30 estimate it from the −5 to −25 dB and −5 to −35 dB parts of the Schroeder curve. (Ch. 5, 6, 8)
- **Tail** — The smooth fading wash after the early echoes; made by the FDN. (Ch. 2)
- **Trim** — Input level on a strip. (Ch. 7)
- **Tuned panel** — A panel that absorbs one narrow bass band (membrane or Helmholtz). (Ch. 6)
- **U87 Ai / SM57** — Two real microphones: a Neumann studio condenser and a Shure dynamic. *Precise:* SM57 × 3 default today; U87 Ai Cardioid planned default in SL7b (D-UI30/38). (Ch. 7)
- VERIFIED — Read in a cited file or measured. (all)
- **Voice** — One of 32 parallel copies of the processing, one per image. (Ch. 5)
- **wasm (WebAssembly)** — Compiled code that runs fast inside a browser. (Ch. 2, 3)
- **Width** — How wide the side signal is spread in MS. *Precise:* replaces trim on route s. (Ch. 7)
- **WORKLET_VER** — The version number of the live engine (51 in build 51). (Ch. 1)
- **XFAIL** — "Expected to fail": a test waiting for a feature that is not built yet. (Ch. 8)
- **XY** — Two directional mics at one point, angled apart. *Precise:* `xy_90`, `xy_120`. (Ch. 7)